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GROMACS 分子模拟基础实验教程

V1.0 版



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第一章 文档说明

1.1 关于本文档

1.1.1 文档目的

GROMACS 分子动力学模拟方向，一般实验学科的初学者基本 Linux 操作不熟练，不能快速判断 Linux 系统相关的报错，或者看英文文档比较费时。因此编写本文档，为相关初学 GROMACS 分子动力学模拟的同学提供参考。

本文档额外补充了一些分子模拟的前置基础知识：比如 Linux 基础操作、在 Linux 上安装 gromacs 软件、介绍分子模拟的理论概念、以及相关结构与模拟的可视化工具软件等。

总的来说，要掌握 GROMACS 分子动力学模拟实验基础的步骤和操作，本身的操作不复杂，重点是要理清清楚操作的细节和步骤，理解每一步操作的目的；

1.1.2 排版约定

正文字体使用微软雅黑；

代码或命令字体使用新宋体，需要执行的操作命令，我们用蓝色字体、灰色背景标记，例如：

```
gmx editconf -f lAKI_processed.gro -o lAKI_newbox.gro -c -d 1.0 -bt cubic
```

1.1.3 面向的对象

本文档没有多少高深的知识理论或技术，仅供有一些Linux操作基础，GROMACS分子模拟没有入门的同学参考学习。

1.2 文档声明

- 本文档免费发布，不做任何商业用途，不对任何操作和结果负责，请勿恶意传播
- 本文档主要来自www.mdtutorials.com，也参考了部分网络资料
- 本文档为个人内部培训讲义，水平有限，疏漏或错字在所难免，但不影响阅读。
- 如发现问题或有疑问，欢迎联系本人讨论交流。

1.3 文档联系人

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微信二维码见下方



生信实战_段老师
上海 黄浦



扫一扫上面的二维码图案，加我为朋友。

第二章 Linux 系统获取和配置概述

由于官方Gromacs是在Linux系统上运行的, 以及部分软件工具也是基于Linux系统的。Linux系统的基础操作和使用成为生物信息学或计算化学等相关专业必备的计算机基础。由于Linux相关的基础资料非常多, 这里我们不赘述, 简单说明一下。

2.1 获取 Linux 环境的方式

2.1.1 使用实验室现成的服务器

如果实验室或单位有现成的Linux服务器使用, 那就直接使用。下面的几种获取Linux的方式可以跳过。

2.1.2 使用 win10 内置 Linux 子系统

这种方式可以在win10系统上启用一个内置的Linux子系统, 相关方法可以查阅资料, 或者在“生信实战”公众号查阅文章。

2.1.3 购买 Linux 云主机

目前IT界各大云计算主流厂商都提供了订购云主机服务, 可以直接花钱订购, 如阿里云、腾讯云、华为云等等。相关方法可以官网资料, 或者在“生信实战”公众号查阅文章。

2.1.4 本地安装 Linux 虚拟机

由于我自己购买了服务器,因此我一般会在本地安装Linux虚拟机进行操作。具体见下一章节。

2.2 本地安装 Ubuntu Linux 虚拟机参考

有人说Ubuntu比较好用、适合入门等,其实不见得,主要是看个人习惯。个人最近这些年一直用的CentOS系列Linux, Ubuntu用的少,但多年前开始学习linux也是用的Ubuntu。

大家习惯就行,这些是次要的。

下面我们就以 Ubuntu 为例,使用 VMWARE 安装虚拟机(相关的资料也很多,可以自己查阅网络资料)

其实我是安装好一个系统后,后面再需要的话就直接虚拟机克隆一份,可以几分钟内就得到一个 Ubuntu Linux 系统。

2.2.1 关于 ubuntu22 系统

GCC was updated to the 11.2.0 release, binutils to 2.38, and glibc to 2.35.

Python  now ships at version 3.10.4, Perl  at version 5.34.0. LLVM now defaults to version 14. go lang defaults to version 1.18.x. rustc defaults to version 1.58. In addition to OpenJDK 11, OpenJDK 18 is now provided (but not used for package builds). Ruby  was updated from v2.7.4 to v3.0.

2.2.2 新建 VMware 虚拟机

先安装好VMware® Workstation 15 Pro（也可以使用其他版本）

准备好Ubuntu22的ISO系统镜像：去官方主页查找下载

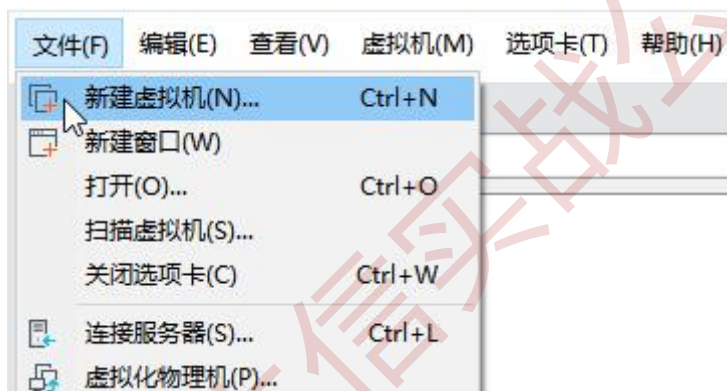
<https://ubuntu.com/download/server>

也可以直接使用这个链接下载（链接后面可能会失效，如果失效建议去ubuntu.com官网找）

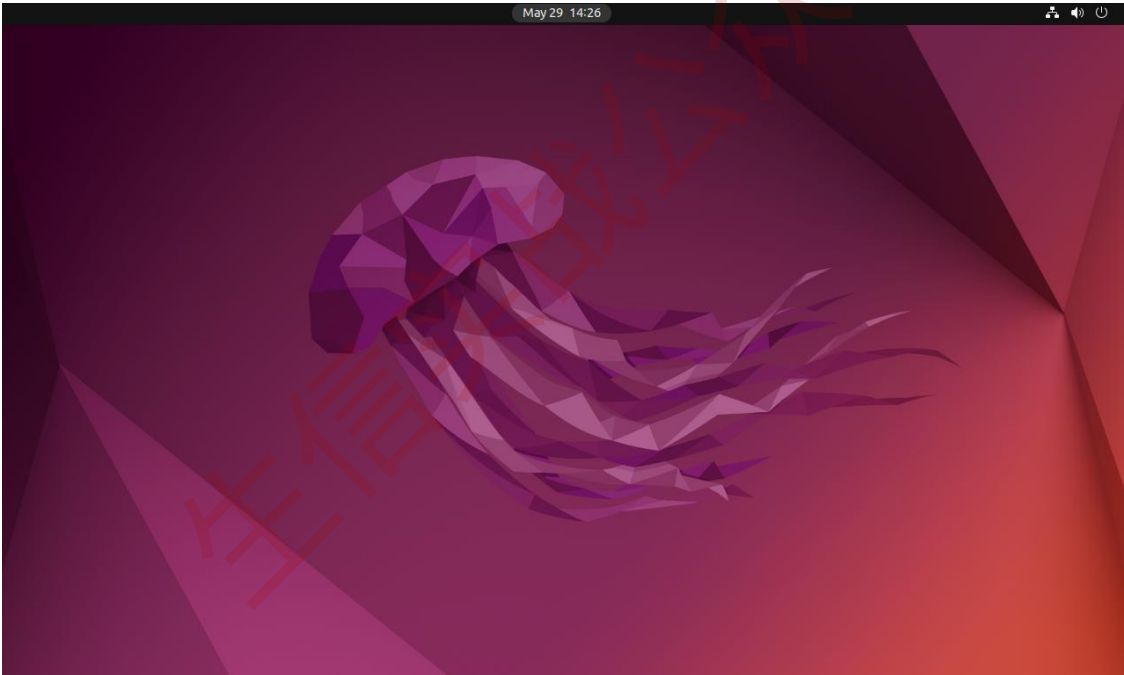
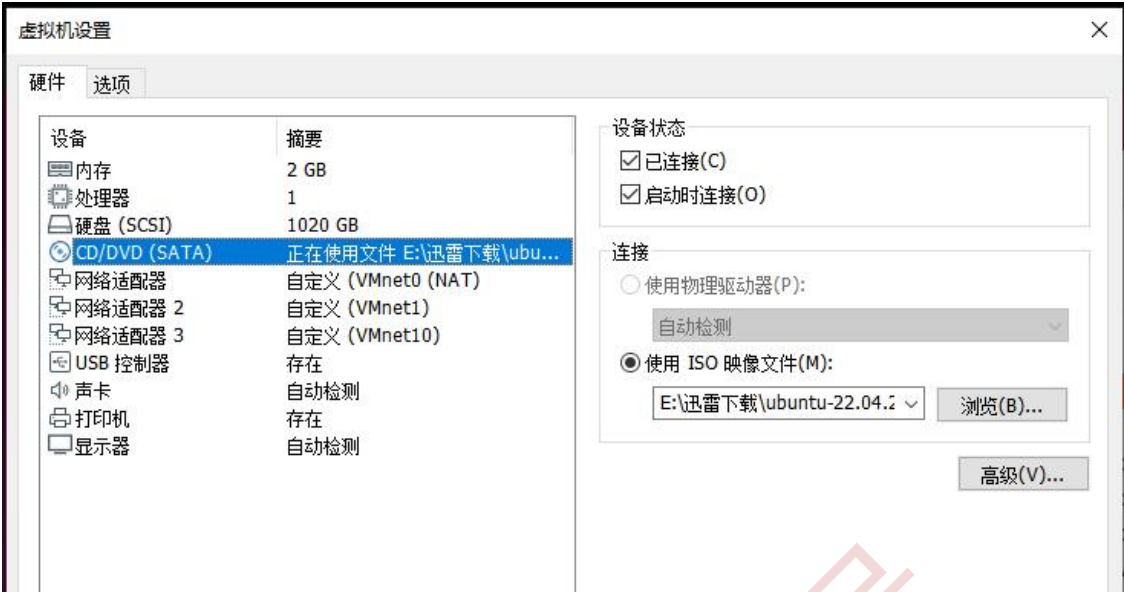
<https://mirrors.cnnic.cn/ubuntu-releases/22.04.2/ubuntu-22.04.2-live-server-amd64.iso>

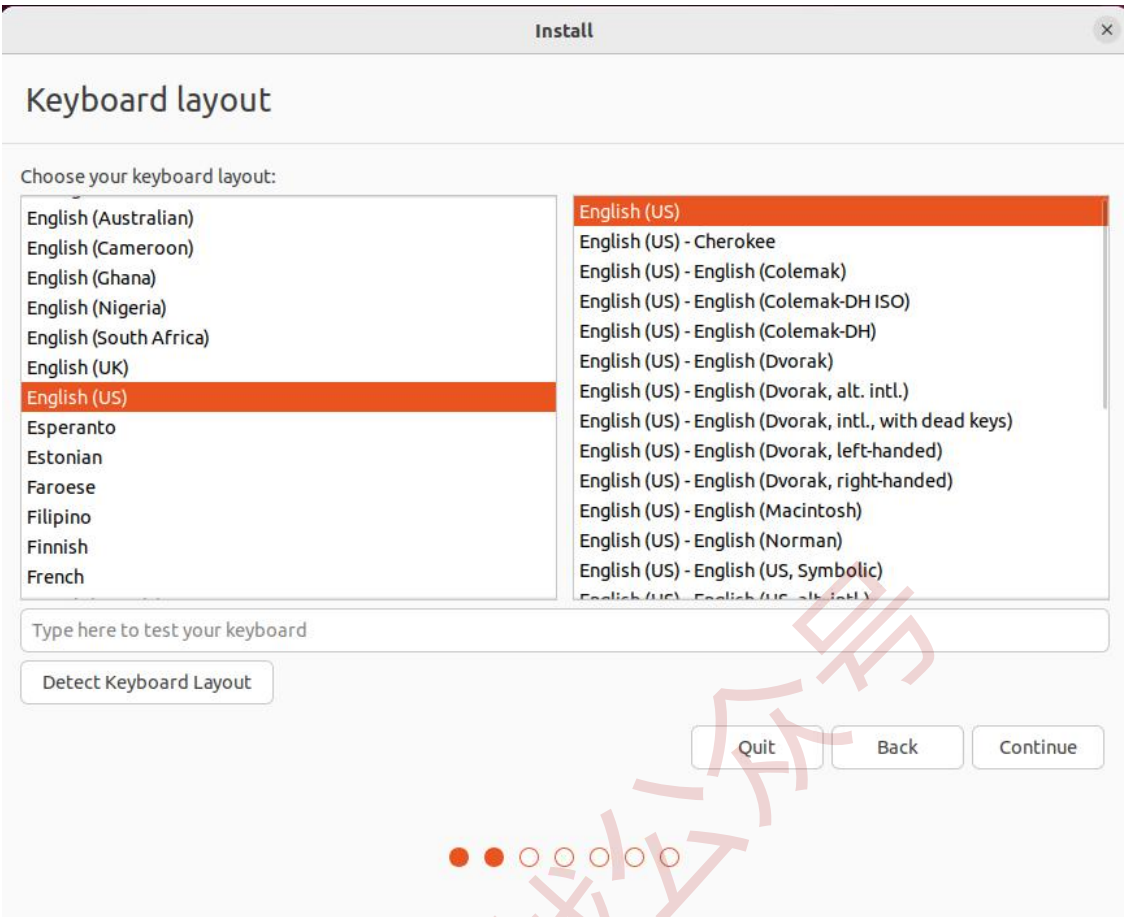
（链接复制到迅雷或浏览器下载工具里，下载即可）

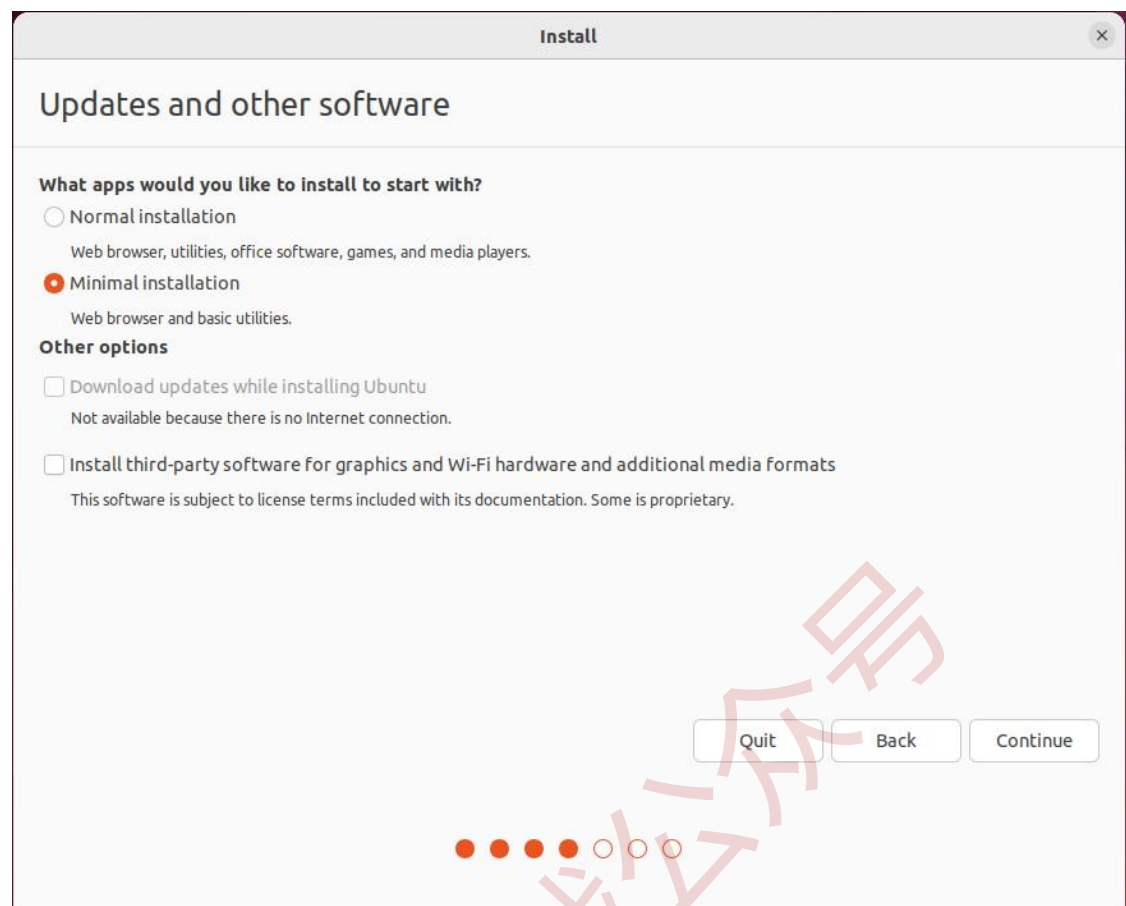
使用VMware新建虚拟机：

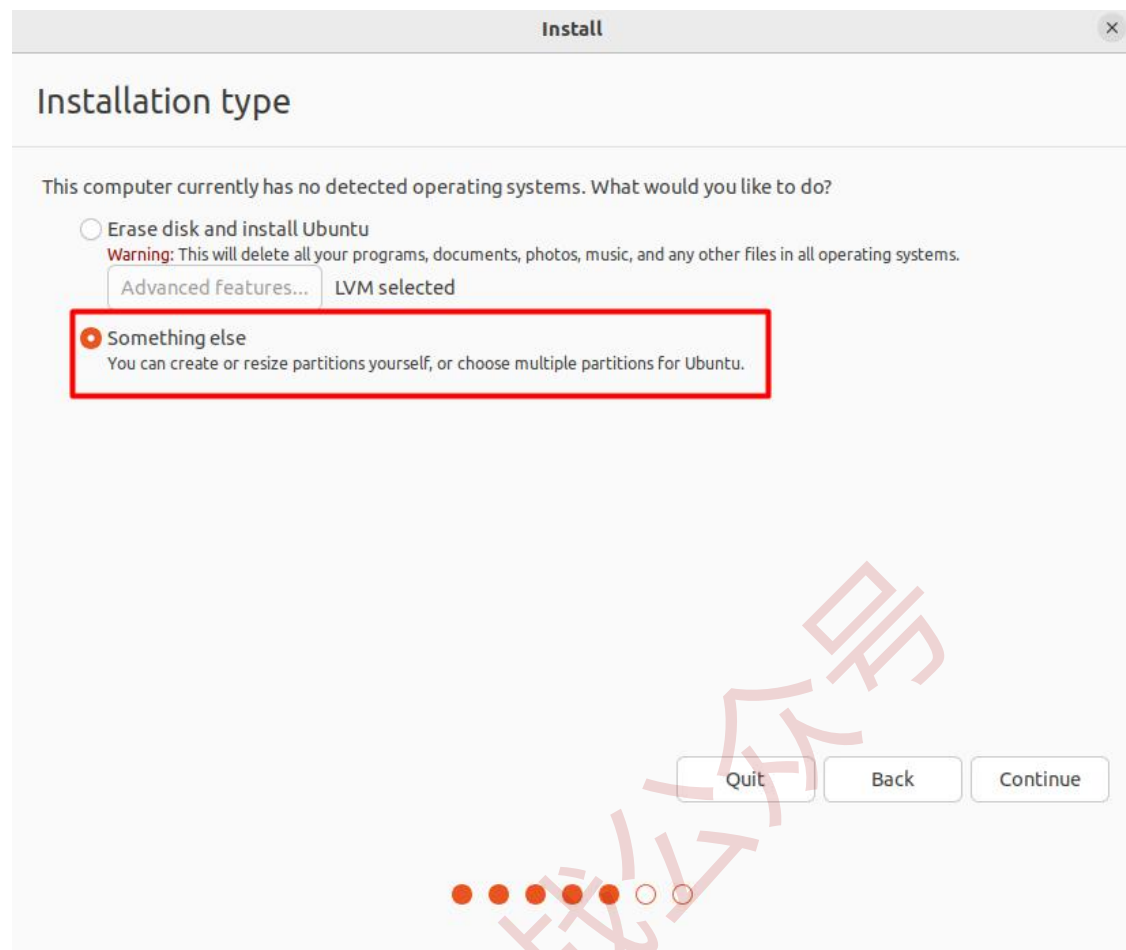


根据下一步的提示进行安装，我配置的虚拟机如下（我这里配置了多个网卡，你选择你自己默认的NAT网卡即可）

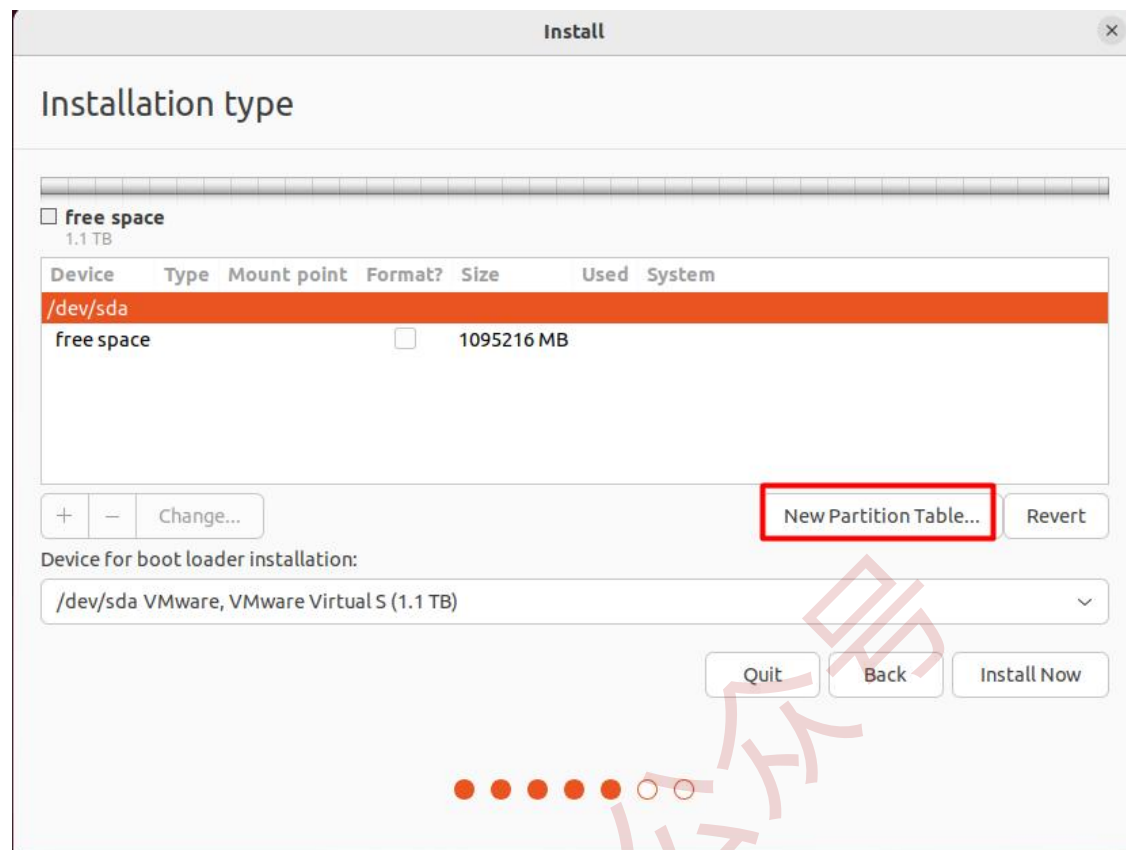


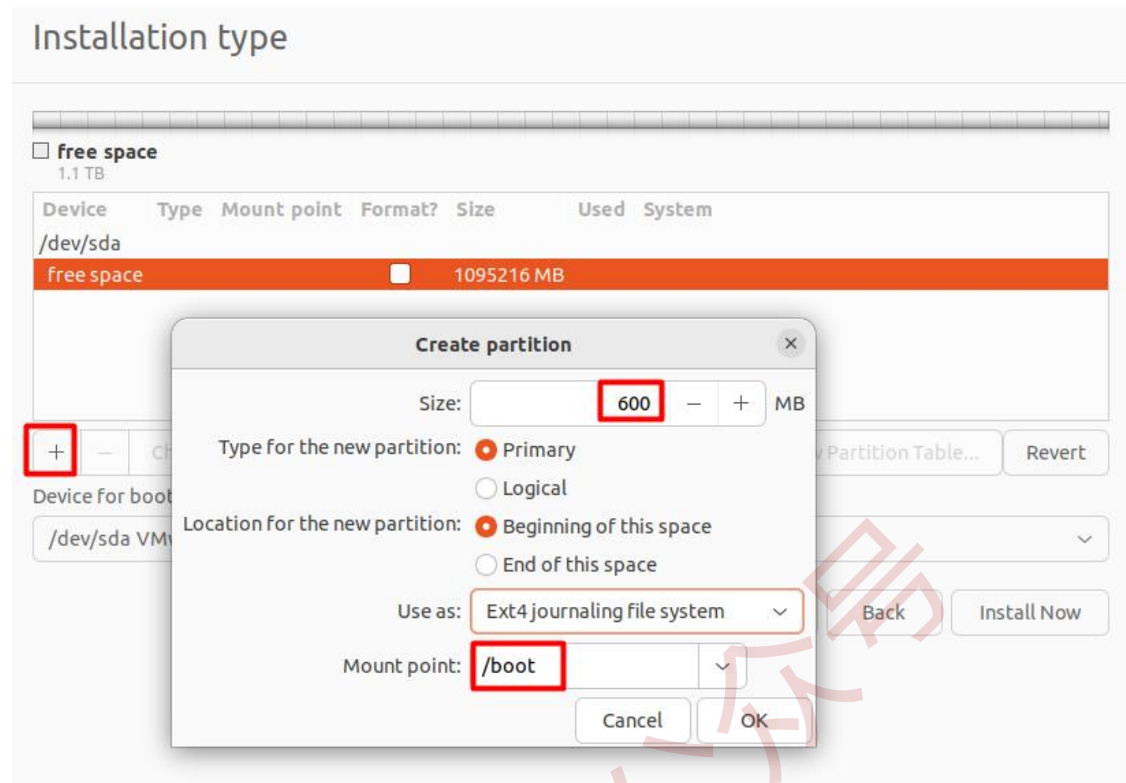




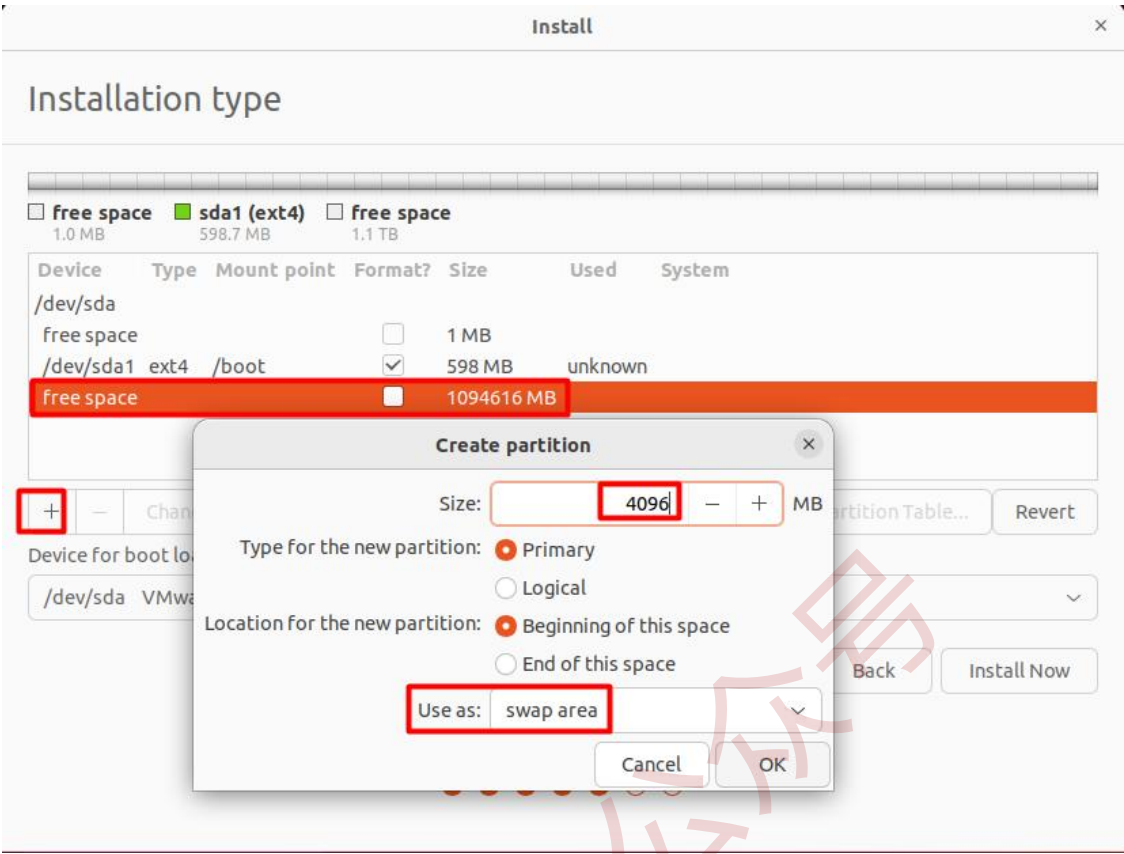


2.2.3 磁盘分区

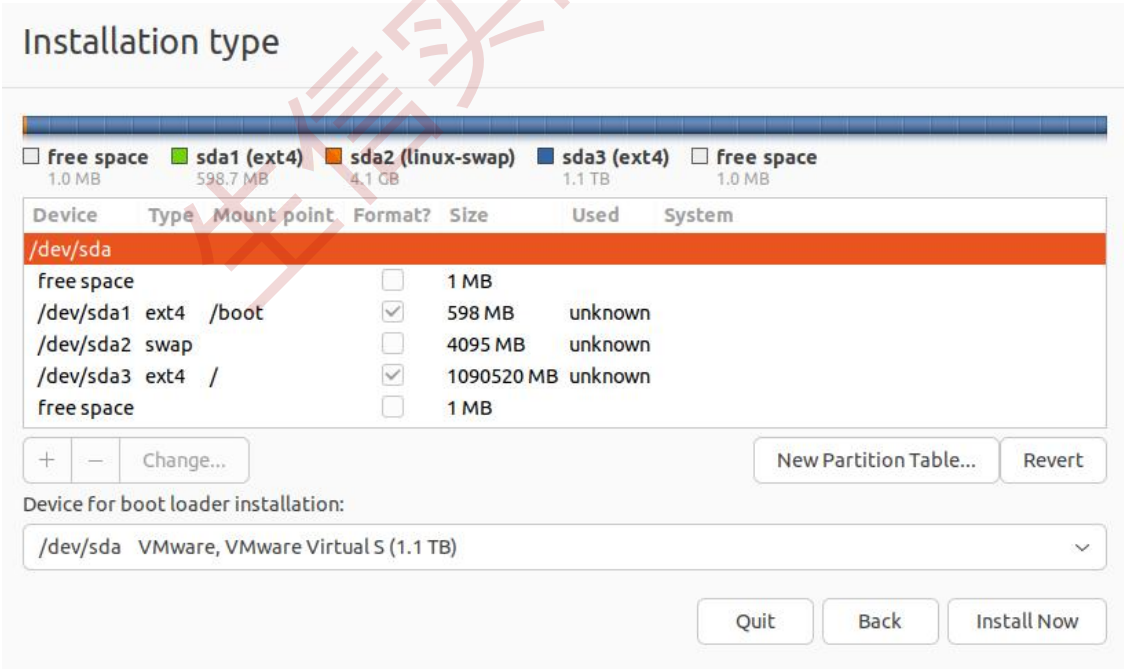
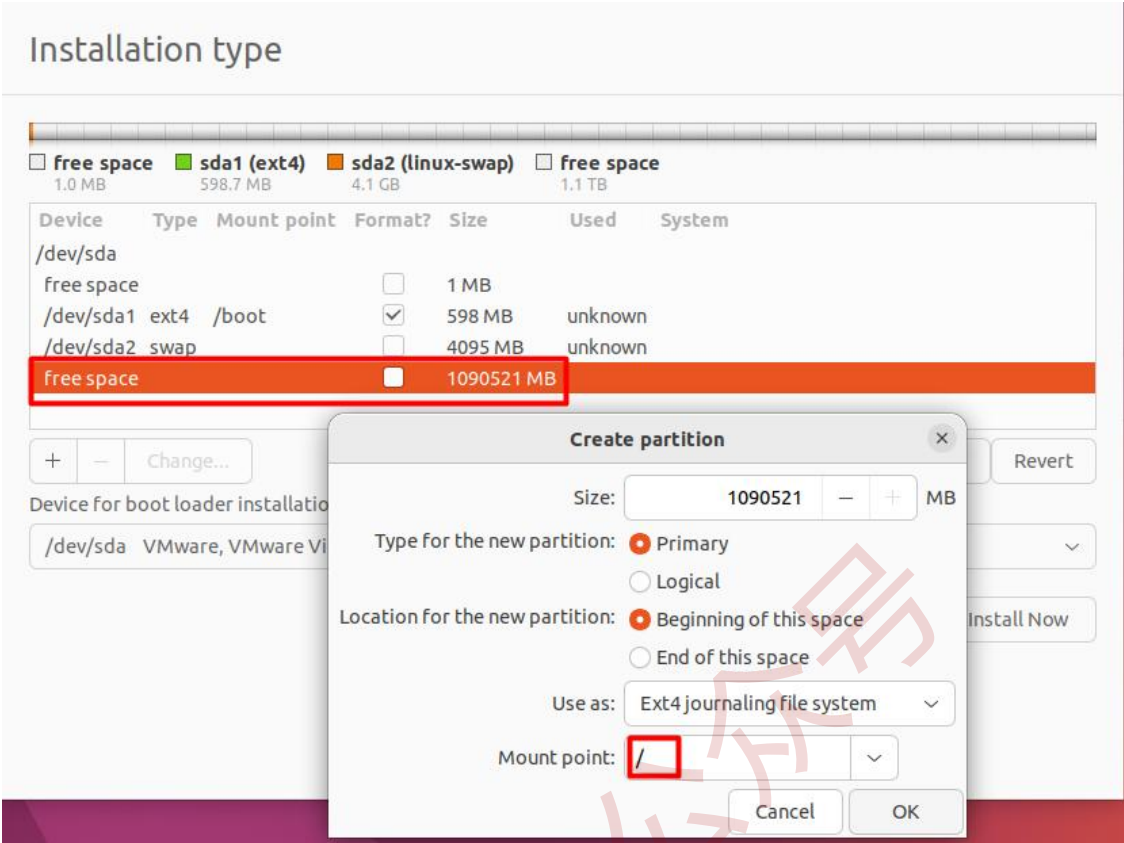




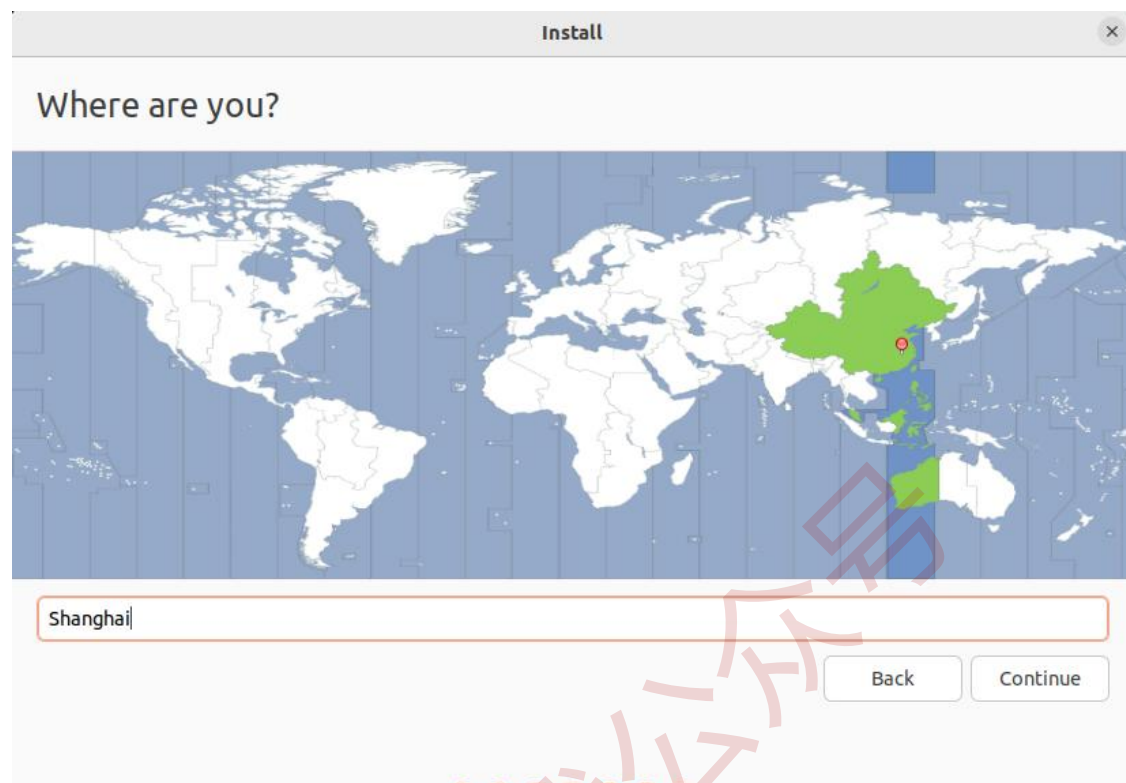
再加一个swap分区



其他都给根分区



2.2.4 配置和安装完成



Install

Who are you?

Your name: asp

Your computer's name: ding-gene95-ubuntu22

The name it uses when it talks to other computers.

Pick a username: asp

Choose a password: Good password

Confirm your password:

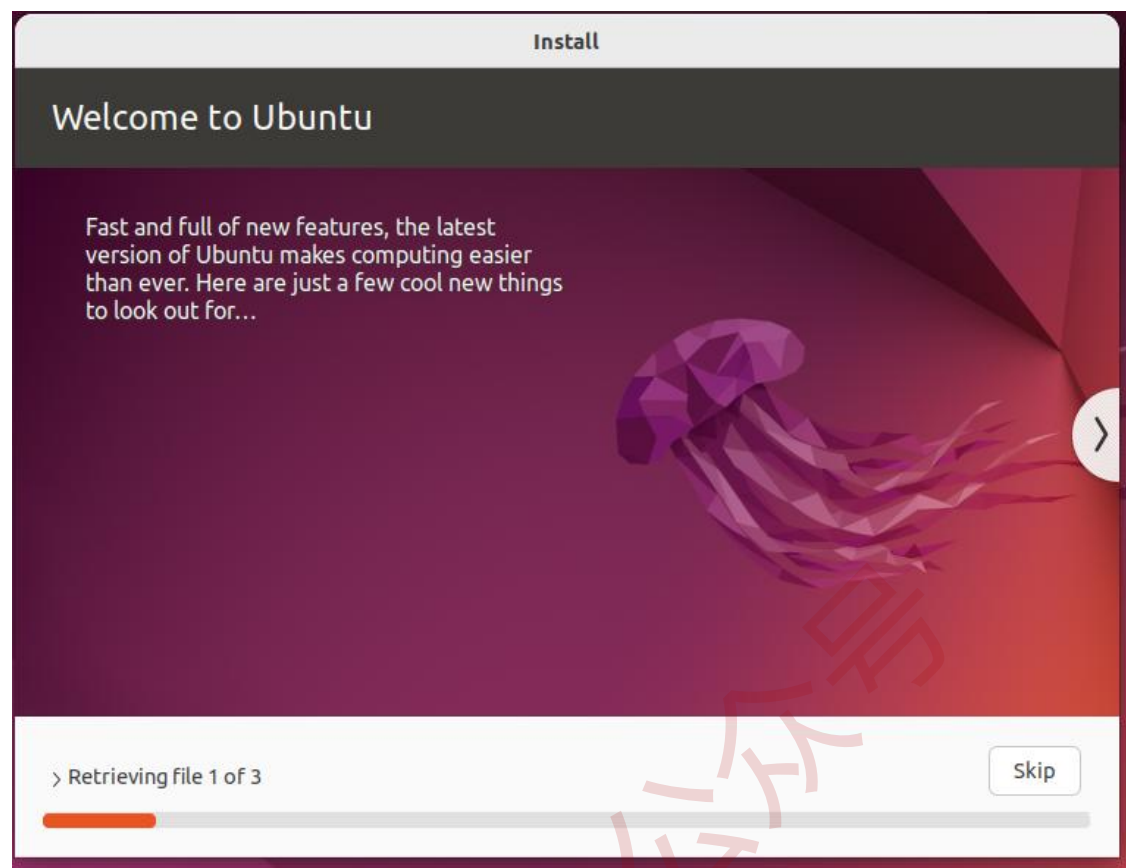
☐ Log in automatically

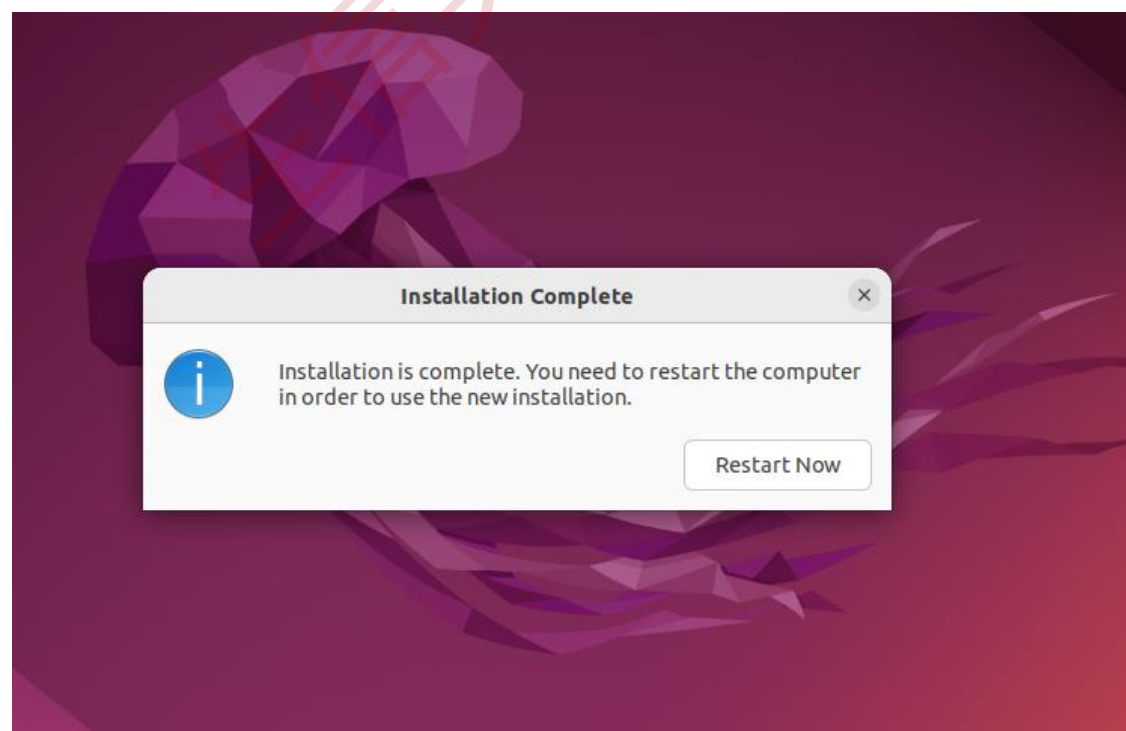
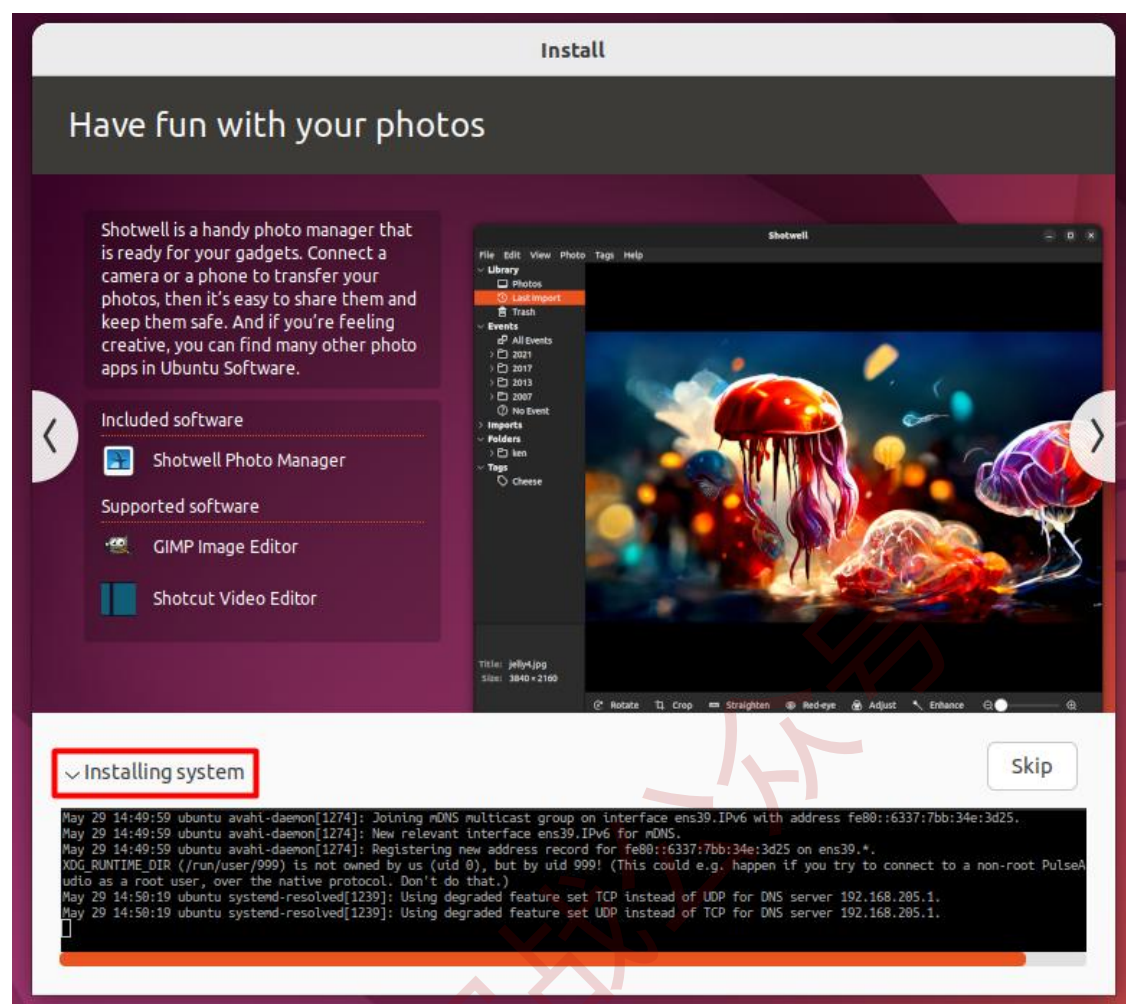
☒ Require my password to log in

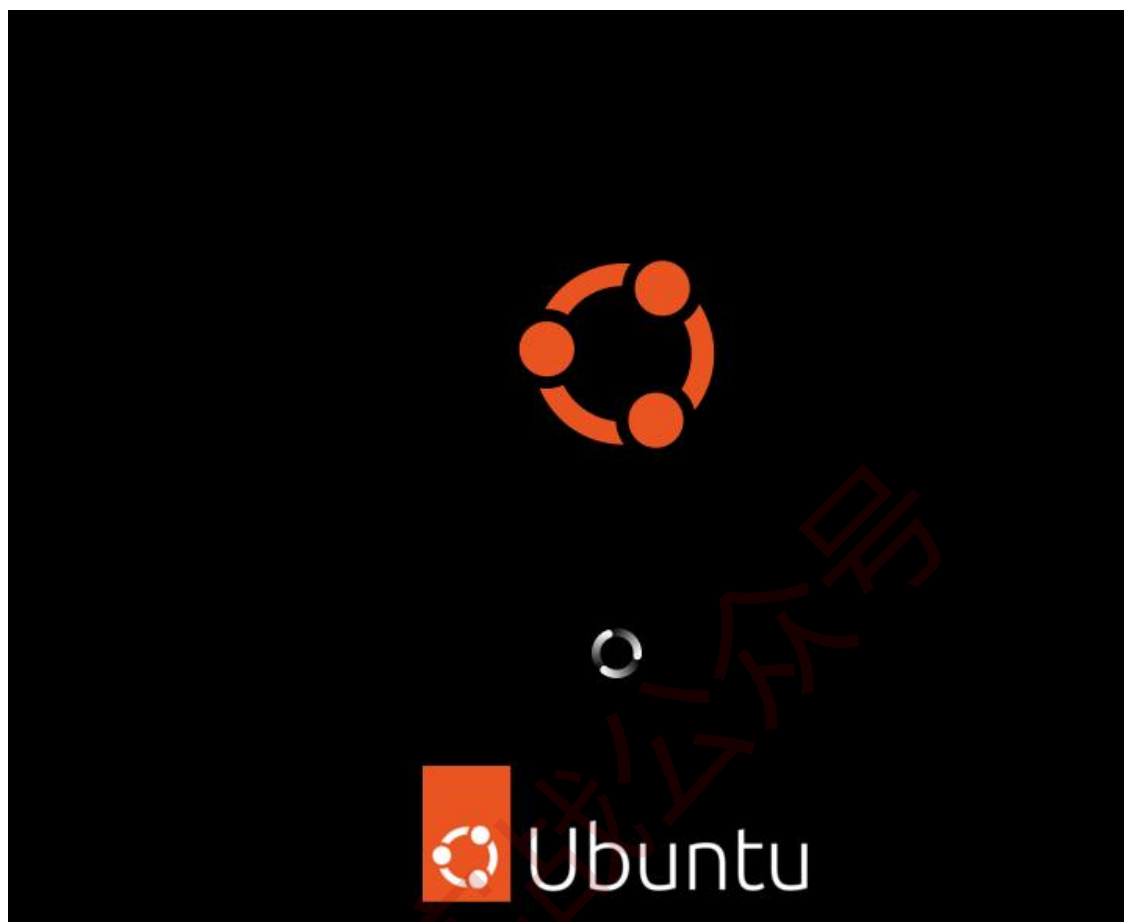
☐ Use Active Directory

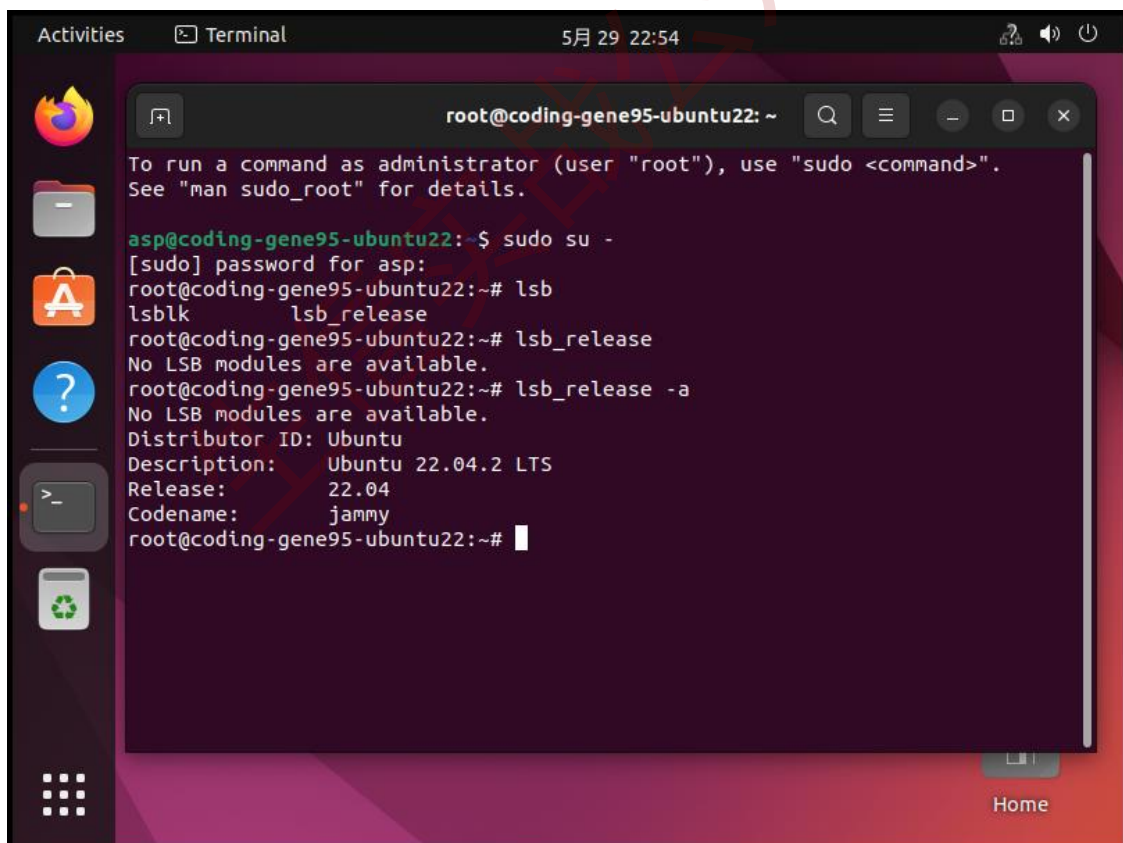
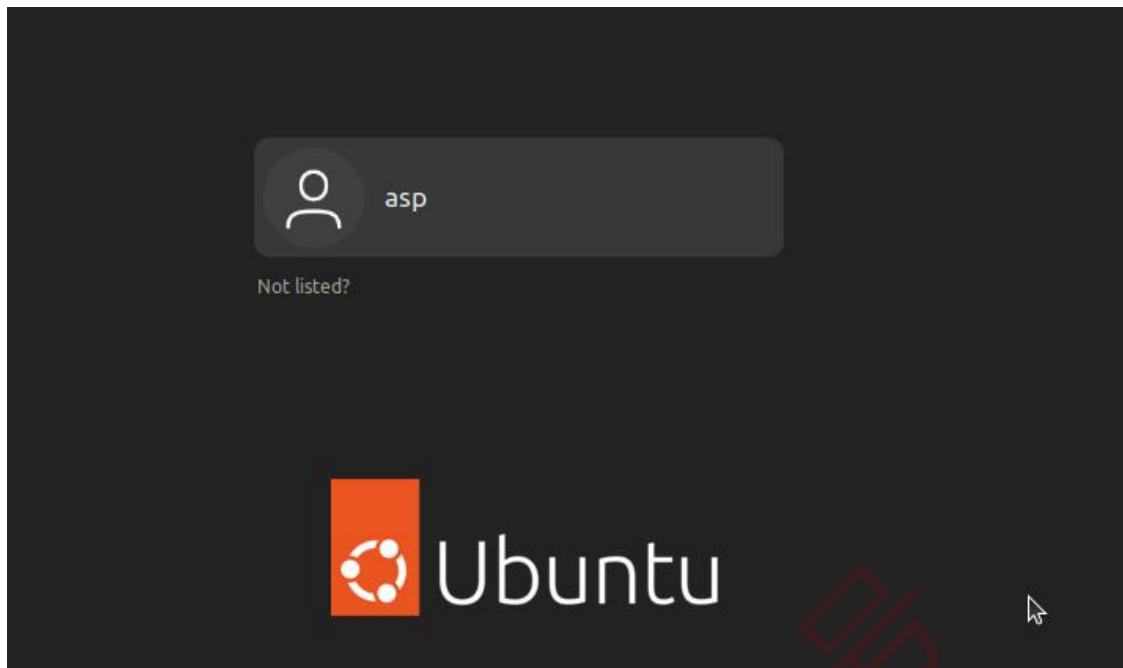
You'll enter domain and other details in the next step.

BackContinue

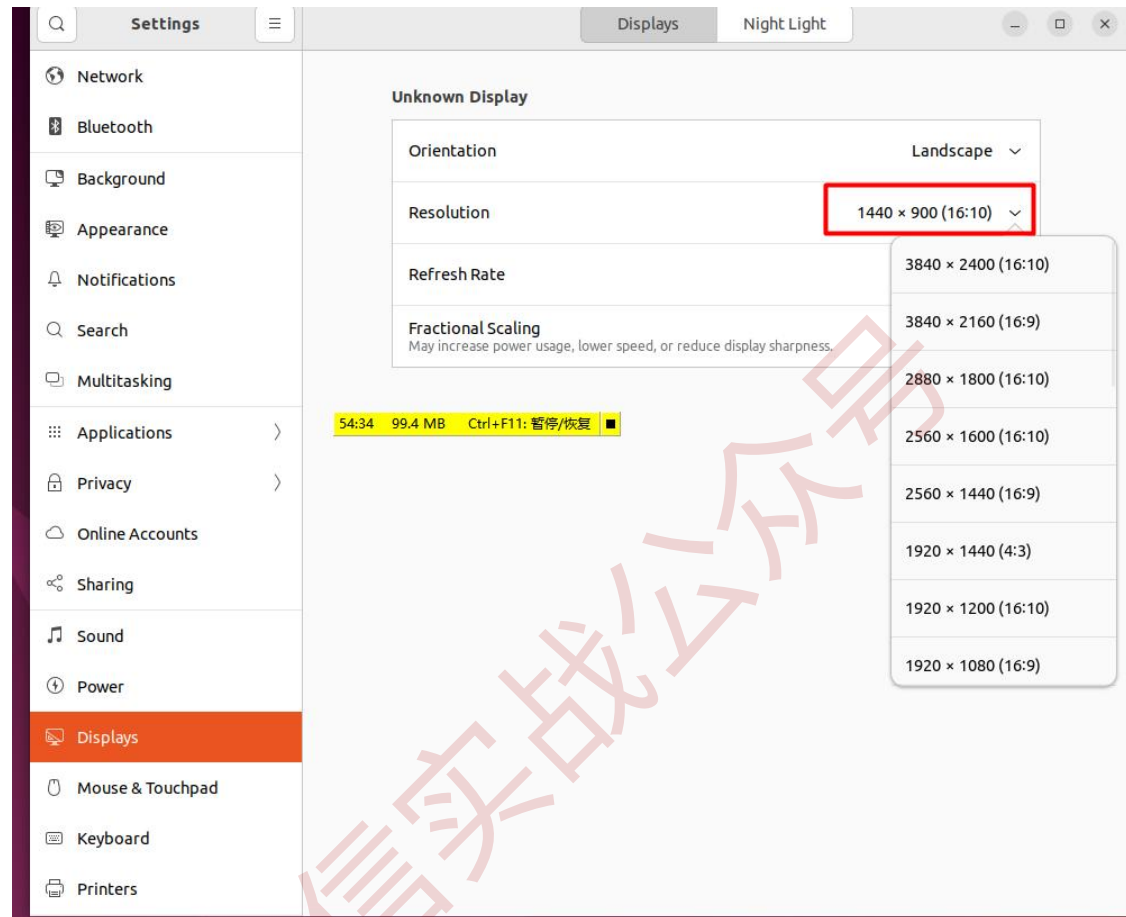






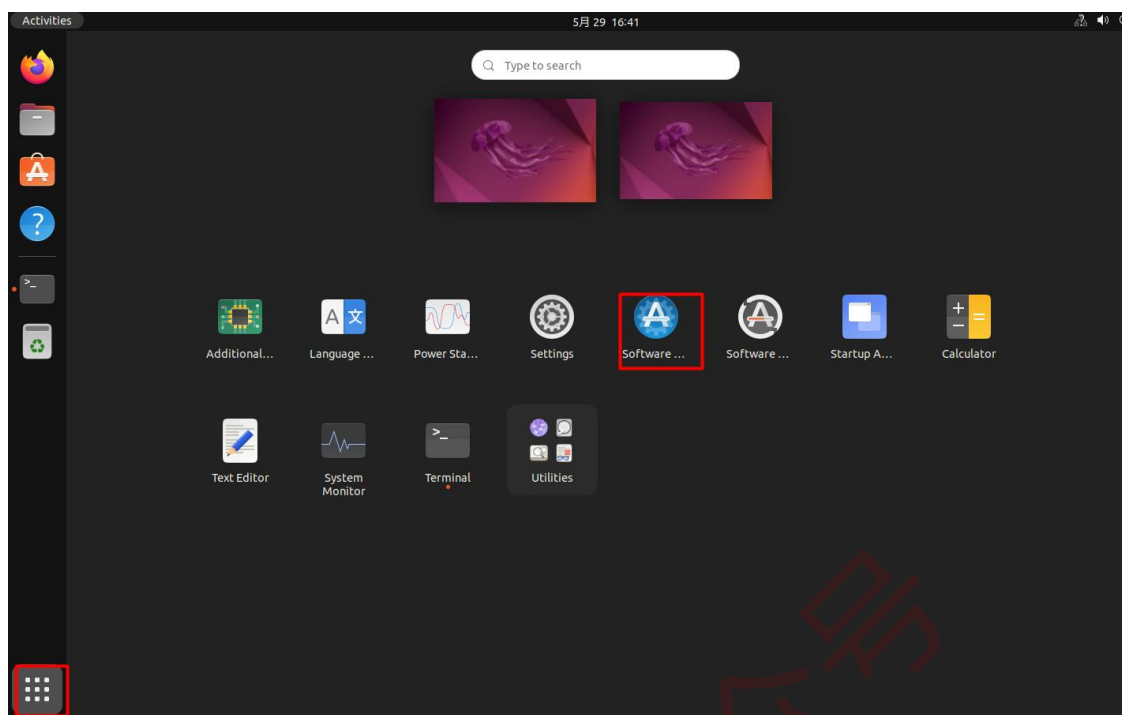


配置分辨率

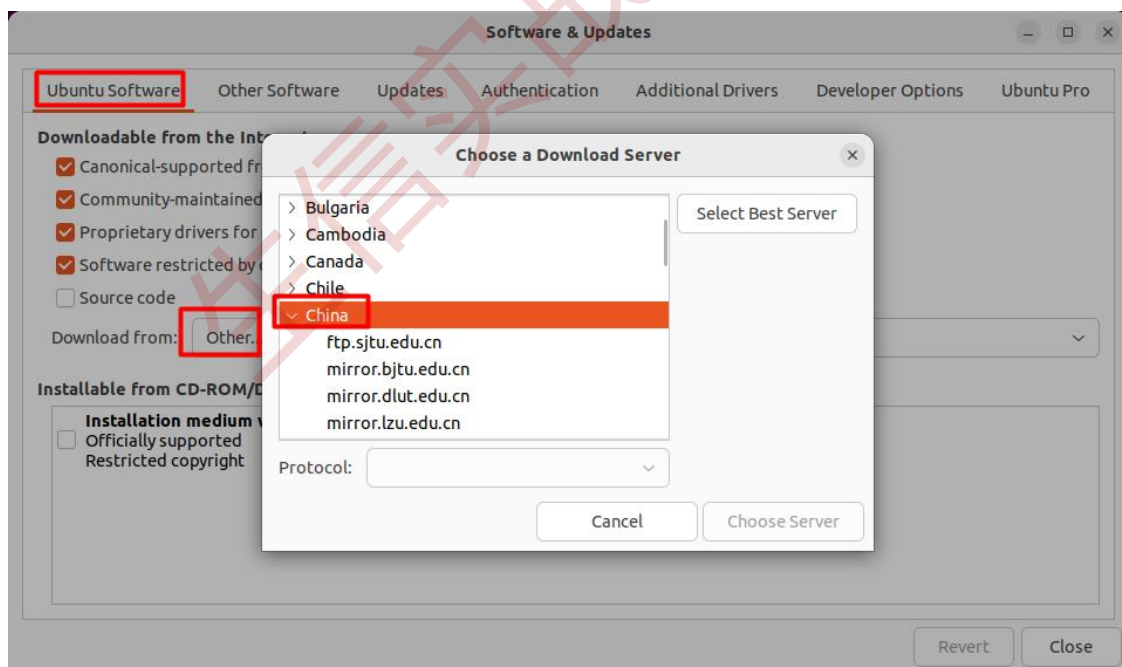


2.2.5 配置 apt 软件源

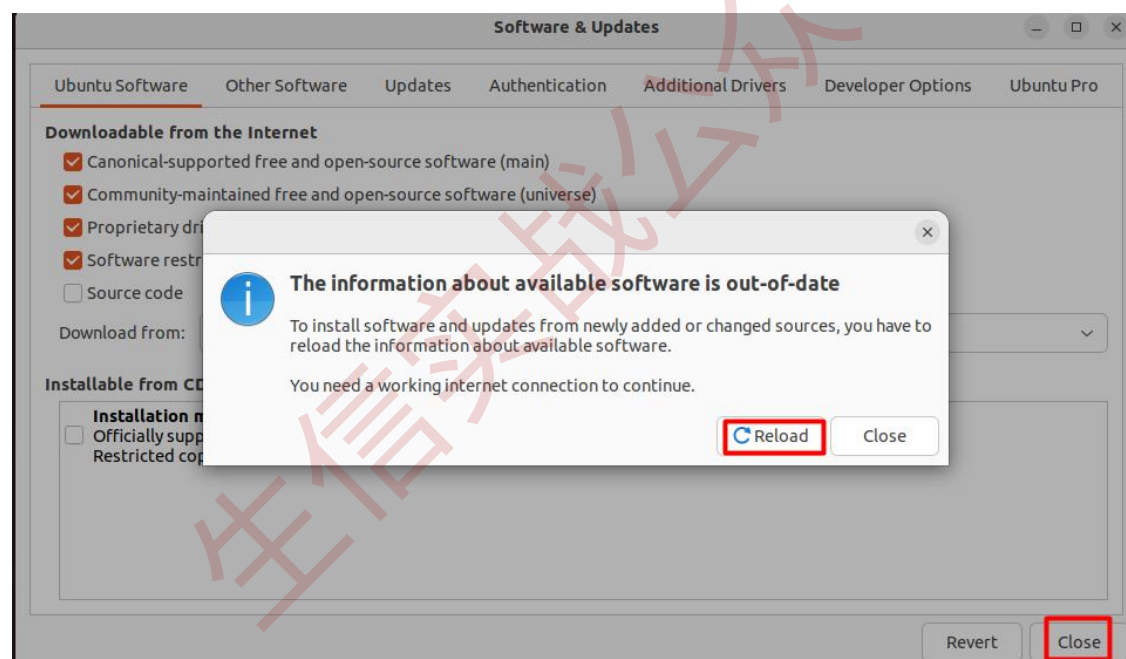
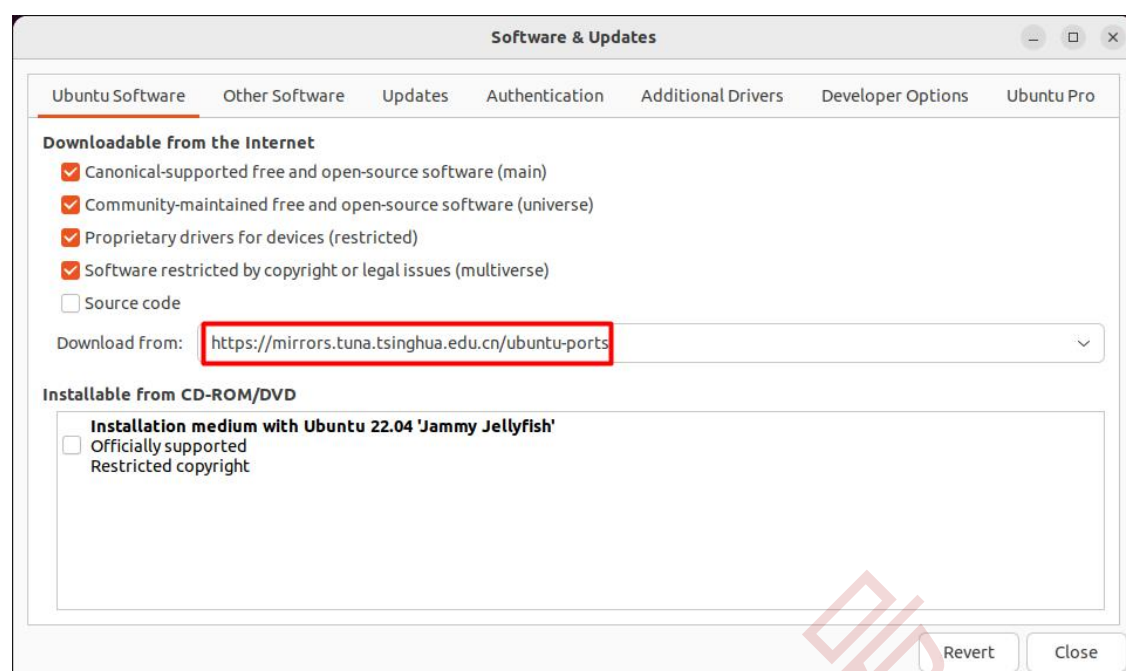
从主菜单，选择Software



选择国内的源



我们选清华大学的这个源



更新源 <https://mirrors.tuna.tsinghua.edu.cn/help/ubuntu/>



AOSP

Adoptium

CPAN

CRAN

CTAN

CocoaPods

NetBSD

OpenBSD

OpenMediaVault

alpine

anaconda

anthon

arch4edu

archlinux

archlinuxarm

archlinuxcn

armbian

bazel-apt

Ubuntu 软件仓库镜像使用帮助

是否使用 HTTPS ☒ 是

是否使用 sudo ☒ 是

本镜像仅包含 32/64 位 x86 架构处理器的软件包，在 ARM(arm64, armhf)、PowerPC(ppc64ell)、RISC-V(riscv64) 和 S390x 等架构的设备上（对应官方源为 ports.ubuntu.com）请使用 [ubuntu-ports 镜像](#)。

Ubuntu 的软件源配置文件是 `/etc/apt/sources.list`。将系统自带的该文件做个备份，将该文件替换为下面内容，即可使用选择的软件源镜像。

Ubuntu 版本:

使用官方安全更新软件源: ☒ 是

启用 proposed: ☐ 否

启用源码镜像: ☐ 否

```
# 默认注释了源码镜像以提高 apt update 速度，如有需要可自行取消注释
deb https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy main restricted universe multiverse
# deb-src https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy main restricted universe multiverse
deb https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy-updates main restricted universe multiverse
# deb-src https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy-updates main restricted universe multiverse
deb https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy-backports main restricted universe multiverse
# deb-src https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy-backports main restricted universe multiverse

# deb https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy-security main restricted universe multiverse
# # deb-src https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy-security main restricted universe multiverse

deb http://security.ubuntu.com/ubuntu/ jammy-security main restricted universe multiverse
# deb-src http://security.ubuntu.com/ubuntu/ jammy-security main restricted universe multiverse

# 预发布软件源，不建议启用
# deb https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy-proposed main restricted universe multiverse
# # deb-src https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy-proposed main restricted universe multiverse
```

因镜像站同步有延迟，可能会导致生产环境系统不能及时检查、安装上最新的安全更新，不建议替换 security 源。

默认注释了源码镜像以提高 apt update 速度，如有需要可自行取消注释

```
deb https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy main restricted universe multiverse
```

```
# deb-src https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy main restricted universe multiverse
```

```
deb https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy-updates main restricted universe multiverse
```

```
# deb-src https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy-updates main restricted universe multiverse
```

```
deb https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy-backports main restricted universe multiverse
```

```
# deb-src https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy-backports main restricted universe multiverse
```

```
# deb https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy-security main restricted universe multiverse
```

```
# # deb-src https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy-security main restricted universe multiverse
```

```
deb http://security.ubuntu.com/ubuntu/ jammy-security main restricted universe
```

```
multiverse
# deb-src http://security.ubuntu.com/ubuntu/ jammy-security main restricted
universe multiverse

# 预发布软件源，不建议启用

# deb https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy-proposed main restricted
universe multiverse
# # deb-src https://mirrors.tuna.tsinghua.edu.cn/ubuntu/ jammy-proposed main
restricted universe multiverse
```

复制粘贴到/etc/apt/sources.list

执行源更新 apt-get update

```
root@coding-gene95-ubuntu22:~# cat /etc/apt/sources.list | egrep -v '^\$|^#'
deb https://mirrors.ustc.edu.cn/ubuntu/ jammy main restricted universe multiverse
deb https://mirrors.ustc.edu.cn/ubuntu/ jammy-security main restricted universe
multiverse
deb https://mirrors.ustc.edu.cn/ubuntu/ jammy-updates main restricted universe
multiverse
deb https://mirrors.ustc.edu.cn/ubuntu/ jammy-backports main restricted universe
multiverse
```

2.2.6 启用 root 用户

```
asp@coding-gene95-ubuntu22:~$ sudo su -
[sudo] password for asp:
```

```
root@coding-gene95-ubuntu22:~# passwd
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
passwd: password updated successfully
```

2.2.7 配置静态 IP

```
root@coding-gene95-ubuntu22:~# cat /etc/netplan/01-network-manager-all.yaml
# Let NetworkManager manage all devices on this system
network:
  renderer: NetworkManager
  ethernets:
    ens33:
      dhcp4: yes
      routes:
        - to: default
          via: 10.10.1.2
      nameservers:
        addresses:
          - 10.10.1.2
    ens38:
      dhcp4: no
      addresses:
        - 192.168.205.95/24
      routes:
        - to: default
version: 2
```

2.2.8 开启 ssh 远程连接并允许 root 登录

```
apt-get install openssh-server
```

```
vi /etc/ssh/sshd_config
```

```
32 #LoginGraceTime 2m
33 #PermitRootLogin prohibit-password
34 #StrictModes yes
35 #MaxAuthTries 6
36 #MaxSessions 10
```

这一行改成：

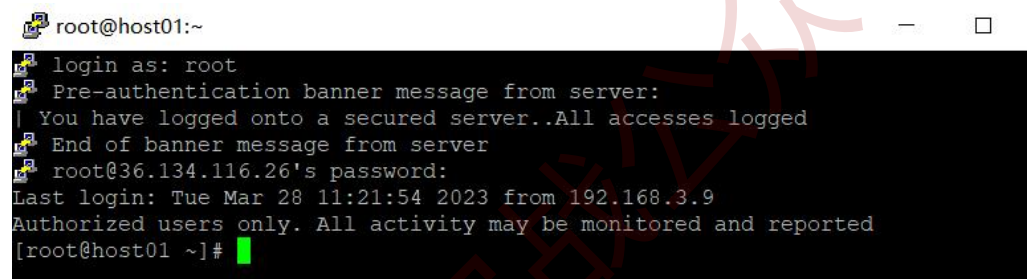
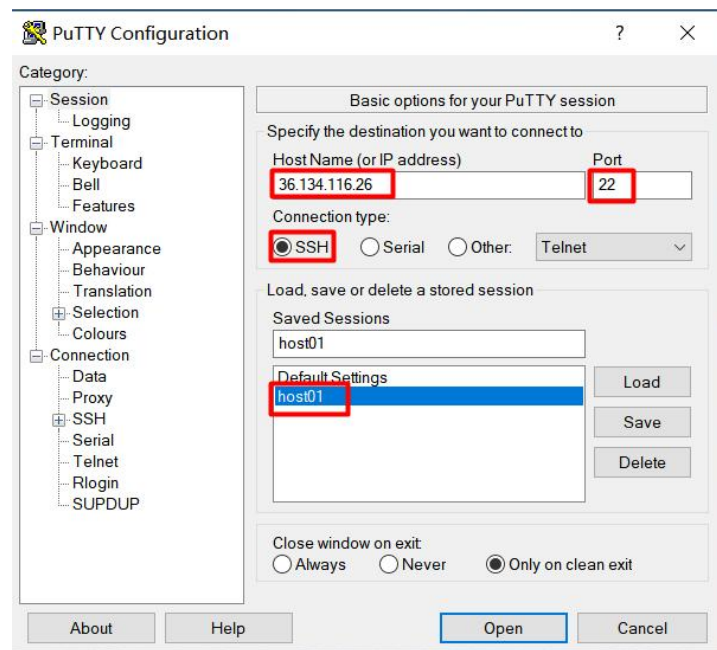
```
PermitRootLogin yes
```

```
/etc/init.d/ssh restart
```

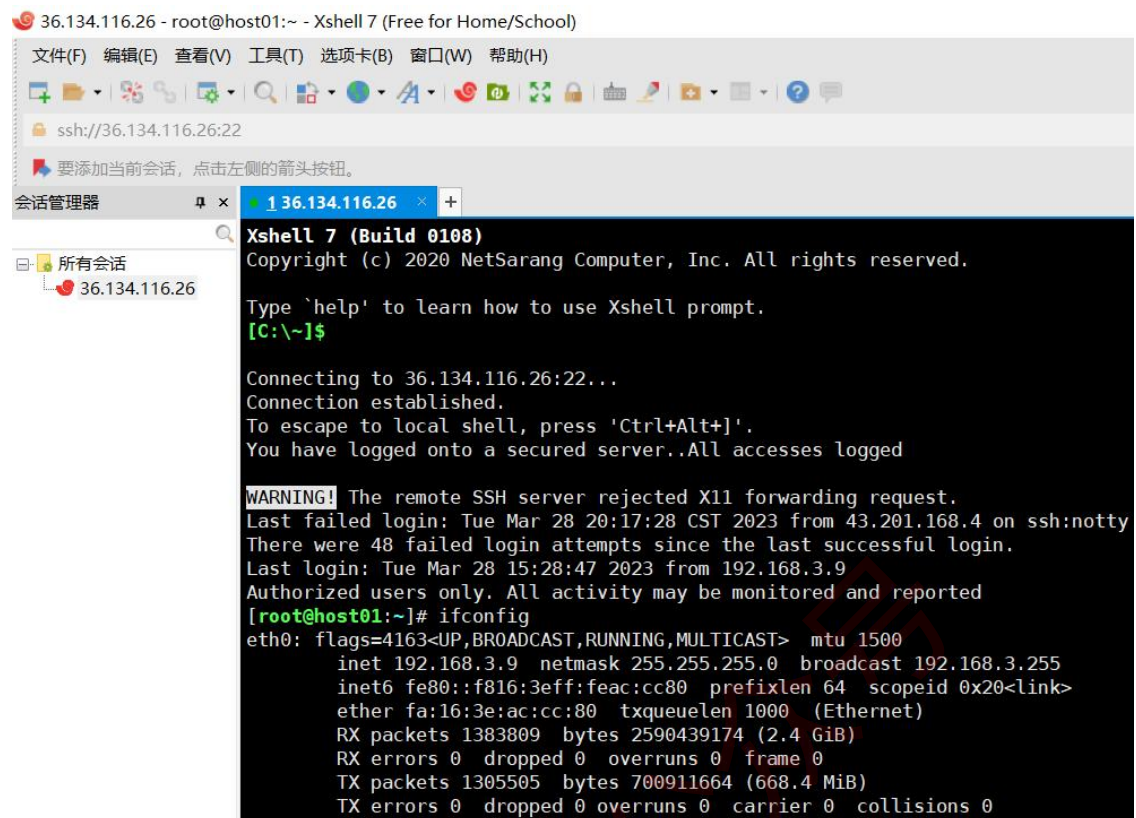
2.3 使用远程客户端连接 Linux

拥有linux系统后，可以通过ssh客户端工具连接服务器：

2.3.1 方式 1-使用 putty



2.3.2 方式 2-使用 Xshell



36.134.116.26 - root@host01:~ - Xshell 7 (Free for Home/School)

文件(F) 编辑(E) 查看(V) 工具(T) 选项卡(B) 窗口(W) 帮助(H)

ssh://36.134.116.26:22

要添加当前会话，点击左侧的箭头按钮。

会话管理器

所有会话

36.134.116.26

Xshell 7 (Build 0108)

Copyright (c) 2020 NetSarang Computer, Inc. All rights reserved.

Type 'help' to learn how to use Xshell prompt.

[C:\>]\$

Connecting to 36.134.116.26:22...

Connection established.

To escape to local shell, press 'Ctrl+Alt+J'.

You have logged onto a secured server..All accesses logged

WARNING! The remote SSH server rejected X11 forwarding request.

Last failed login: Tue Mar 28 20:17:28 CST 2023 from 43.201.168.4 on ssh:notty

There were 48 failed login attempts since the last successful login.

Last login: Tue Mar 28 15:28:47 2023 from 192.168.3.9

Authorized users only. All activity may be monitored and reported

[root@host01:~]# ifconfig

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500

inet 192.168.3.9 netmask 255.255.255.0 broadcast 192.168.3.255

inet6 fe80::f816:3eff:feac:cc80 prefixlen 64 scopeid 0x20<link>

ether fa:16:3e:ac:cc:80 txqueuelen 1000 (Ethernet)

RX packets 1383809 bytes 2590439174 (2.4 GiB)

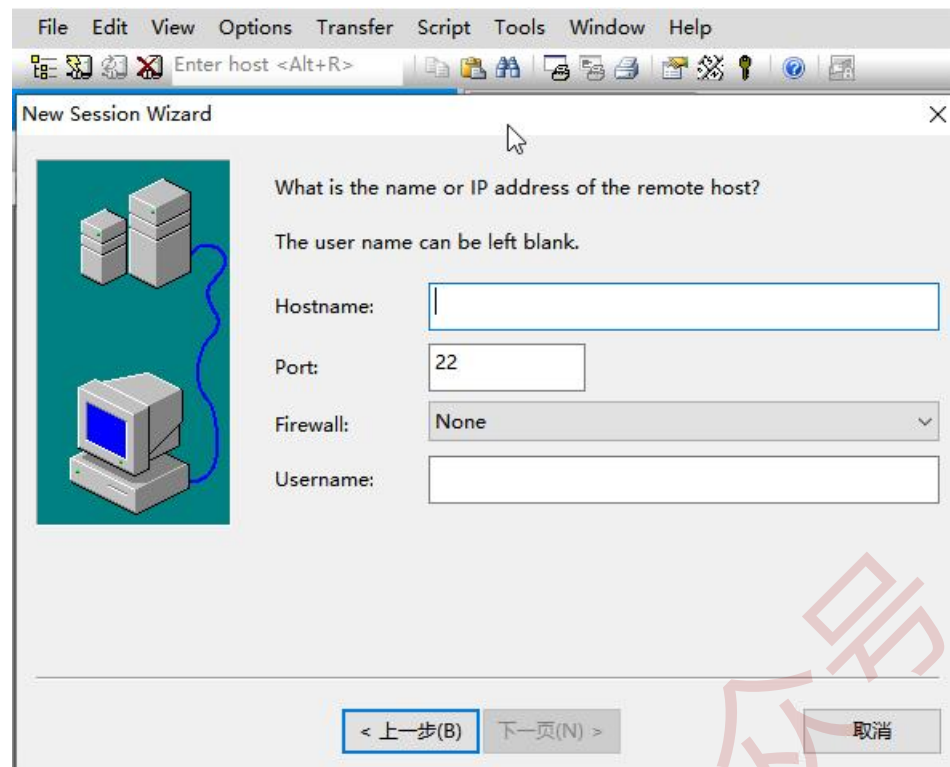
RX errors 0 dropped 0 overruns 0 frame 0

TX packets 1305505 bytes 700911664 (668.4 MiB)

TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

也可以使用其他客户端软件，建议Xshell

2.3.3 方式 3-使用 SecureCRT



2.3.4 文件上传下载

如果需要将Windows本地的文件上传到Linux服务器，方法也有很多种。

可以使用XShell配套的Xftp工具



或者使用FileZilla Client工具



2.4 我的 Ubuntu 虚拟机定制配置

管理员初始化操作（现有的服务器就不用去设置了）

2、查看ubuntu系统版本: Ubuntu 16.04.6 LTS

```
asp@coding_gene94_ubuntu16:~$ lsb_release -a
```

```
No LSB modules are available.
```

```
Distributor ID: Ubuntu
```

```
Description:    Ubuntu 16.04.6 LTS
```

```
Release:        16.04
```

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```
Codename: xenial
```

另外关于版本的问题，据说ubuntu里还设置了一个代号：

这个Codename:xenial就是对应版本的代号，我们添加国内源的时候，可以复核一下：

2.4.1 设定主机名

```
hostnamectl set-hostname coding_gene94_ubuntu16
```

2.4.2 用户和目录规划

使用普通用户asp安装，提前规划部署路径，并授权asp用户：

使用管理员权限新建如下用户和目录

```
useradd -g apps -d /home/apps apps
```

```
mkdir -p /apps/svr
```

【这个目录可以自己指定，保持统一就行，我个人习惯将所有软件安装在/apps/svr】

授予asp用户对应的权限：

```
chown -R asp:asp /apps/svr/
```

```
chmod -R 775 /apps/
```

【这个很重要，后面我们规划都安装到】

2.4.3 服务器连公网

一般大家使用 VMware NAT 网络就可以链接公网

```
asp@coding_gene94_ubuntu16:~$ ping 126.com
PING 126.com.localdomain (123.126.96.181) 56(84) bytes of data.
64 bytes from 123.126.96.181: icmp_seq=1 ttl=128 time=30.1 ms
64 bytes from 123.126.96.181: icmp_seq=2 ttl=128 time=30.5 ms
（因为Ubuntu对网络依赖性较强，需要随时联网安装缺少的依赖包，如果没有公网的可以内部配置局域网的源）
```

如果NAT不能连接外网，需要配置域名解析：

```
root@coding_gene_ubuntu16:~# route -n
Kernel IP routing table
Destination      Gateway          Genmask         Flags Metric Ref    Use Iface
0.0.0.0          10.10.1.2       0.0.0.0         UG    0      0      0 ens33
10.10.1.0        0.0.0.0         255.255.255.0   U      0      0      0 ens33
192.168.205.0    0.0.0.0         255.255.255.0   U      0      0      0 ens34
```

```
root@coding_gene_ubuntu16:~# cat /etc/resolv.conf
# Dynamic resolv.conf(5) file for glibc resolver(3) generated by resolvconf(8)
#     DO NOT EDIT THIS FILE BY HAND -- YOUR CHANGES WILL BE OVERWRITTEN
```

```
root@coding_gene_ubuntu16:~# echo "nameserver 10.10.1.2" >> /etc/resolv.conf
```

10.10.1.2是我的VMWARE的NAT网络的网关。

```
root@coding_gene_ubuntu16:~# ping 126.com
PING 126.com (123.126.96.181) 56(84) bytes of data.
64 bytes from mail-m96181.mail.126.com (123.126.96.181): icmp_seq=1 ttl=128
time=37.4 ms
64 bytes from mail-m96181.mail.126.com (123.126.96.181): icmp_seq=2 ttl=128
time=37.9 ms
64 bytes from mail-m96181.mail.126.com (123.126.96.181): icmp_seq=3 ttl=128
time=35.8 ms
^C
--- 126.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2018ms
rtt min/avg/max/mdev = 35.822/37.099/37.998/0.941 ms
```

5、更改网络软件源

默认的国外源下载会比较慢，可以考虑使用国内的apt软件源，更改步骤参考如下：

默认的官方源查看：`cat /etc/apt/sources.list | egrep -v '^$|^#'`

```
asp@coding_gene94:~$ cat /etc/apt/sources.list | egrep -v '^$|^#'  
deb http://us.archive.ubuntu.com/ubuntu/ xenial main restricted  
deb http://us.archive.ubuntu.com/ubuntu/ xenial-updates main restricted  
deb http://us.archive.ubuntu.com/ubuntu/ xenial universe  
deb http://us.archive.ubuntu.com/ubuntu/ xenial-updates universe  
deb http://us.archive.ubuntu.com/ubuntu/ xenial multiverse  
deb http://us.archive.ubuntu.com/ubuntu/ xenial-updates multiverse  
deb http://us.archive.ubuntu.com/ubuntu/ xenial-backports main restricted universe multiverse  
deb http://security.ubuntu.com/ubuntu xenial-security main restricted  
deb http://security.ubuntu.com/ubuntu xenial-security universe  
deb http://security.ubuntu.com/ubuntu xenial-security multiverse
```

上面这些源我们就不要了。修改之前先备份一份配置：

```
cp /etc/apt/sources.list ./
```

然后把原来的源先移走或删除，重新新建一个空的，加入我们需要的源：

```
root@coding_gene94_ubuntu16:~# mv /etc/apt/sources.list /tmp  
root@coding_gene94_ubuntu16:~# vi /etc/apt/sources.list
```

按进入编辑模式，粘贴如下内容 按键盘左上角Esc键，输入:wq退出编辑。

```
deb-src http://archive.ubuntu.com/ubuntu xenial main restricted  
deb http://mirrors.aliyun.com/ubuntu/ xenial main restricted  
deb-src http://mirrors.aliyun.com/ubuntu/ xenial main restricted multiverse  
universe  
deb http://mirrors.aliyun.com/ubuntu/ xenial-updates main restricted  
deb-src http://mirrors.aliyun.com/ubuntu/ xenial-updates main restricted  
multiverse universe  
deb http://mirrors.aliyun.com/ubuntu/ xenial universe  
deb http://mirrors.aliyun.com/ubuntu/ xenial-updates universe  
deb http://mirrors.aliyun.com/ubuntu/ xenial multiverse  
deb http://mirrors.aliyun.com/ubuntu/ xenial-updates multiverse  
deb http://mirrors.aliyun.com/ubuntu/ xenial-backports main restricted universe  
multiverse  
deb-src http://mirrors.aliyun.com/ubuntu/ xenial-backports main restricted  
universe multiverse  
deb http://archive.canonical.com/ubuntu xenial partner  
deb-src http://archive.canonical.com/ubuntu xenial partner  
deb http://mirrors.aliyun.com/ubuntu/ xenial-security main restricted  
deb-src http://mirrors.aliyun.com/ubuntu/ xenial-security main restricted  
multiverse universe  
deb http://mirrors.aliyun.com/ubuntu/ xenial-security universe  
deb http://mirrors.aliyun.com/ubuntu/ xenial-security multiverse
```

(如果vi不熟的，可以使用文本编辑器：`gedit /etc/apt/sources.list` 也需要提前安装apt

install gedit ，并在图形化界面下操作)

保存后/etc/apt/sources.list内容后，执行`apt-get update`

第三章 Linux 上安装 Gromacs

3.1 掌握 Linux 的基础操作

关于Linux的基本操作，这里不详细说，一些基础的Linux要先掌握，

如cd pwd ls vi grep 等命令的操作和使用

3.1.1 基本文件目录操作

```
cd /apps/svr/
```

```
ls -l
```

```
pwd
```

```
ls -l gromacs/
```

```
cat /etc/lsb-release
```

```
more 查看文件
```

```
.....
```

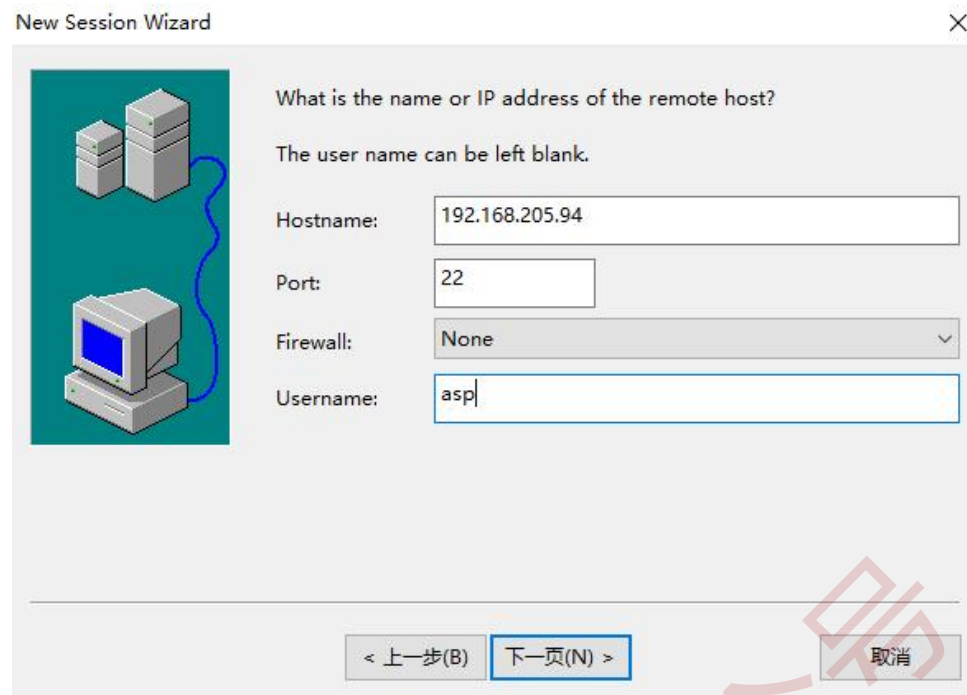
3.1.2 文件编辑操作

数量使用vi编辑器等，能对文件进行删除、修改和保存等。

3.2 安装 gromacs 前准备

3.2.1 远程连接 Linux 系统

使用Xshell或SecureCRT远程客户端工具,这里我使用SecureCRT7.3客户端。



```
Welcome to Ubuntu 16.04.6 LTS (GNU/Linux 4.4.0-142-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/advantage

201 packages can be updated.
147 updates are security updates.

New release '18.04.6 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu May 25 08:41:22 2023 from 192.168.205.1
root@coding_gene94_ubuntu16:~# ifconfig
ens33      Link encap:Ethernet  HWaddr 00:0c:29:9f:a4:28
           inet addr:10.10.1.216  Bcast:10.10.1.255  Mask:255.255.255.0
           inet6 addr: fe80::20c:29ff:fe9f:a428/64 Scope:Link
           inet6 addr: fd15:4ba5:5a2b:1000:20c:29ff:fe9f:a428/64 Scope:Global
           UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
           RX packets:5233 errors:0 dropped:0 overruns:0 frame:0
           TX packets:843 errors:0 dropped:0 overruns:0 carrier:0
           collisions:0 txqueuelen:1000
           RX bytes:1517968 (1.5 MB)  TX bytes:62596 (62.5 KB)

ens34      Link encap:Ethernet  HWaddr 00:0c:29:9f:a4:32
           inet addr:192.168.205.94  Bcast:192.168.205.255  Mask:255.255.255.0
           inet6 addr: fe80::20c:29ff:fe9f:a432/64 Scope:Link
           UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
           RX packets:1888 errors:0 dropped:0 overruns:0 frame:0
           TX packets:1127 errors:0 dropped:0 overruns:0 carrier:0
           collisions:0 txqueuelen:1000
           RX bytes:179772 (179.7 KB)  TX bytes:172526 (172.5 KB)
```

3.2.2配置 linux 终端显示配置

su - asp用户

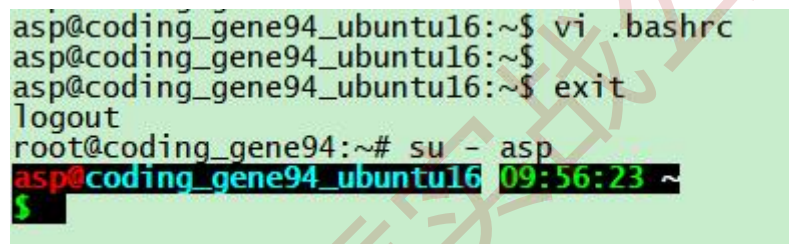
vi .bashrc环境变量

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加入配置：

```
if [ "$TERM" = "xterm" ]
then
    export PS1="\[\e[31;1m\]\u@\[\e[0m\]\[\e[36;1m\]\h\[\e[0m\] \[\e[32;1m\]\t
\[\e[0m\]\[\e[37;1m\]\w\[\e[0m\]\n\[\e[32;1m\]$ \[\e[0m\]"
# export PS1="\[\e[33;1m\]\u\[\e[0m\] \[\e[35;1m\]\t
\[\e[0m\]\[\e[34;1m\]\w\[\e[0m\]\n\[\e[32;1m\]$ \[\e[0m\]"
else
    export PS1="\[\e[31;1m\]\u@\[\e[0m\]\[\e[36;1m\]\h\[\e[0m\] \[\e[32;1m\]\t
\[\e[0m\]\[\e[37;1m\]\w\[\e[0m\]\n\[\e[32;1m\]$ \[\e[0m\]"
fi
```

保存退出后重新登录一下，或者source .bashrc即可看到显示效果：



```
asp@coding_gene94_ubuntu16:~$ vi .bashrc
asp@coding_gene94_ubuntu16:~$
asp@coding_gene94_ubuntu16:~$ exit
logout
root@coding_gene94:~# su - asp
asp@coding_gene94_ubuntu16 09:56:23 ~
$
```

这样操作起来还是比较舒服一些，能通过颜色区分，看到操作的时间

3.2.3 安装 gromacs 参考资料

<https://wiki.archlinux.org/title/GROMACS#top-page>

<https://manual.gromacs.org/current/install-guide/index.html>

3.2.4 获取 gromacs 软件包

可以直接去官方页面下载。也可以参照我们的公众号文章“分子模拟基础-获取Gromacs软件”

3.3 在 Ubuntu16 安装 gromacs2020 版

准备需要的软件（对应的软件包可关注生信实战公众号查阅对应的文章）

上传以下软件包到ubuntu服务器/home/asp/software/md目录:

```
ADH_bench_systems.tar.gz
```

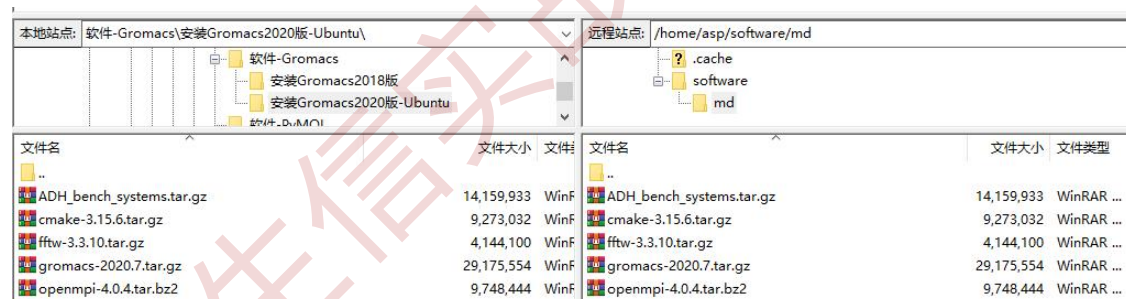
```
cmake-3.15.6.tar.gz
```

```
fftw-3.3.10.tar.gz
```

```
gromacs-2020.7.tar.gz
```

```
openmpi-4.0.4.tar.bz2
```

可以使用FileZilla Client、WinSCP、xftp等sftp工具上传，我这里使用的是FileZilla



3.3.1 安装 gcc 和 g++

一般系统自带gcc，或者建议直接在线安装一个

```
apt-get install gcc g++
```


另外还需要先安装必要的依赖包：

```
apt-get install build-essential python-sphinx doxygen graphviz
```

下面开始编译安装相关软件包，在编译之前先确定使用root或sudo权限用户新建了如下目录：

```
mkdir -p /apps/svr
```

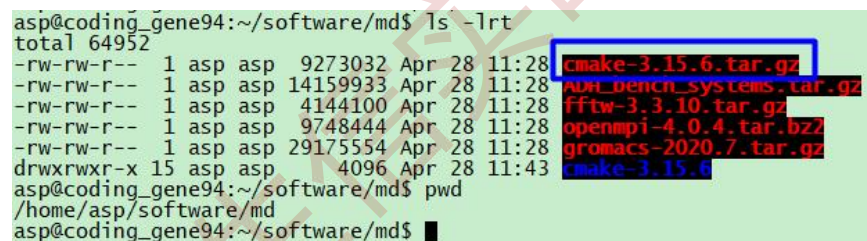
```
chown -R asp:asp /apps/svr/
```

```
chmod -R 775 /apps/
```

【当然，如果你不安装到/apps/svr目录，而是使用其他已有权限的目录，如你的/home/xxuser/install目录，就不用新建，后面编译的时候统一使用指定的这个目录】

3.3.2 安装 cmake

文件如下：我上传到了ubuntu服务器/home/asp/software/md目录



A terminal window showing the contents of the directory /home/asp/software/md. The command 'ls -lrt' is executed, displaying a list of files with permissions, owner, group, size, date, and time. The files listed are cmake-3.15.6.tar.gz, cmake-3.15.6.tar.gz, cmake-3.15.6.tar.gz, cmake-3.15.6.tar.gz, cmake-3.15.6.tar.gz, and cmake-3.15.6.tar.gz. The terminal prompt is asp@coding_gene94:~/software/md\$.

```
cd /home/asp/software/md
```

```
tar -zxf cmake-3.15.6.tar.gz
```

```
cd cmake-3.15.6/
```

```
./bootstrap --prefix=/apps/svr/cmake --parallel=4
```

```
make -j4
```

```
make install
```



```
asp@coding_gene94:~$ cd software/md/
asp@coding_gene94:~/software/md$ ll
total 64956
drwxrwxr-x 2 asp asp 4096 Apr 28 11:28 ./
drwxrwxr-x 3 asp asp 4096 Apr 28 11:27 ../
-rw-rw-r-- 1 asp asp 14159933 Apr 28 11:28 ADH_bench_systems.tar.gz
-rw-rw-r-- 1 asp asp 9273032 Apr 28 11:28 cmake-3.15.6.tar.gz
-rw-rw-r-- 1 asp asp 4144100 Apr 28 11:28 fftw-3.3.10.tar.gz
-rw-rw-r-- 1 asp asp 29175554 Apr 28 11:28 gromacs-2020.7.tar.gz
-rw-rw-r-- 1 asp asp 9748444 Apr 28 11:28 openmpi-4.0.4.tar.bz2
asp@coding_gene94:~/software/md$
asp@coding_gene94:~/software/md$
asp@coding_gene94:~/software/md$ tar -zxf cmake-3.15.6.tar.gz
asp@coding_gene94:~/software/md$ cd cmake-3.15.6/
asp@coding_gene94:~/software/md/cmake-3.15.6$ ./bootstrap --prefix=/apps/avr/cmake --parallel=4
```

加入环境变量CMAKE_HOME如下:

```
vi ~/.bashrc
```

到最后一行

按i进入编辑模式, 粘贴如下内容 按键盘左上角Esc键, 输入:wq退出编辑。

```
CMAKE_HOME=/apps/avr/cmake
```

```
PATH=$PATH:$HOME/bin:$CMAKE_HOME/bin
```

```
CMAKE_HOME=/apps/avr/cmake
PATH=$PATH:$HOME/bin:/apps/avr/bin:$CMAKE_HOME/bin
:wq
```

vi模式下, 按Esc键后, 再输入:wq保存内容, 回车退出

更新环境变量:

```
asp@coding_gene94_ubuntu16:~/software/md/cmake-3.15.6$ source /home/asp/.bashrc
```

显示CMAKE_HOME环境变量:

```
asp@coding_gene94_ubuntu16:~/software/md/cmake-3.15.6$ which cmake
```

```
/apps/avr/cmake/bin/cmake
```

如果上面的步骤忘记安装apt-get install build-essential

那么./bootstrap --prefix=/apps/avr/cmake --parallel=4

这一步可能会遇到下面的错误，如果遇到了，安装下build-essential就可以解决

```
asp@coding_gene94:~/software/md/cmake-3.15.6$ ./bootstrap --prefix=/apps/avr/cmake --parallel=4
-----
CMake 3.15.6, Copyright 2000-2019 Kitware, Inc. and Contributors
Found GNU toolchain
C compiler on this system is: gcc
C++ compiler on this system is: g++ -std=gnu++17
-----
Error when bootstrapping CMake:
Cannot find appropriate Makefile processor on this system.
Please specify one using environment variable MAKE.
-----
Log of errors: /home/asp/software/md/cmake-3.15.6/Bootstrap.cmk/cmake_bootstrap.log
-----
```

3.3.3安装 fftw

fftw3是gromacs必须的数学运行库

文件如下：我上传到了ubuntu服务器/home/asp/software/md目录

```
asp@coding_gene94:~/software/md$ ls -lrt
total 64952
-rw-rw-r-- 1 asp asp 9273032 Apr 28 11:28 cmake-3.15.6.tar.gz
-rw-rw-r-- 1 asp asp 14159933 Apr 28 11:28 ADI_bench_systems.tar.gz
-rw-rw-r-- 1 asp asp 4144100 Apr 28 11:28 fftw-3.3.10.tar.gz
-rw-rw-r-- 1 asp asp 9748444 Apr 28 11:28 openmpi-4.0.4.tar.bz2
-rw-rw-r-- 1 asp asp 29175554 Apr 28 11:28 gromacs-2020.7.tar.gz
drwxrwxr-x 15 asp asp 4096 Apr 28 11:46 cmake-3.15.6
asp@coding_gene94:~/software/md$ pwd
/home/asp/software/md
asp@coding_gene94:~/software/md$
```

```
cd /home/asp/software/md
```

```
tar -zxf fftw-3.3.10.tar.gz
```

```
cd fftw-3.3.10/
```

```
./configure --prefix=/apps/avr/fftw3 --enable-float --enable-shared
```

```
make
```

```
make install
```

这个包路径不用加入环境变量，后面安装gromacs的时候调用一下就行

检查安装：ls -l /apps/svr/fftw3/

```
asp@coding_gene94:~/software/md$ ls -l /apps/svr/fftw3/
total 16
drwxrwxr-x 2 asp asp 4096 Apr 28 12:00 bin
drwxrwxr-x 2 asp asp 4096 Apr 28 12:00 include
drwxrwxr-x 4 asp asp 4096 Apr 28 12:00 lib
drwxrwxr-x 4 asp asp 4096 Apr 28 12:00 share
asp@coding_gene94:~/software/md$
```

3.3.4 安装 openmpi

文件如下：我上传到了ubuntu服务器/home/asp/software/md目录

```
asp@coding_gene94:~/software/md$ ls -l
total 64956
-rw-rw-r-- 1 asp asp 14159933 Apr 28 11:28 ADH_bench_systems.tar.gz
drwxrwxr-x 15 asp asp 4096 Apr 28 11:46 cmake-3.15.6
-rw-rw-r-- 1 asp asp 9273032 Apr 28 11:28 cmake-3.15.6.tar.gz
drwxr-xr-x 18 asp asp 4096 Apr 28 12:00 fftw-3.3.10
-rw-rw-r-- 1 asp asp 4144100 Apr 28 11:28 fftw-3.3.10.tar.gz
-rw-rw-r-- 1 asp asp 29175554 Apr 28 11:28 gromacs-2020.7.tar.gz
-rw-rw-r-- 1 asp asp 9748444 Apr 28 11:28 openmpi-4.0.4.tar.bz2
asp@coding_gene94:~/software/md$
```

```
tar jxvf openmpi-4.0.4.tar.bz2
```

```
cd openmpi-4.0.4/
```

```
mkdir build
```

```
cd build/
```

```
../configure --prefix=/apps/svr/openmpi
```

```
make -j4
```

```
make install
```

配置环境变量vi /home/asp/.bashrc

按i进入编辑模式，粘贴如下内容 按键盘左上角Esc键，输入:wq退出编辑。

```
#CMAKE_HOME=/apps/avr/cmake
```

```
#PATH=$PATH:$HOME/bin:$CMAKE_HOME/bin
```

#之前加的这2行可以注释掉，新的环境变量如下：

```
CMAKE_HOME=/apps/avr/cmake
```

```
OPENMPI_HOME=/apps/avr/openmpi
```

```
PATH=$PATH:$HOME/bin:$CMAKE_HOME/bin:$OPENMPI_HOME/bin
```

```
export PATH
```

```
export LD_LIBRARY_PATH=$OPENMPI_HOME/lib:$CMAKE_HOME/lib:$LD_LIBRARY_PATH
```

```
# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
  if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi

#CMAKE_HOME=/apps/avr/cmake
#PATH=$PATH:$HOME/bin:/apps/bin:$CMAKE_HOME/bin

CMAKE_HOME=/apps/avr/cmake
OPENMPI_HOME=/apps/avr/openmpi
PATH=$PATH:$HOME/bin:$CMAKE_HOME/bin:$OPENMPI_HOME/bin

export PATH
export LD_LIBRARY_PATH=$OPENMPI_HOME/lib:$CMAKE_HOME/lib:$LD_LIBRARY_PATH
```

之前的CMAKE_HOME可以先注释掉；

验证配置：

```
asp@coding_gene94_ubuntu16:~/software/md/openmpi-4.0.4/build$ source
```

```
/home/asp/.bashrc
```

```
asp@coding_gene94_ubuntu16:~/software/md/openmpi-4.0.4/build$ echo $OPENMPI_HOME
```

```
/apps/avr/openmpi
```

```
asp@coding_gene94_ubuntu16:~/software/md/openmpi-4.0.4/build$ which mpicc
```

```
/apps/avr/openmpi/bin/mpicc
```

3.3.5 安装 gromacs2020

文件如下：我上传到了ubuntu服务器/home/asp/software/md目录

```
asp@coding_gene94:~/software/md$ ll
total 64968
drwxrwxr-x  5 asp asp      4096 Apr 28 12:23 ./
drwxrwxr-x  3 asp asp      4096 Apr 28 11:27 ../
-rw-rw-r--  1 asp asp 14159933 Apr 28 11:28 ADH_bench_systems.tar.gz
drwxrwxr-x 15 asp asp      4096 Apr 28 11:46 cmake-3.15.6/
-rw-rw-r--  1 asp asp   9273032 Apr 28 11:28 cmake-3.15.6.tar.gz
drwxr-xr-x 18 asp asp      4096 Apr 28 12:00 fftw-3.3.10/
-rw-rw-r--  1 asp asp   4144100 Apr 28 11:28 fftw-3.3.10.tar.gz
-rw-rw-r--  1 asp asp 29175554 Apr 28 11:28 gromacs-2020.7.tar.gz
drwxrwxr-x 11 asp asp      4096 Apr 28 12:24 openmpi-4.0.4/
-rw-rw-r--  1 asp asp   9748444 Apr 28 11:28 openmpi-4.0.4.tar.bz2
asp@coding_gene94:~/software/md$
```

这里我规划2个目录

```
asp@coding_gene94_ubuntu16:~/software/md$ mkdir -p /apps/avr/gromacs/gmx2020
```

```
asp@coding_gene94_ubuntu16:~/software/md$ mkdir -p /apps/avr/gromacs/gmx2020_mpi
```

/apps/avr/gromacs/gmx2020 用来安装单机版

/apps/avr/gromacs/gmx2020_mpi用来安装并行版

安装单机版：

```
tar -zxf gromacs-2020.7.tar.gz
```

```
cd gromacs-2020.7
```

```
mkdir build_single
```

```
cd build_single
```

```
cmake .. \
```

```
-DCMAKE_INSTALL_PREFIX=/apps/avr/gromacs/gmx2020 \
```

```
-DGMX_FFT_LIBRARY=fftw3 \
```

```
-DFFTW_LIBRARY=/apps/avr/fftw3/lib/libfftw3f.so \
```

```
-DFFTW_INCLUDE_DIR=/apps/avr/fftw3/include/ \
```

```
-DGMX_BINARY_SUFFIX=_2020.7 \
```

```
-DGMX_LIBS_SUFFIX=_2020.7
```

```
make -j4
```

```
#make check （可选）
```

```
make install
```

验证一下命令：`/apps/avr/gromacs/gmx2020/bin/gmx --version`

```
asp@coding_gene94:~/software/md/gromacs-2020.7/build_single$ /apps/svr/gromacs/gmx2020/bin/gmx --version
:~) GROMACS - gmx, 2020.7 (-:

GROMACS is written by:
  Emile Apol      Rossen Apostolov    Paul Bauer      Herman J.C. Berendsen
  Par Bjelkmar    Christian Blau      Viacheslav Boilykh    Kevin Boyd
  Aldert van Buuren  Rudi van Drunen    Anton Feenstra      Alan Gray
  Gerrit Groenhof  Anca Hamuraru      Vincent Hindriksen    M. Eric Irrgang
  Aleksei Iupinov  Christoph Junghans  Joe Jordan          Dimitrios Karkoulis
  Peter Kasson     Jiri Kraus          Carsten Kutzner      Per Larsson
  Justin A. Lemkul  Viveca Lindahl      Magnus Lundborg      Erik Marklund
  Pascal Merz       Pieter Meulenhoff    Teemu Murtola        Szilard Pall
  Sander Pronk      Roland Schulz        Michael Shirts       Alexey Shvetsov
  Alfons Sijbers    Peter Tieleman       Jon Vincent          Teemu Virolainen
  Christian Wennberg  Maarten Wolf        Artem Zhmurov

and the project leaders:
  Mark Abraham, Berk Hess, Erik Lindahl, and David van der Spoel

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check out http://www.gromacs.org for more information.

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under the terms of the GNU Lesser General Public License
as published by the Free Software Foundation; either version 2.1
of the License, or (at your option) any later version.

GROMACS:      gmx, version 2020.7
Executable:    /apps/svr/gromacs/gmx2020/bin/gmx
Data prefix:   /apps/svr/gromacs/gmx2020
Working dir:    /home/asp/software/md/gromacs-2020.7/build_single
Command line:   gmx --version

GROMACS version: 2020.7
Verified release checksum is 761fa697fdcd80bd396d0dd395d20957171ad38eb6e080f0760c9511f862dbb8
Precision:      single
Memory model:   64 bit
MPI library:    thread_mpi
OpenMP support: enabled (GMX_OPENMP_MAX_THREADS = 64)
GPU support:    disabled
SIMD instructions: AVX2_256
FFT library:    fftw-3.3.10
RDTSCP usage:   enabled
FNG support:    enabled
hwloc support:  disabled
Tracing support: disabled
C compiler:     /usr/bin/cc GNU 5.4.0
C compiler flags: -mavx2 -mfma -fexcess-precision=fast -funroll-all-loops -O3 -DNDEBUG
C++ compiler:   /usr/bin/c++ GNU 5.4.0
C++ compiler flags: -mavx2 -mfma -fexcess-precision=fast -funroll-all-loops -fopenmp -O3 -DNDEBUG
```

安装并行版:

```
cd gromacs-2020.7 (实际目录是/home/asp/software/md/gromacs-2020.7)
```

```
mkdir build_mpi #注意是准备重新编译, 因此新建了一个上面单机版不一样的目录
```

```
cd build_mpi
```

```
#在cmake之前先安装: apt-get install python-sphinx
```

#要不然cmake可能遇到错误:

```
ImportError: No module named 'pygments'
```

```
-- Could NOT find Sphinx (missing: SPHINX_EXECUTABLE pygments) (Required is at least
version "1.6.1")
```

```
另外安装: apt-get install doxygen graphviz
```

#要不然cmake可能遇到错误:

```
-- Could NOT find Doxygen (missing: DOXYGEN_EXECUTABLE)
```

```
-- Doxygen not found. Documentation targets will not be generated.
```

#注意cmake .. 后面有个空格

```
cmake .. \  
-DCMAKE_INSTALL_PREFIX=/apps/avr/gromacs/gmx2020_mpi \  
-DGMX_FFT_LIBRARY=fftw3 \  
-DFFTW_LIBRARY=/apps/avr/fftw3/lib/libfftw3f.so \  
-DFFTW_INCLUDE_DIR=/apps/avr/fftw3/include/ \  
-DCMAKE_C_COMPILER=/apps/avr/openmpi/bin/mpicc \  
-DCMAKE_CXX_COMPILER=/apps/avr/openmpi/bin/mpicxx \  
-DGMX_MPI=on \  
-DGMX_OPENMP=ON \  
-DGMX_BINARY_SUFFIX=_2020.7_s \  
-DGMX_LIBS_SUFFIX=_2020.7_s
```

#如果需要跨节点计算加上-DGMX_MPI=on

#需要GPU支持加上-DGMX_GPU=on, 需要提前配置好mpi和cuda环境

#-DGMX_BINARY_SUFFIX=_5.0.7_s 指定执行文件后缀 默认为gmx (串行单精度) ,

gmx_mpi (并行单精度) , gmx_mpi_d(并行双精度) 根据实际版本调整

#-DGMX_LIBS_SUFFIX=_5.0.7_s 库文件后缀, 根据实际版本调整

```
make -j4
```

```
make check
```

```
make install
```

执行检查命令:

```
/apps/avr/gromacs/gmx2020_mpi/bin/gmx_mpi --version
```



```
asp@coding_gene94:~/software/md/gromacs-2020.7/build_mpi$ /apps/avr/gromacs/gmx2020_mpi/bin/gmx_mpi --version
:~) GROMACS - gmx_mpi, 2020.7 (-:

GROMACS is written by:
  Emile Apol          Rossen Apostolov    Paul Bauer          Herman J.C. Berendsen
  Par Bjelkmar        Christian Blau       Viacheslav Bolnykh  Kevin Boyd
  Aldert van Buuren   Rudi van Drunen    Anton Feenstra      Alan Gray
  Gerrit Groenhof     Anca Hamuraru      Vincent Hindriksen  M. Eric Irrgang
  Aleksei Iupinov     Christoph Junghans  Joe Jordan           Dimitrios Karkoulis
  Peter Kasson        Jiri Kraus          Carsten Kutzner      Per Larsson
  Justin A. Lemkul     Viveca Lindahl      Magnus Lundborg      Erik Marklund
  Pascal Merz          Pieter Meulenhoff   Teemu Murtola        Szilard Pall
  Sander Pronk         Roland Schulz       Michael Shirts       Alexey Shvetsov
  Alfons Sijbers      Peter Tieleman      Jon Vincent          Teemu Virolainen
  Christian Wennberg   Maarten Wolf        Artem Zhmurov

and the project leaders:
  Mark Abraham, Berk Hess, Erik Lindahl, and David van der Spoel

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as published by the Free Software Foundation; either version 2.1
of the License, or (at your option) any later version.

GROMACS:          gmx_mpi, version 2020.7
Executable:       /apps/avr/gromacs/gmx2020_mpi/bin/gmx_mpi
Data prefix:      /apps/avr/gromacs/gmx2020_mpi
Working dir:      /home/asp/software/md/gromacs-2020.7/build_mpi
Command line:
  gmx_mpi --version

GROMACS version:  2020.7
Verified release checksum is 761fa697fdcd80bd396d0dd395d20957171ad38eb6e080f0760c9511f862dbb8
Precision:        single
Memory model:     64 bit
MPI library:      MPI
OpenMP support:   enabled (GMX_OPENMP_MAX_THREADS = 64)
GPU support:      disabled
SIMD instructions: AVX2_256
FFT library:      fftw-3.3.10
RDTSCP usage:     enabled
TNG support:      enabled
Hwloc support:    disabled
Tracing support:  disabled
C compiler:       /apps/avr/openmpi/bin/mpicc GNU 5.4.0
C compiler flags: -mavx2 -mfma -fexcess-precision=fast -funroll-all-loops -O3 -DNDEBUG
C++ compiler:     /apps/avr/openmpi/bin/mpicxx GNU 5.4.0
C++ compiler flags: -mavx2 -mfma -fexcess-precision=fast -funroll-all-loops -fopenmp -O3 -DNDEBUG
```

3.3.6跑一下模拟试试

使用这个测试文件ADH_bench_systems.tar.gz

```
asp@coding_gene94:~/software/md$ ll
total 64972
drwxrwxr-x 6 asp asp 4096 Apr 28 12:52 /
drwxrwxr-x 3 asp asp 4096 Apr 28 11:27 /
-rw-rw-r-- 1 asp asp 14159933 Apr 28 11:28 ADH_bench_systems.tar.gz
drwxrwxr-x 15 asp asp 4096 Apr 28 11:46 cmake-3.15.6
-rw-rw-r-- 1 asp asp 9273032 Apr 28 11:28 cmake-3.15.6.tar.gz
drwxr-xr-x 18 asp asp 4096 Apr 28 12:00 fftw-3.3.10
-rw-rw-r-- 1 asp asp 4144100 Apr 28 11:28 fftw-3.3.10.tar.gz
drwxr-xr-x 12 asp asp 4096 Apr 28 13:34 gromacs-2020.7
-rw-rw-r-- 1 asp asp 29175554 Apr 28 11:28 gromacs-2020.7.tar.gz
drwxrwxr-x 11 asp asp 4096 Apr 28 12:24 openmpi-4.0.4
-rw-rw-r-- 1 asp asp 9748444 Apr 28 11:28 openmpi-4.0.4.tar.bz2
```

tar xvf ADH_bench_systems.tar.gz

cd ADH

cd adh_cubic

```
asp@coding_gene94:~/software/md/ADH/adh_cubic$ pwd
/home/asp/software/md/ADH/adh_cubic
asp@coding_gene94:~/software/md/ADH/adh_cubic$ ls -l
total 15248
-rw-r--r-- 1 asp asp 9258297 Jul 3 2012 conf.gro
-rw-r--r-- 1 asp asp 706 Oct 8 2016 pme_verlet.mdp
-rw-r--r-- 1 asp asp 712 Oct 8 2016 rf_verlet.mdp
-rw-r--r-- 1 asp asp 1582031 Jul 3 2012 topol_Protein_chain_A.itp
-rw-r--r-- 1 asp asp 1582034 Jul 3 2012 topol_Protein_chain_B.itp
-rw-r--r-- 1 asp asp 1582031 Jul 3 2012 topol_Protein_chain_C.itp
-rw-r--r-- 1 asp asp 1582034 Jul 3 2012 topol_Protein_chain_D.itp
-rw-r--r-- 1 asp asp 1264 Jul 3 2012 topol.top
```

使用单机版执行:

```
/apps/svr/gromacs/gmx2020/bin/gmx grompp -f rf_verlet.mdp
```

会发现多了2个文件:

```
asp@coding_gene94:~/software/md/ADH/adh_cubic$ ls -lrt
total 23304
-rw-r--r-- 1 asp asp 1264 Jul 3 2012 topol.top
-rw-r--r-- 1 asp asp 1582034 Jul 3 2012 topol_Protein_chain_D.itp
-rw-r--r-- 1 asp asp 1582031 Jul 3 2012 topol_Protein_chain_C.itp
-rw-r--r-- 1 asp asp 1582034 Jul 3 2012 topol_Protein_chain_B.itp
-rw-r--r-- 1 asp asp 1582031 Jul 3 2012 topol_Protein_chain_A.itp
-rw-r--r-- 1 asp asp 9258297 Jul 3 2012 conf.gro
-rw-r--r-- 1 asp asp 706 Oct 8 2016 pme_verlet.mdp
-rw-r--r-- 1 asp asp 712 Oct 8 2016 rf_verlet.mdp
-rw-rw-r-- 1 asp asp 10764 Apr 28 13:55 mdout.mdp
-rw-rw-r-- 1 asp asp 8234108 Apr 28 13:55 topol.tpr
```

执行模拟:

使用单机版: `/apps/svr/gromacs/gmx2020/bin/gmx mdrun`

或使用并行版: `/apps/svr/gromacs/gmx2020_mpi/bin/gmx_mpi mdrun`

```
asp@coding_gene94:~/software/md/ADH/adh_cubic$ /apps/svr/gromacs/gmx2020_mpi/bin/gmx_mpi mdrun
:-) GROMACS - gmx mdrun, 2020.7 (-:

      GROMACS is written by:
      Emile Apol          Rossen Apostolov      Paul Bauer          Herman J.C. Berendsen
      Par Bjeckmar        Christian Blau        Viacheslav Bolnykh   Kevin Boyd
      Aldert van Buuren   Rudi van Drunen     Anton Feenstra       Alan Gray
      Gerrit Groenhof     Anca Hamuraru       Vincent Hindriksen   M. Eric Irrgang
      Aleksei Iupinov     Christoph Junghans   Joe Jordan           Dimitrios Karkoulis
      Peter Kasson        Jiri Kraus          Carsten Kutzner      Per Larsson
      Justin A. Lemkul     Viveca Lindahl      Magnus Lundborg      Erik Marklund
      Pascal Merz         Pieter Meulenhoff    Teemu Murtola        Szilard Pall
      Sander Pronk        Roland Schulz        Michael Shirts       Alexey Shvetsov
      Alfons Sijbers      Peter Tieleman       Jon Vincent          Teemu Virolainen
      Christian Wennberg   Maarten Wolf        Artem Zhmurov
      and the project leaders:
      Mark Abraham, Berk Hess, Erik Lindahl, and David van der Spoel

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Uppsala University, Stockholm University and
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of the License, or (at your option) any later version.

GROMACS:      gmx mdrun, version 2020.7
Executable:    /apps/svr/gromacs/gmx2020_mpi/bin/gmx_mpi
Data prefix:   /apps/svr/gromacs/gmx2020_mpi
Working dir:    /home/asp/software/md/ADH/adh_cubic
Command line:  gmx_mpi mdrun

Reading file topol.tpr, VERSION 2020.7 (single precision)
Changing nstlist from 10 to 20, rlist from 0.966 to 1.068

Using 1 MPI process
Using 8 OpenMP threads

starting mdrun 'NADP-DEPENDENT ALCOHOL DEHYDROGENASE in water'
10000 steps,      20.0 ps.
```

top命令查看CPU使用率

```
Tasks: 220 total, 1 running, 219 sleeping, 0 stopped, 0 zombie
%Cpu0  : 98.7 us, 0.3 sy, 0.0 ni, 1.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu1  : 99.0 us, 0.3 sy, 0.0 ni, 0.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu2  : 99.0 us, 0.3 sy, 0.0 ni, 0.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu3  : 99.3 us, 0.3 sy, 0.0 ni, 0.3 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu4  : 99.0 us, 0.3 sy, 0.0 ni, 0.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu5  : 99.3 us, 0.0 sy, 0.0 ni, 0.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu6  : 98.7 us, 0.3 sy, 0.0 ni, 1.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu7  : 99.0 us, 0.3 sy, 0.0 ni, 0.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
KiB Mem : 8156784 total, 5269312 free, 260228 used, 2627244 buff/cache
KiB Swap: 0 total, 0 free, 0 used, 7513480 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
 94219 asp        20   0   892508 169008 16504 S   792.4   2.1   1:03.84 gmx_mpi
```

完成后的提示:


```
Reading file topol.tpr, VERSION 2020.7 (single precision)
Changing nstlist from 10 to 20, rlist from 0.966 to 1.068

Using 1 MPI process
Using 8 OpenMP threads

starting mdrun 'NADP-DEPENDENT ALCOHOL DEHYDROGENASE in water'
10000 steps,      20.0 ps.

Writing final coordinates.

NOTE: 11 % of the run time was spent in pair search,
      you might want to increase nstlist (this has no effect on accuracy)

      Time:      Core t (s)      Wall t (s)      (%)
              937.450      117.181      800.0
              (ns/day)      (hour/ns)
Performance:   14.748      1.627

GROMACS reminds you: "I don't believe in astrology; I'm a Sagittarian and we
```

剩下的就是进一步分析模拟结果了。

3.4 在 CentOS7.6 安装 gromacs2018

一步一步在Linux7.6编译安装Gromacs并行版

因为不同的linux版本、不同的软件版本操作不完全一样，所以这里我提前说明一下

Linux版本：CentOS7.6（可通过cat /etc/*release命令查看）

Gromacs版本：gromacs-2018.8

本文测试用的root用户安装和操作，真正使用建议结合实际情况使用普通用户

具体软件清单如下：

[cmake-3.15.6.tar.gz](#)

[fftw-3.3.10.tar.gz](#)

[gromacs-2018.8.tar.gz](#)

《生信实战》微信公众号

[openmpi-4.0.4.tar.bz2](#)

```
total 65672
-rw-r--r-- 1 root root 14159933 Apr 22 22:03 ADH_bench_systems.tar.gz
-rw-r--r-- 1 root root 9273032 Apr 22 20:48 cmake-3.15.6.tar.gz
-rw-r--r-- 1 root root 4144100 Apr 22 14:57 fftw-3.3.10.tar.gz
-rw-r--r-- 1 root root 29913703 Apr 22 14:15 gromacs-2018.8.tar.gz
-rw-r--r-- 1 root root 9748444 Apr 22 21:16 openmpi-4.0.4.tar.bz2
```

首先准备软件包和测试文件，并上传到指定目录，例如我这里放到了~/software/md

其中ADH_bench_systems.tar.gz是做模拟的测试文件。

部署目录规划：

```
/apps/avr/
├── cmake
├── fftw3
├── gromacs
└── openmpi
```

3.4.1 确认 gcc 编译器

关于gcc，建议就使用系统自带的gcc

```
yum install -y gcc gcc-c++ gcc-gfortran texinfo
```

```
[root@bigdata2bioinfo ~/software/md]# rpm -qa | grep gcc
```

```
libgcc-4.8.5-44.el7.x86_64
```

gcc-c++-4.8.5-44.el7.x86_64

gcc-4.8.5-44.el7.x86_64

gcc-gfortran-4.8.5-44.el7.x86_64

```
[root@bigdata2bioinfo ~]/software/md]# gcc --version
```

gcc (GCC) 4.8.5 20150623 (Red Hat 4.8.5-44)

Copyright (C) 2015 Free Software Foundation, Inc.

【后续我再补充离线安装gcc等的步骤】

3.4.2 安装 cmake

#先提前安装

```
yum install python-sphinx -y
```

#获取cmake软件: <https://cmake.org/files/v3.15/>

#也可以直接使用提供的离线包

#执行安装步骤:

```
tar -zxf cmake-3.15.6.tar.gz
```

```
cd cmake-3.15.6
```

```
./bootstrap --prefix=/apps/avr/cmake --parallel=4
```

```
make -j4
```

```
make install
```

#安装完成后在.bash_profile加入环境变量CMAKE_HOME如下:

```
CMAKE_HOME=/apps/avr/cmake
```

```
PATH=$PATH:$HOME/bin:/apps/bin:$CMAKE_HOME/bin
```

```
export PATH
```

```
[root@bigdata2bioinfo ~/software/md/cmake-3.15.6]# which cmake
```

```
/apps/avr/cmake/bin/cmake
```

3.4.3 安装 fftw

#fftw3是gromacs运行需要的傅立叶变换数学库

#执行安装步骤:

```
tar -zxf fftw-3.3.10.tar.gz
```

```
cd fftw-3.3.10
```

```
./configure --prefix=/apps/avr/fftw3 --enable-float --enable-shared
```

```
make
```

```
make install
```

《生信实战》微信公众号

#这个不用加入环境变量，后面安装gromacs的时候调用一下就行

3.4.4 安装 openmpi

#openmpi主要是用来做并行计算

#执行安装步骤：

```
tar jxvf openmpi-4.0.4.tar.bz2
```

```
cd openmpi-4.0.4
```

```
mkdir build
```

```
cd build/
```

```
../configure --prefix=/apps/avr/openmpi
```

```
make -j4
```

```
make install
```

#vi配置环境变量.bash_profile，新增OPENMPI_HOME和对应的PATH

```
CMAKE_HOME=/apps/avr/cmake
```

```
OPENMPI_HOME=/apps/avr/openmpi
```

```
PATH=$PATH:$HOME/bin:/apps/avr/bin:$CMAKE_HOME/bin:$OPENMPI_HOME/bin
```

```
export PATH
```



```
export
```

```
LD_LIBRARY_PATH=$OPENMPI_HOME/lib:$CMKAE_HOME/lib:$LD_LIBRARY_PATH
```

#更新环境变量，检查安装的命令：

```
source /root/.bash_profile
```

```
[root@bigdata2bioinfo ~/software/md/openmpi-4.0.4/build]# which mpicc
```

```
/apps/svr/openmpi/bin/mpicc
```

3.4.5 安装 gromacs

关于Gromacs的安装，官方文档有简单说明，可以参考。

Quick and dirty installation

Get the latest version of your C and C++ compilers.
Check that you have CMake version 3.18.4 or later.
Get and unpack the latest version of the GROMACS tarball.
Make a separate build directory and change to it.
Run cmake with the path to the source as an argument
Run make, make check, and make install
Source GMXRC to get access to GROMACS
Or, as a sequence of commands to execute:

```
tar xzf gromacs-2023.1.tar.gz
cd gromacs-2023.1
mkdir build
cd build
cmake .. -DGMX_BUILD_OWN_FFTW=ON -DREGRESSIONTEST_DOWNLOAD=ON
make
make check
sudo make install
source /usr/local/gromacs/bin/GMXRC
```

#本次执行安装步骤

```
tar -zxf gromacs-2018.8.tar.gz
```

```
cd gromacs-2018.8
```

```
mkdir build
```

```
cd build
```

```
cmake .. \
```

```
-DCMAKE_INSTALL_PREFIX=/apps/avr/gromacs/gmx2018 \
```

```
-DGMX_FFT_LIBRARY=fftw3 \
```

```
-DFFTW_LIBRARY=/apps/avr/fftw3/lib/libfftw3f.so \
```

```
-DFFTW_INCLUDE_DIR=/apps/avr/fftw3/include/ \
```

```
-DCMAKE_C_COMPILER=/apps/avr/openmpi/bin/mpicc \
```

```
-DCMAKE_CXX_COMPILER=/apps/avr/openmpi/bin/mpicxx \
```

```
-DGMX_MPI=on \
```

```
-DGMX_OPENMP=ON
```

#cmake比较关键

#注意cmake .. 后面有个空格

#-DCMAKE_INSTALL_PREFIX 用于指定自定义的安装路径

#-DFFTW_LIBRARY和-DFFTW_INCLUDE_DIR分别指定已经安装的FFTW库文件

#如果需要跨节点计算加上-DGMX_MPI=on

#如果需要GPU支持加上-DGMX_GPU=on, 需要配置好mpi和cuda环境（这里我没有GPU就不加）

cmake配置完成后，继续安装和验证：

```
make -j
```

```
make check
```

```
make install
```

#更新加载Gromacs相关环境变量

```
source /apps/avr/gromacs/gmx2018/bin/GMXRC
```

```
[root@bigdata2bioinfo ~/software/md/gromacs-2018.8/build]# echo $GMXPREFIX  
/apps/avr/gromacs/gmx2018
```

3.4.6 命令执行模拟测试

上传测试文件ADH_bench_systems.tar.gz

```
tar xvf ADH_bench_systems.tar.gz
```

```
cd ADH
```

```
cd adh_cubic
```

```
ls -l
```

里面包含一些测试用的md配置和拓扑文件等。

```
total 15248  
-rw-r--r-- 1 1038 dialout 9258297 Jul 3 2012 conf.gro  
-rw-r--r-- 1 1038 dialout 706 Oct 8 2016 pme_verlet.mdp  
-rw-r--r-- 1 1038 dialout 712 Oct 8 2016 rf_verlet.mdp  
-rw-r--r-- 1 1038 dialout 1582031 Jul 3 2012 topol_Protein_chain_A.itp  
-rw-r--r-- 1 1038 dialout 1582034 Jul 3 2012 topol_Protein_chain_B.itp  
-rw-r--r-- 1 1038 dialout 1582031 Jul 3 2012 topol_Protein_chain_C.itp  
-rw-r--r-- 1 1038 dialout 1582034 Jul 3 2012 topol_Protein_chain_D.itp  
-rw-r--r-- 1 1038 dialout 1264 Jul 3 2012 topol.top
```

```
执行: gmx_mpi grompp -f rf_verlet.mdp
```

(这个gmx_mp执行的是/apps/avr/gromacs/gmx2018/bin/gmx_mpi, 因为加了环境变量, 可以直接调用)

显示如下:

```
Command line:
gmx_mpi grompp -f rf_verlet.mdp

Replacing old mdp entry 'nstxtcout' by 'nstxout-compressed'
Setting the LD random seed to -1874764158
Generated 2145 of the 2145 non-bonded parameter combinations
Generating 1-4 interactions: fudge = 0.5
Generated 2145 of the 2145 1-4 parameter combinations
Excluding 3 bonded neighbours molecule type 'Protein_chain_A'
turning H bonds into constraints...
Excluding 3 bonded neighbours molecule type 'Protein_chain_B'
turning H bonds into constraints...
Excluding 3 bonded neighbours molecule type 'Protein_chain_C'
turning H bonds into constraints...
Excluding 3 bonded neighbours molecule type 'Protein_chain_D'
turning H bonds into constraints...
Excluding 2 bonded neighbours molecule type 'SOL'
turning H bonds into constraints...
Excluding 2 bonded neighbours molecule type 'SOL'
Excluding 2 bonded neighbours molecule type 'SOL'
Excluding 2 bonded neighbours molecule type 'SOL'
Excluding 1 bonded neighbours molecule type 'NA'
turning H bonds into constraints...
Removing all charge groups because cutoff-scheme=Verlet
Analysing residue names:
There are: 1408 Protein residues
There are: 37611 Water residues
There are: 12 Ion residues
Analysing Protein...
Analysing residues not classified as Protein/DNA/RNA/Water and splitting into groups...
Number of degrees of freedom in T-Coupling group System is 278935.00
Determining Verlet buffer for a tolerance of 0.005 kJ/mol/ps at 300 K
Calculated rlist for 1x1 atom pair-list as 0.992 nm, buffer size 0.092 nm
Set rlist, assuming 4x4 atom pair-list, to 0.966 nm, buffer size 0.066 nm
Note that mdrun will redetermine rlist based on the actual pair-list setup
This run will generate roughly 10 Mb of data

GROMACS reminds you: "I Wrapped a Newspaper Round My Head" (F. Zappa)
```

执行gmx_mpi mdrun进行计算：如果能够正常计算，说明gromacs安装成功。


```
[root@node61 ~/software/md/ADH/adh_cubic]# gmx_mpi mdrun
:-) GROMACS - gmx mdrun, 2018.8 (-:

GROMACS is written by:
    Emile Apol          Rossen Apostolov      Paul Bauer          Herman J.C. Berendsen
    Par Bjeelkmar       Aldert van Buuren      Rudi van Drunen     Anton Feenstra
    Gerrit Groenhof     Aleksei Iupinov        Christoph Junghans   Anca Hamuraru
    Vincent Hindriksen  Dimitrios Karkoulis    Peter Kasson         Jiri Kraus
    Carsten Kutzner     Per Larsson            Justin A. Lemkul      Viveca Lindahl
    Magnus Lundborg     Pieter Meulenhoff      Erik Marklund        Teemu Murtola
    Szilard Pall        Sander Pronk           Roland Schulz         Alexey Shvetsov
    Michael Shirts      Alfons Sijbers         Peter Tieleman        Teemu Virolainen
    Christian Wennberg   Maarten Wolf

    and the project leaders:
    Mark Abraham, Berk Hess, Erik Lindahl, and David van der Spoel

Copyright (c) 1991-2000, University of Groningen, The Netherlands.
Copyright (c) 2001-2017, The GROMACS development team at
Uppsala University, Stockholm University and
the Royal Institute of Technology, Sweden.
check out http://www.gromacs.org for more information.

GROMACS is free software; you can redistribute it and/or modify it
under the terms of the GNU Lesser General Public License
as published by the Free Software Foundation; either version 2.1
of the License, or (at your option) any later version.

GROMACS:          gmx mdrun, version 2018.8
Executable:       /apps/avr/gromacs/gmx2018/bin/gmx_mpi
Data prefix:      /apps/avr/gromacs/gmx2018
Working dir:      /root/software/md/ADH/adh_cubic
Command line:     gmx_mpi mdrun

Compiled SIMD: None, but for this host/run AVX2_256 might be better (see log).
The current CPU can measure timings more accurately than the code in
gmx mdrun was configured to use. This might affect your simulation
speed as accurate timings are needed for load-balancing.
Please consider rebuilding gmx mdrun with the GMX_USE_RDTSCP=ON CMake option.
Reading file topol.tpr, VERSION 2018.8 (single precision)
Changing nstlist from 10 to 20, rlist from 0.966 to 1.068

Using 1 MPI process
Using 4 OpenMP threads

WARNING: Using the slow plain C kernels. This should
not happen during routine usage on supported platforms.
starting mdrun 'NADP-DEPENDENT ALCOHOL DEHYDROGENASE in water'
10000 steps,      20.0 ps.
```

我本地测试服务器只有4核，通过linux top命令可以看到CPU都使用上了，使用率比较高

```
top - 23:14:22 up 114 days, 5:53, 3 users, load average: 4.29, 3.28, 1.75
Tasks: 144 total, 1 running, 98 sleeping, 0 stopped, 0 zombie
%Cpu0  : 91.1 us, 2.0 sy, 0.3 ni, 5.6 id, 0.0 wa, 0.7 hi, 0.3 si, 0.0 st
%Cpu1  : 93.7 us, 3.0 sy, 0.0 ni, 2.7 id, 0.0 wa, 0.7 hi, 0.0 si, 0.0 st
%Cpu2  : 93.7 us, 2.3 sy, 0.0 ni, 3.0 id, 0.0 wa, 0.7 hi, 0.3 si, 0.0 st
%Cpu3  : 91.4 us, 2.3 sy, 0.3 ni, 5.0 id, 0.0 wa, 0.7 hi, 0.3 si, 0.0 st
KiB Mem : 15896844 total, 3619068 free, 1114636 used, 11163140 buff/cache
KiB Swap: 0 total, 0 free, 0 used, 14451348 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
553794 root        20   0 654588 140092 14912 S 359.8   0.9 17:55.93 gmx_mpi
```

还可以进一步查看运行日志md.log

```
[root@node61 ~/software/md/ADH/adh_cubic]# ls -lrt
total 29720
-rw-r--r-- 1 1038 dialout 1264 Jul 3 2012 topol.top
-rw-r--r-- 1 1038 dialout 1582034 Jul 3 2012 topol_Protein_chain_D.itp
-rw-r--r-- 1 1038 dialout 1582031 Jul 3 2012 topol_Protein_chain_C.itp
-rw-r--r-- 1 1038 dialout 1582034 Jul 3 2012 topol_Protein_chain_B.itp
-rw-r--r-- 1 1038 dialout 1582031 Jul 3 2012 topol_Protein_chain_A.itp
-rw-r--r-- 1 1038 dialout 9258297 Jul 3 2012 conf.gro
-rw-r--r-- 1 1038 dialout 706 Oct 8 2016 pme_verlet.mdp
-rw-r--r-- 1 1038 dialout 712 Oct 8 2016 rf_verlet.mdp
-rw-r--r-- 1 root root 11730 Apr 22 23:09 mdout.mdp
-rw-r--r-- 1 root root 8327872 Apr 22 23:09 topol.tpr
-rw-r--r-- 1 root root 15555 Apr 22 23:39 md.log
-rw-r--r-- 1 root root 3221744 Apr 22 23:39 state.cpt
-rw-r--r-- 1 root root 3221744 Apr 22 23:39 state_prev.cpt
-rw-r--r-- 1 root root 10156 Apr 22 23:50 ener.edr
[root@node61 ~/software/md/ADH/adh_cubic]#
[root@node61 ~/software/md/ADH/adh_cubic]#
[root@node61 ~/software/md/ADH/adh_cubic]# tail md.log
2.85895e+05 1.99951e+05 -2.24905e+06 -1.63240e+06 3.36085e+05
Total Energy Conserved En. Temperature Pressure (bar) Constr. rmsd
-1.29632e+06 -1.29632e+06 2.89829e+02 2.17242e+03 3.84565e-06

Writing checkpoint, step 3320 at Sat Apr 22 23:24:21 2023

Writing checkpoint, step 6620 at Sat Apr 22 23:39:21 2023
```

运行完成后的提示信息如下：

```
-----
Total 2723.575 25056.870 100.0
-----
Time: Core t (s) Wall t (s) (%)
      10894.300 2723.575 400.0
      (ns/day) 45:23
Performance: 0.635 37.824
Finished mdrun on rank 0 Sat Apr 22 23:54:39 2023
```

本文用到的软件包获取：公众号后台回复 安装gmx

```
total 65672
-rw-r--r-- 1 root root 14159933 Apr 22 22:03 ADH_bench_systems.tar.gz
-rw-r--r-- 1 root root 9273032 Apr 22 20:48 cmake-3.15.6.tar.gz
-rw-r--r-- 1 root root 4144100 Apr 22 14:57 fftw-3.3.10.tar.gz
-rw-r--r-- 1 root root 29913703 Apr 22 14:15 gromacs-2018.8.tar.gz
-rw-r--r-- 1 root root 9748444 Apr 22 21:16 openmpi-4.0.4.tar.bz2
```

安装的时候可能会遇到一些小错误，仔细阅读错误提示，不难解决。

另外由于是测试环境，本文用的root安装，实际建议使用普通用户安装和操作。

生信实战公众号

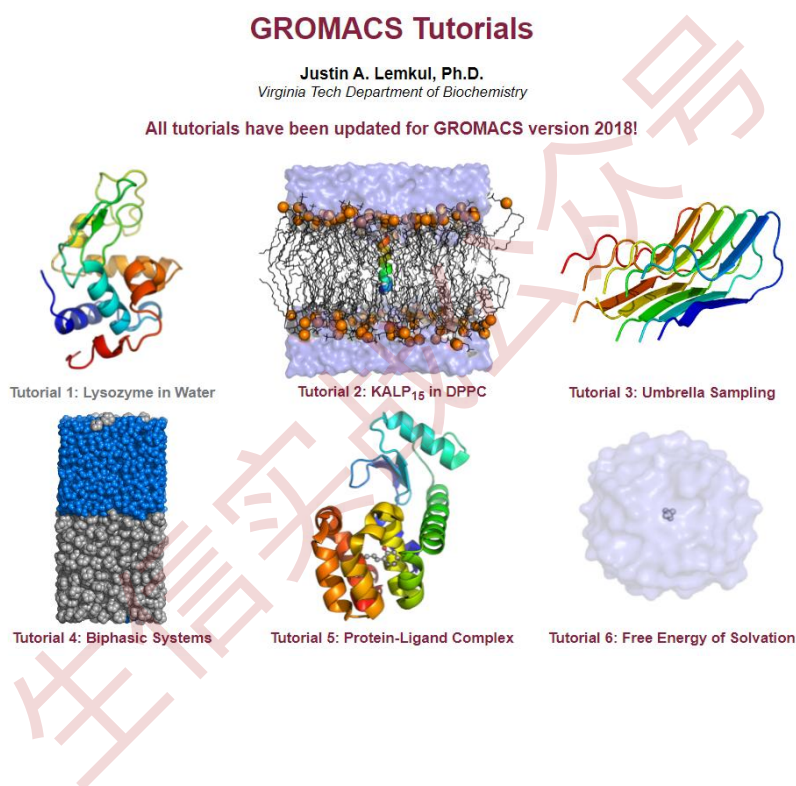
第四章 Gromacs 实验准备

4.1 Gromacs 实验 tutorial 说明

4.1.1 示例在线文档

这里包括了GROMACS的多个场景的基础实验，可以参照学习

<http://www.mdtutorials.com/gmx/index.html>



4.1.2 蛋白质入门的在线例子

本文档主要是参照第1个lysozyme in water的实验场景，

<http://www.mdtutorials.com/gmx/lysozyme/index.html>

但绝对不是简单的翻译：

- ✧ 为新手新增补充了大量的Linux基础技能、
- ✧ 本文档包含了详细的安装GROMACS的过程
- ✧ 补充了相关理论知识与实操技能

《生信实战》微信公众号

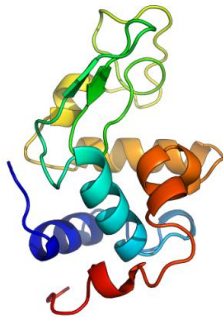
✧ 对相关细节和难点进行了说明

✧

GROMACS Tutorial

Lysozyme in Water

Justin A. Lemkul, Ph.D.
Virginia Tech Department of Biochemistry



This example will guide a new user through the process of setting up a simulation system containing a protein (lysozyme) in a box of water, with ions. Each step will contain an explanation of input and output, using typical settings for general use. This tutorial assumes you are using a GROMACS version in the 2018.x series.

4.2 再次强调掌握 linux 系统的基础操作

如cd pwd ls vi grep 等命令的操作和使用，例如会在linux上做如下基本操作：

```
cd /apps/svr/
```

```
ls -l
```

```
pwd
```

```
ls -l gromacs/
```

```
cat /etc/lsb-release
```

```
.....
```

```
asp@coding_gene94:~$ cd /apps/avr/
asp@coding_gene94:/apps/avr$ ls -l
total 16
drwxrwxr-x 5 asp asp 4096 Apr 28 11:46 cmake
drwxrwxr-x 6 asp asp 4096 Apr 28 12:00 fftw3
drwxrwxr-x 4 asp asp 4096 Apr 28 13:50 gromacs
drwxrwxr-x 7 asp asp 4096 Apr 28 12:42 openmpi
asp@coding_gene94:/apps/avr$ pwd
/apps/avr
asp@coding_gene94:/apps/avr$ ls -l gromacs/
total 8
drwxrwxr-x 6 asp asp 4096 Apr 28 13:30 gmx2020
drwxrwxr-x 6 asp asp 4096 Apr 28 13:51 gmx2020_mpi
asp@coding_gene94:/apps/avr$ cat /etc/lsb-release
DISTRIB_ID=Ubuntu
DISTRIB_RELEASE=16.04
DISTRIB_CODENAME=xenial
DISTRIB_DESCRIPTION="Ubuntu 16.04.6 LTS"
```

熟练使用vi或vim编辑器，进行文本内容的增删改、保存等操作。

基本的Linux操作这里也不赘述，请大家自行掌握。

4.3 安装好 Gromacs2018 以上版本

具体安装可以参考教程文章：

检查gromacs的安装，可以执行gmx_mpi help grompp或gmx_mpi grompp -h命令，能

正常执行看到帮助信息基本代表程序包安装没问题；

```
source /apps/avr/gromacs/gmx2020/bin/GMXRC.bash
```

```
which gmx
```

gmx --version能显示对应的版本信息

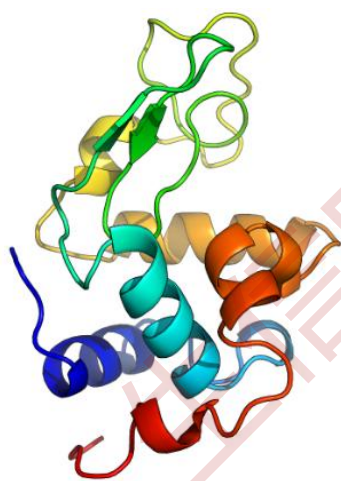
```
asp@coding_gene94:~$ ls -l /apps/avr/gromacs/gmx2020/bin
total 280
-rwxr-xr-x 1 asp asp 3382 Feb 3 2022 demux.pl
-rwxr-xr-x 1 asp asp 102656 Apr 28 13:00 gmx
-rw-r--r-- 1 asp asp 139642 Feb 3 2022 gmx-completion.bash
-rw-r--r-- 1 asp asp 37 Apr 28 12:53 gmx-completion-gmx.bash
-rwxr-xr-x 1 asp asp 580 Apr 28 12:53 GMXRC
-rwxr-xr-x 1 asp asp 3146 Apr 28 12:53 GMXRC.bash
-rwxr-xr-x 1 asp asp 3048 Apr 28 12:53 GMXRC.csh
-rwxr-xr-x 1 asp asp 125 Apr 28 12:53 GMXRC.zsh
-rwxr-xr-x 1 asp asp 9008 Feb 3 2022 xplor2gmx.pl
asp@coding_gene94:~$
asp@coding_gene94:~$
asp@coding_gene94:~$ source /apps/avr/gromacs/gmx2020/bin/GMXRC.bash
asp@coding_gene94:~$ which gmx
/apps/avr/gromacs/gmx2020/bin/gmx
```

```
GROMACS:      gmx, version 2020.7
Executable:    /apps/avr/gromacs/gmx2020/bin/gmx
Data prefix:   /apps/avr/gromacs/gmx2020
Working dir:   /home/asp
Command line:  gmx --version

GROMACS version: 2020.7
Verified release checksum is 761fa697fcd80bd396d0dd395d20957171ad38eb6e080f0760c9511f862dbb8
Precision:      single
Memory model:   64 bit
MPI library:    thread_mpi
OpenMP support: enabled (GMX_OPENMP_MAX_THREADS = 64)
GPU support:    disabled
SIMD instructions: AVX2_256
FFT library:    fftw-3.3.10
RDTSOP usage:   enabled
TNG support:    enabled
Hwloc support:  disabled
Tracing support: disabled
C compiler:     /usr/bin/cc GNU 5.4.0
C compiler flags: -mavx2 -mfma -fexcess-precision=fast -funroll-all-loops -O3 -DNDEBUG
C++ compiler:   /usr/bin/c++ GNU 5.4.0
C++ compiler flags: -mavx2 -mfma -fexcess-precision=fast -funroll-all-loops -fopenmp -O3 -DNDEBUG
```

4.4 可访问 PDB 数据库网站

<https://www.rcsb.org>, 获取编号为1AKI的PDB格式的文件, 作为本次实验的结构示例文件:

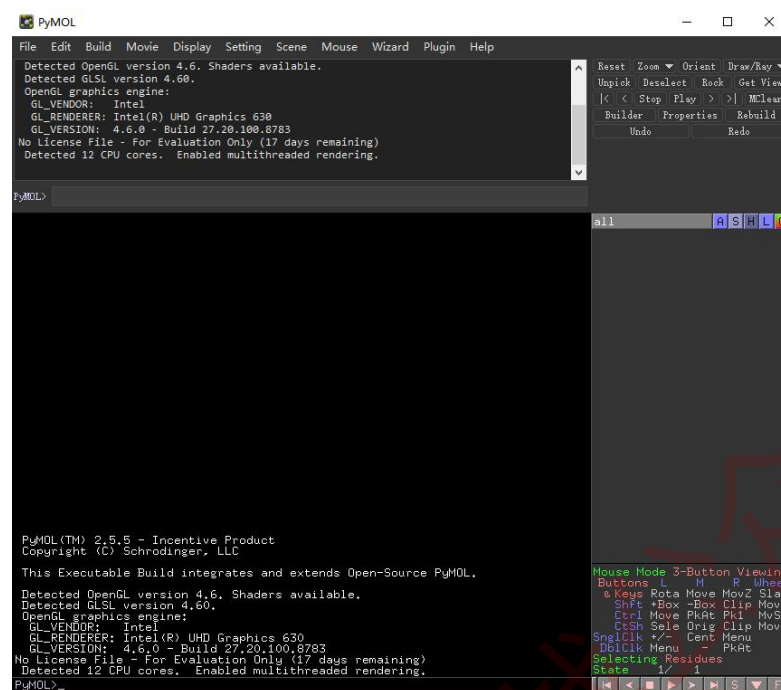


4.5 安装好 PyMol 软件

https://pymol.org/installers/PyMOL-2.5.5-Windows-x86_64.exe

下载链接后面可能会失效, 可以去官方网站pymol.org去查询下载。

下载后直接下一步--下一步.....进行安装，安装后双击图标，可以启动如下图形界面

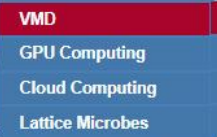


4.6 安装好 VMD 软件

下载地址<http://www.ks.uiuc.edu/Research/vmd/>

NIH CENTER FOR MACROMOLECULAR MODELING & BIOINFORMATICS | UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN

THEORETICAL *and* COMPUTATIONAL BIOPHYSICS GROUP

Home	Research	Publications	Software	Instruction	News	Galleries	Facilities	About Us
Home		NAMD		Download				
Overview		VMD		Plugins				
Publications		GPU Computing		User's Guide				
Research		Cloud Computing		Mailing List				
Software		Lattice Microbes						
		Atomic Resolution Brownian Dynamics						
		MDF						
		QwikMD						
		VND (Neuronal)						
		Other						

VMD is a molecular visualization program ([more details...](#)) designed for visualizing large biomolecular systems using

Spotlight

VMD provides a **sequence viewer** used to view a structure's sequence, secondary structure code, and other information in VMD's OpenGL graphics window.

下载Windows版本后，需要做一些简单的注册，根据提示操作。

下载后也是下一步--下一步完成安装即可

New User Registration

New User Registration for 'coding_gene':

First and Last Name:

Email Address:

Affiliation:
☒ Academic ☐ Government ☐ Industrial ☐ Other (specify)

The number of people using TCBG software at my site is:
☐ 1 ☒ 2-4 ☐ 5-10 ☐ 11-20 ☐ 21 or more

I use TCBG software primarily for:
☒ Research ☐ Teaching ☐ Commerce ☐ Personal

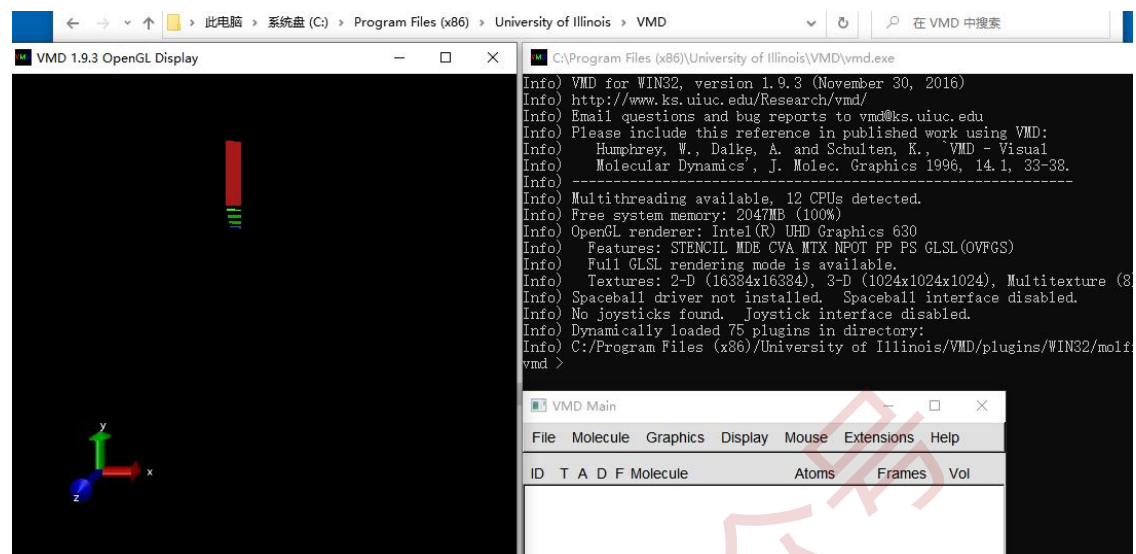
The work I do with TCBG software is funded (at least partially) by NIH:
☒ Yes ☐ No

Re-enter password for confirmation:

然后获取到下载链接

这里我安装的是VMD 1.9.3 for Microsoft Windows 2000/XP/Vista/7/8/10

完成本地安装：



第五章 Gromacs 蛋白质结构模拟简明过程

5.1 准备蛋白结构文件和 MD 参数

准备实验需要的蛋白质结构文件1aki.pdb，和相关MD参数。以下PDB结构文件和参数文件可在公众号后台回复关键词“简明gmx”

PDB数据库获取1aki.pdb文件

```
-rw-r--r-- 1 asp asp 116316 Jun 24 23:16 1aki.pdb
```

相关参数文件：

```
-rw-rw-r-- 1 asp asp 1045 Jun 24 23:16 ions.mdp
```

```
-rw-rw-r-- 1 asp asp 2741 Jun 24 23:16 md.mdp
```

```
-rw-rw-r-- 1 asp asp 10865 Jun 24 23:16 mdout.mdp
```

```
-rw-rw-r-- 1 asp asp 1045 Jun 24 23:16 minim.mdp
```

```
-rw-rw-r-- 1 asp asp 2591 Jun 24 23:16 npt.mdp
```

```
-rw-rw-r-- 1 asp asp 2369 Jun 24 23:16 nvt.mdp
```

使用GROMACS2018以上版本，如果还没有安装好gromacs，也可以参照文章：

5.2 步骤 1-从 PDB 结构生成拓扑文件

PDB数据库获取1aki.pdb文件，去掉本实验中不需要的水分子

```
grep -v HOH 1aki.pdb > 1AKI_clean.pdb
```

生成拓扑，执行如下命令：

```
gmx pdb2gmx -f 1AKI_clean.pdb -o 1AKI_processed.gro -water spce
```

执行后输入15选择力场

15: OPLS-AA/L all-atom force field (2001 aminoacid dihedrals)

执行后记录下体系的电荷情况，例如本例子：Total charge 8.000 e

5.3 步骤 2-检查和理解拓扑文件

查看和理解拓扑文件

```
more topol.top
```

查看相关力场文件

```
find /apps/svr/gromacs/gmx2020 -name forcefield.itp
```

```
find /apps/svr/gromacs/gmx2020 -name forcefield.itp | grep oplsa
```

5.4 步骤 3-定义模拟体系的尺寸并加水

使用editconf模块定义晶胞容器的尺寸

```
gmx editconf -f 1AKI_processed.gro -o 1AKI_newbox.gro -c -d 1.0 -bt cubic
```

使用solvate调用溶剂模块向晶胞容器中填充水

```
gmx solvate -cp 1AKI_newbox.gro -cs spc216.gro -o 1AKI_solv.gro -p topol.top
```

目前是10644个水分子

5.5 步骤 4-为模拟体系添加离子

使用下面的命令来产生ions.tpr这个.tpr文件:

```
gmx grompp -f ions.mdp -c 1AKI_solv.gro -p topol.top -o ions.tpr
```

加入中和离子

```
gmx genion -s ions.tpr -o 1AKI_solv_ions.gro -p topol.top -pname NA -nname CL -neutral
```

选择13后回车

检查结果输出: 添加了8个CL-离子, 中和系统的8个正电荷

Replacing 8 solute molecules in topology file (topol.top) by 0 NA and 8 CL ions.

topol.top文件最后是:

SOL 10636

CL 8

5.6 步骤 5-体系能量最小化

生成能量最小化的拓扑文件:

```
gmx grompp -f minim.mdp -c 1AKI_solv_ions.gro -p topol.top -o em.tpr
```

执行能量最小化 (一般会在1000步内达到能量最小化)

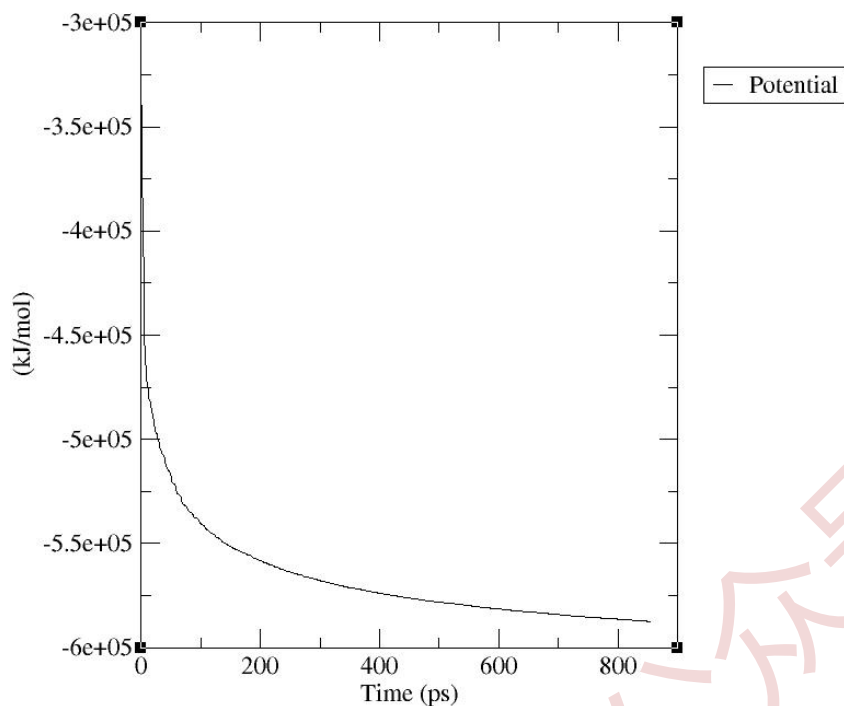
```
gmx mdrun -v -deffnm em
```

gmx energy模块来分析能量最小化的结果

```
gmx energy -f em.edr -o potential.xvg
```

提示时, 输入10 0来选择势能Potential(10), 并用零(0)来结束输入

查看能量最小化的势能变化: xmgrace potential.xvg



5.7 步骤 6-NVT 温度平衡

第1步: `gmx grompp -f nvt.mdp -c em.gro -r em.gro -p topol.top -o nvt.tpr`

第2步: `gmx mdrun -deffnm nvt`

(需要一点时间完成平衡, 此时Linux服务器CPU使用率会比较高, 我的12核CPU大概10分钟运行完成, 运行时间可以根据自己的CPU数量参考)

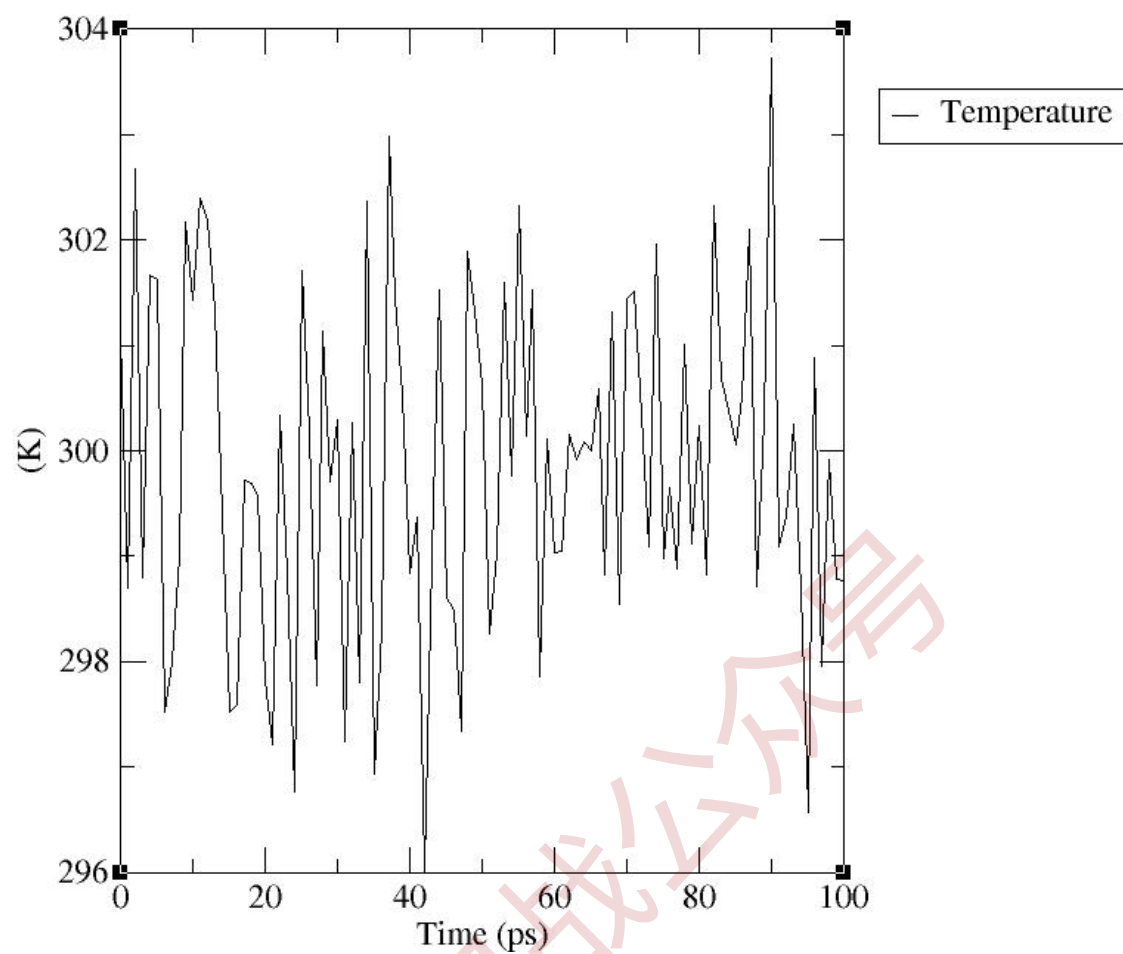
平衡结果分析

`gmx energy -f nvt.edr -o temperature.xvg`

在提示符下键入“16 0”, 选择温度代号16, 选择0退出

查看温度平衡情况

`xmgrace temperature.xvg`



5.8 步骤 7-NPT 压力平衡

第1步: `gmx grompp -f npt.mdp -c nvt.gro -r nvt.gro -t nvt.cpt -p topol.top -o npt.tpr`

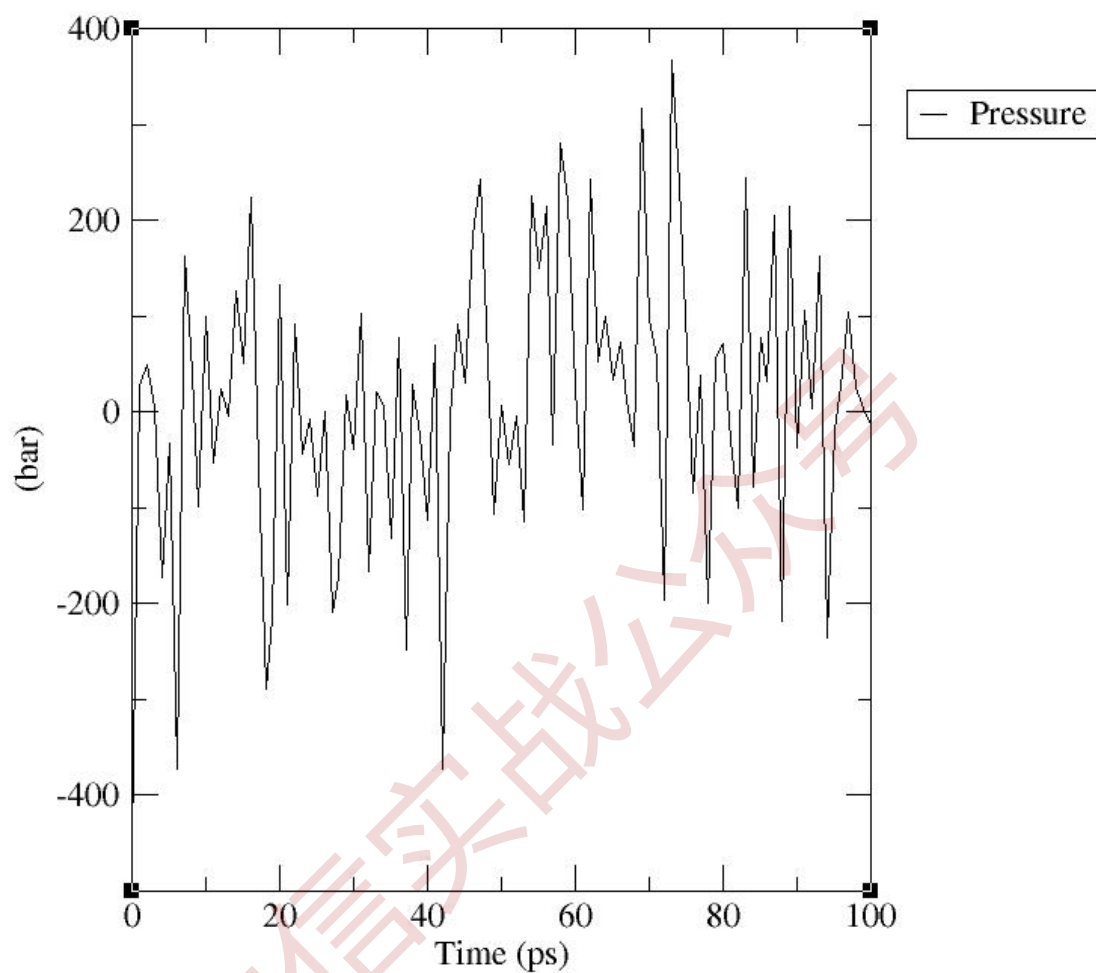
第2步: `gmx mdrun -deffnm npt`

压力分析:

`gmx energy -f npt.edr -o pressure.xvg`

根据提示输入18 0 获得压力的数据结果

通过xmgrace pressure.xvg查看



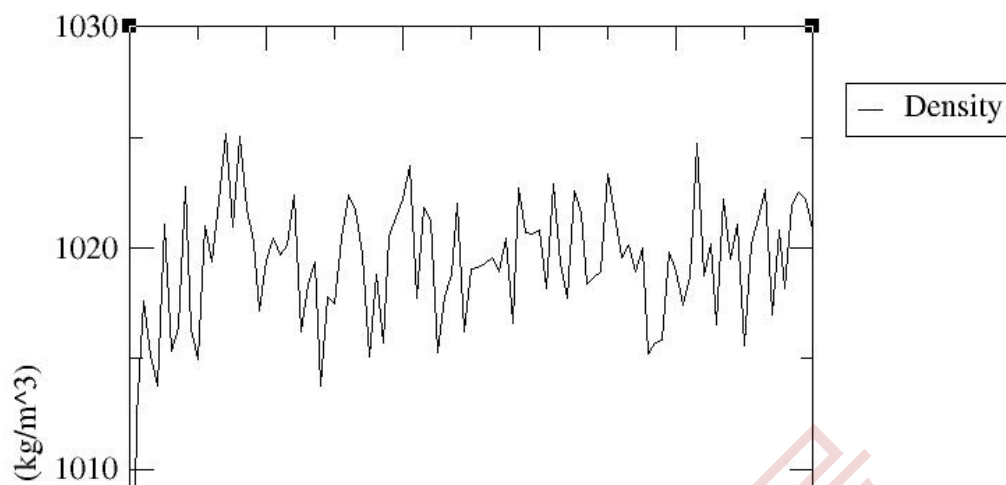
密度分析:

```
gmx energy -f npt.edr -o density.xvg
```

根据提示 输入 24 0 获得结果

执行xmgrace density.xvg查看:

GROMACS Energies



5.9 步骤 8-正式的动力学模拟

执行grompp生成模拟的tpr文件:

```
gmx grompp -f md.mdp -c npt.gro -t npt.cpt -p topol.top -o md_0_1.tpr
```

执行模拟

```
gmx mdrun -deffnm md_0_1
```

5.10 步骤 9-分子动力学模拟结果分析

使用下面的工具来进行修复:

```
gmx trjconv -s md_0_1.tpr -f md_0_1.xtc -o md_0_1_noPBC.xtc -pbc mol -center
```

选择1 (Select 1 ("Protein") as the group to be centered and 0 ("System") for output)

完成后生成md_0_1_noPBC.xtc, 我们将基于这个修复的数据进行后续的数据分析

计算RMSD:

```
gmx rms -s md_0_1.tpr -f md_0_1_noPBC.xtc -o rmsd.svg -tu ns
```

Choose 4 ("Backbone") for both the least-squares fit and the group for RMSD

calculation

计算相对于模拟之前晶体的结构差异，可以使用下面的命令：

```
gmx rms -s em.tpr -f md_0_1_noPBC.xtc -o rmsd_xtal.svg -tu ns
```

也是选择4 ("Backbone")

可以把两个RMSD拟合一下查看：

```
xmrgace rmsd.svg rmsd_xtal.svg
```

计算回旋半径：

Linux执行：

```
gmx gyrate -s md_0_1.tpr -f md_0_1_noPBC.xtc -o gyrate.svg
```

选择group 1 (Protein)，执行完成后新生成了gyrate.svg

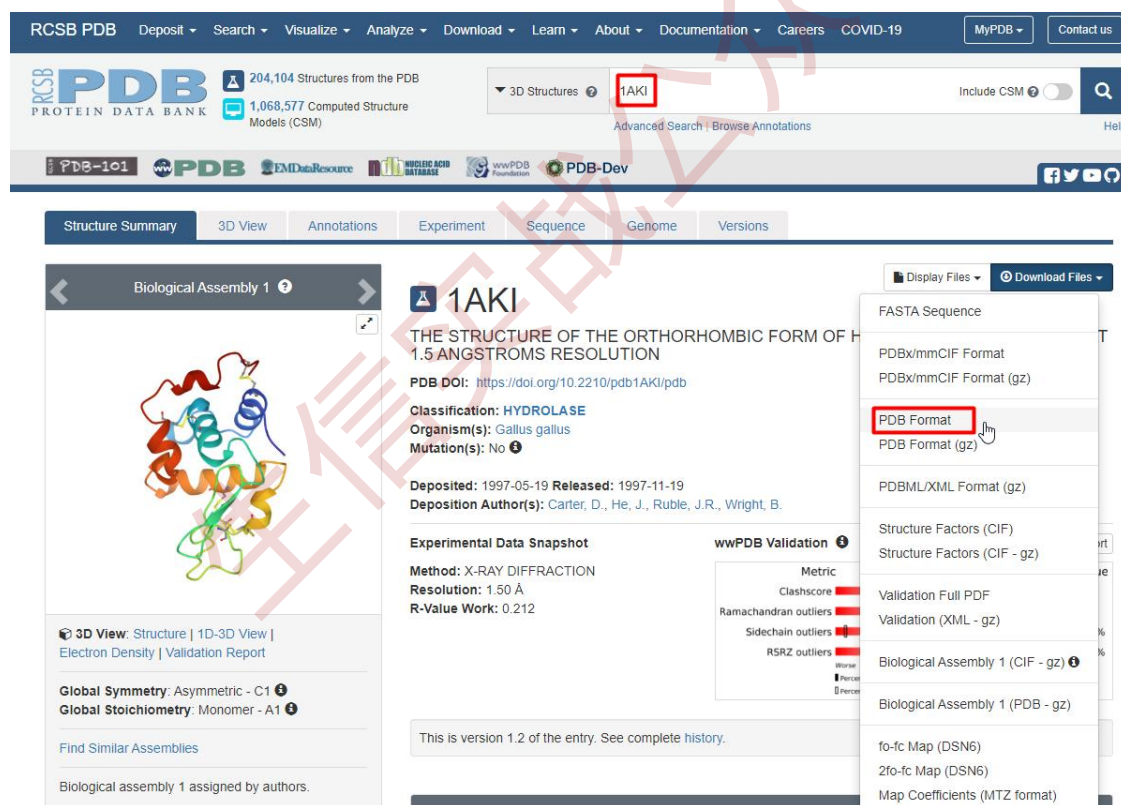
使用xmrgace gyrate.svg查看

第六章 GROMACS 实验基础详细教程

6.1 步骤 1-从 PDB 结构生成拓扑文件

6.1.1 结构准备与处理

打开PDB蛋白数据库<https://www.rcsb.org/> 输入编号1AKI进行搜索:



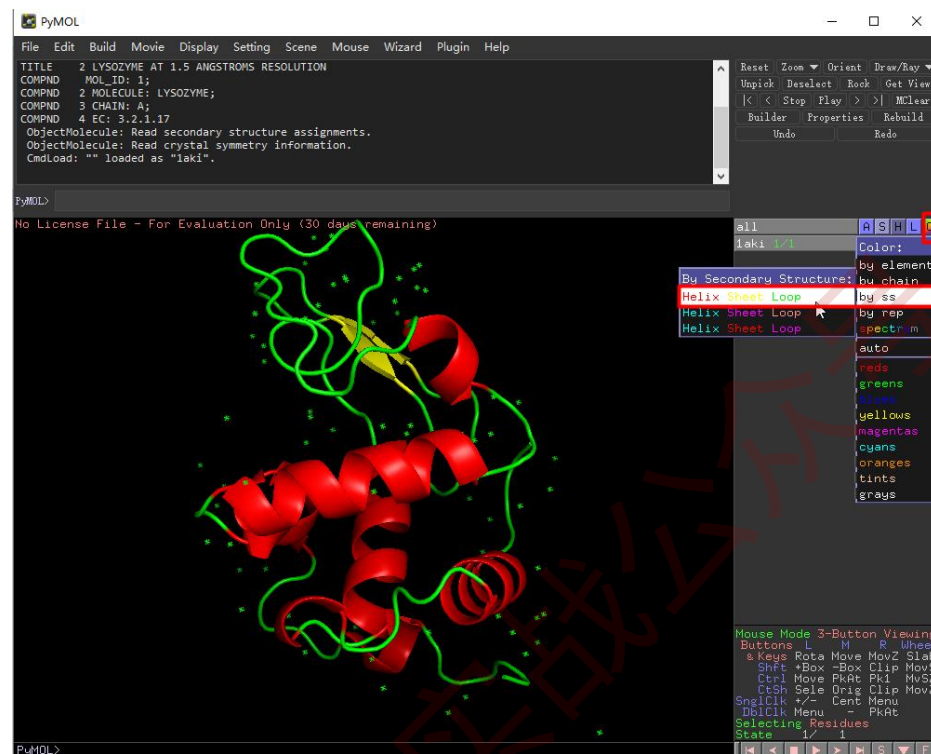
The screenshot shows the RCSB PDB website interface. The search bar at the top contains the ID '1AKI'. The main content area displays the protein structure of 1AKI, which is a hydrolase from Gallus gallus. The page includes a 3D view of the structure and a table of experimental data. A dropdown menu is open, showing options to download the structure in various formats, with 'PDB Format' highlighted.

Metric	Value
Clashscore	1.0
Ramachandran outliers	0.0
Sidechain outliers	0.0
RSRZ outliers	0.0

下载后, 可以在本地使用PyMOL软件打开



在PyMol中视图模式可以选择 by ss



图中绿色的点点，就是水分子，我们本次实验要去掉水分子。

6.1.2实操 3-去掉结构中的水分子

去除水分子可以在linux系统下执行：

```
grep -v HOH laki.pdb > 1AKI_clean.pdb
```

说明：

去除水分子并不是常规操作，某些和水分子有紧密联系或活性功能的场景是需要水分子的。

但我们本次的实验场景不需要水分子

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2、从PDB文件中去除水分子有多种方式，可以使用一般的文本编辑器进行编辑删除HOH，也可以使用linux的命令`grep -v HOH 1aki.pdb > 1AKI_clean.pdb`进行快速处理

去除水分子后的1AKI_clean.pdb可视化结构如下：



6.1.3 拓扑转换说明和准备

注意检查.pdb文件中MISSING注释下面列出的项，这些项代表了在晶体结构文件中缺失的那些原子或残基。在模拟中，末端区域的缺失可能并不会引起问题，但缺失原子的不完整内部序列或任何氨基酸残基都将导致pdb2gmx程序运行失败。必须使用其他软件对这些缺失的原子/残基进行建模并补充完整

注意pdb2gmx不是全能的，它无法为任意分子生成拓扑文件，而只能用于力场中已经定义的残基

pdb2gmx的主要目的是生成三类文件：

- ✓ The topology for the molecule.
- ✓ A position restraint file.
- ✓ A post-processed structure file.

翻译出来就是：

- ✓ 分子拓扑文件 .top
- ✓ 位置限制文件 .itp
- ✓ 后处理结构文件 .gro

检查安装环境：

```
asp@coding_gene94_ubuntu16:~/mdtest/demol$ ls -l /apps/avr/gromacs/gmx2020/bin/  
total 280  
-rwxr-xr-x 1 asp asp 3382 Feb 3 2022 demux.pl  
-rwxr-xr-x 1 asp asp 102656 Apr 28 13:00 gmx  
-rw-r--r-- 1 asp asp 139642 Feb 3 2022 gmx-completion.bash  
-rw-r--r-- 1 asp asp 37 Apr 28 12:53 gmx-completion-gmx.bash  
-rwxr-xr-x 1 asp asp 580 Apr 28 12:53 GMXRC  
-rwxr-xr-x 1 asp asp 3146 Apr 28 12:53 GMXRC.bash  
-rwxr-xr-x 1 asp asp 3048 Apr 28 12:53 GMXRC.csh  
-rwxr-xr-x 1 asp asp 125 Apr 28 12:53 GMXRC.zsh  
-rwxr-xr-x 1 asp asp 9008 Feb 3 2022 xplor2gmx.pl
```

```
asp@coding_gene94_ubuntu16:~/mdtest/demol$ source  
/apps/avr/gromacs/gmx2020/bin/GMXRC.bash  
asp@coding_gene94_ubuntu16:~/mdtest/demol$ which gmx  
/apps/avr/gromacs/gmx2020/bin/gmx
```

source后可以直接敲gmx命令运行了，当然如果不source环境变量，也可以直接敲全路径：

```
/apps/avr/gromacs/gmx2020/bin/gmx
```

总之操作之前，确认会使用linux操作、安装好了gromacs、gmx命令能调用。避免后续实验
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遇到麻烦。

6.1.4 实操 4-生成拓扑文件

下面开始生成拓扑，执行如下命令：

```
gmx pdb2gmx -f 1AKI_clean.pdb -o 1AKI_processed.gro -water spce
```

例如我的环境显示如下：

```
GROMACS:      gmx pdb2gmx, version 2020.7
Executable:    /apps/svr/gromacs/gmx2020/bin/gmx
Data prefix:    /apps/svr/gromacs/gmx2020
Working dir:    /home/asp/mdtest/demo1
Command line:   gmx pdb2gmx -f 1AKI_clean.pdb -o 1AKI_processed.gro -water spce

Select the Force Field:
From '/apps/svr/gromacs/gmx2020/share/gromacs/top':
 1: AMBER03 protein, nucleic AMBER94 (Duan et al., J. Comp. Chem. 24, 1999-2012, 2003)
 2: AMBER94 force field (Cornell et al., JACS 117, 5179-5197, 1995)
 3: AMBER96 protein, nucleic AMBER94 (Kollman et al., Acc. Chem. Res. 29, 461-469, 1996)
 4: AMBER99 protein, nucleic AMBER94 (Wang et al., J. Comp. Chem. 21, 1049-1074, 2000)
 5: AMBER99SB protein, nucleic AMBER94 (Hornak et al., Proteins 65, 712-725, 2006)
 6: AMBER99SB-ILDN protein, nucleic AMBER94 (Lindorff-Larsen et al., Proteins 78, 1950-58, 2010)
 7: AMBERGS force field (Garcia & Sanbonmatsu, PNAS 99, 2782-2787, 2002)
 8: CHARMM27 all-atom force field (CHARM22 plus CMAP for proteins)
 9: GROMOS96 43a1 force field
10: GROMOS96 43a2 force field (improved alkane dihedrals)
11: GROMOS96 45a3 force field (Schuler JCC 2001 22 1205)
12: GROMOS96 53a5 force field (JCC 2004 vol 25 pag 1656)
13: GROMOS96 53a6 force field (JCC 2004 vol 25 pag 1656)
14: GROMOS96 54a7 force field (Eur. Biophys. J. (2011), 40,, 843-856, DOI: 10.1007/s00249-011-0700-9)
15: OPLS-AA/L all-atom force field (2001 aminoacid dihedrals)
```

Select the Force Field:

From '/apps/svr/gromacs/gmx2020/share/gromacs/top':

- 1: AMBER03 protein, nucleic AMBER94 (Duan et al., J. Comp. Chem. 24, 1999-2012, 2003)
- 2: AMBER94 force field (Cornell et al., JACS 117, 5179-5197, 1995)
- 3: AMBER96 protein, nucleic AMBER94 (Kollman et al., Acc. Chem. Res. 29, 461-469, 1996)
- 4: AMBER99 protein, nucleic AMBER94 (Wang et al., J. Comp. Chem. 21, 1049-1074, 2000)
- 5: AMBER99SB protein, nucleic AMBER94 (Hornak et al., Proteins 65, 712-725, 2006)
- 6: AMBER99SB-ILDN protein, nucleic AMBER94 (Lindorff-Larsen et al., Proteins 78, 1950-58, 2010)
- 7: AMBERGS force field (Garcia & Sanbonmatsu, PNAS 99, 2782-2787, 2002)
- 8: CHARMM27 all-atom force field (CHARM22 plus CMAP for proteins)
- 9: GROMOS96 43a1 force field
- 10: GROMOS96 43a2 force field (improved alkane dihedrals)
- 11: GROMOS96 45a3 force field (Schuler JCC 2001 22 1205)
- 12: GROMOS96 53a5 force field (JCC 2004 vol 25 pag 1656)
- 13: GROMOS96 53a6 force field (JCC 2004 vol 25 pag 1656)
- 14: GROMOS96 54a7 force field (Eur. Biophys. J. (2011), 40,, 843-856, DOI:

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10. 1007/s00249-011-0700-9)

15: OPLS-AA/L all-atom force field (2001 aminoacid dihedrals)

这个力场的选择很重要,必须选择适合自己的研究体系的力场。这个实验里我们选择all-atom

OPLS力场, **输入15然后回车。**

完整的执行日志如下:

```
GROMACS:      gmx pdb2gmx, version 2020.7
Executable:    /apps/avr/gromacs/gmx2020/bin/gmx
Data prefix:    /apps/avr/gromacs/gmx2020
Working dir:    /home/asp/mdtest/demo1
Command line:
  gmx pdb2gmx -f 1AKI_clean.pdb -o 1AKI_processed.gro -water spce

Select the Force Field:
From '/apps/avr/gromacs/gmx2020/share/gromacs/top':
  1: AMBER03 protein, nucleic AMBER94 (Duan et al., J. Comp. Chem. 24, 1999-2012, 2003)
  2: AMBER94 force field (Cornell et al., JACS 117, 5179-5197, 1995)
  3: AMBER96 protein, nucleic AMBER94 (Kollman et al., Acc. Chem. Res. 29, 461-469, 1996)
  4: AMBER99 protein, nucleic AMBER94 (Wang et al., J. Comp. Chem. 21, 1049-1074, 2000)
  5: AMBER99SB protein, nucleic AMBER94 (Hornak et al., Proteins 65, 712-725, 2006)
  6: AMBER99SB-ILDN protein, nucleic AMBER94 (Lindorff-Larsen et al., Proteins 78, 1950-58, 2010)
  7: AMBERGS force field (Garcia & Sanbonmatsu, PNAS 99, 2782-2787, 2002)
  8: CHARMM27 all-atom force field (CHARM22 plus CMAP for proteins)
  9: GROMOS96 43a1 force field
 10: GROMOS96 43a2 force field (improved alkane dihedrals)
 11: GROMOS96 45a3 force field (Schuler JCC 2001 22 1205)
 12: GROMOS96 53a5 force field (JCC 2004 vol 25 pag 1656)
 13: GROMOS96 53a6 force field (JCC 2004 vol 25 pag 1656)
 14: GROMOS96 54a7 force field (Eur. Biophys. J. (2011), 40,, 843-856, DOI:
10.1007/s00249-011-0700-9)
15: OPLS-AA/L all-atom force field (2001 aminoacid dihedrals)
15
```

Using the Oplsaa force field in directory oplsa. ff

```
going to rename oplsa. ff/aminoacids.r2b
Opening force field file
/apps/avr/gromacs/gmx2020/share/gromacs/top/oplsaa. ff/aminoacids.r2b
Reading 1AKI_clean.pdb...
```



```
WARNING: all CONECT records are ignored
Read 'LYSOZYME', 1001 atoms
Analyzing pdb file
Splitting chemical chains based on TER records or chain id changing.
There are 1 chains and 0 blocks of water and 129 residues with 1001 atoms
```

chain	#res	#atoms
1 'A'	129	1001

```
All occupancies are one
Opening force field file
/apps/avr/gromacs/gmx2020/share/gromacs/top/oplsaa.ff/atomtypes.atp
Reading residue database... (Oplsaa)
Opening force field file
/apps/avr/gromacs/gmx2020/share/gromacs/top/oplsaa.ff/aminoacids.rtp
Opening force field file
/apps/avr/gromacs/gmx2020/share/gromacs/top/oplsaa.ff/aminoacids.hdb
Opening force field file
/apps/avr/gromacs/gmx2020/share/gromacs/top/oplsaa.ff/aminoacids.n.tdb
Opening force field file
/apps/avr/gromacs/gmx2020/share/gromacs/top/oplsaa.ff/aminoacids.c.tdb
Processing chain 1 'A' (1001 atoms, 129 residues)
Analysing hydrogen-bonding network for automated assignment of histidine
protonation. 213 donors and 184 acceptors were found.
There are 255 hydrogen bonds
Will use HISE for residue 15
Identified residue LYS1 as a starting terminus.
Identified residue LEU129 as a ending terminus.
8 out of 8 lines of specbond.dat converted successfully
Special Atom Distance matrix:
```

		CYS6	MET12	HIS15	CYS30	CYS64	CYS76	CYS80
		SG48	SD87	NE2118	SG238	SG513	SG601	SG630
MET12	SD87	1.166						
HIS15	NE2118	1.776	1.019					
CYS30	SG238	1.406	1.054	2.069				
CYS64	SG513	2.835	1.794	1.789	2.241			
CYS76	SG601	2.704	1.551	1.468	2.116	0.765		
CYS80	SG630	2.959	1.951	1.916	2.391	0.199	0.944	
CYS94	SG724	2.550	1.407	1.382	1.975	0.665	0.202	0.855
MET105	SD799	1.827	0.911	1.683	0.888	1.849	1.461	2.036
CYS115	SG889	1.576	1.084	2.078	0.200	2.111	1.989	2.262
CYS127	SG981	0.197	1.072	1.721	1.313	2.799	2.622	2.934
		CYS94	MET105	CYS115				

```

          SG724   SD799   SG889
MET105   SD799   1.381
CYS115   SG889   1.853   0.790
CYS127   SG981   2.475   1.686   1.483
Linking CYS-6 SG-48 and CYS-127 SG-981...
Linking CYS-30 SG-238 and CYS-115 SG-889...
Linking CYS-64 SG-513 and CYS-80 SG-630...
Linking CYS-76 SG-601 and CYS-94 SG-724...
Start terminus LYS-1: NH3+
End terminus LEU-129: COO-
Checking for duplicate atoms....
Generating any missing hydrogen atoms and/or adding termini.
Now there are 129 residues with 1960 atoms
Making bonds...
Number of bonds was 1984, now 1984
Generating angles, dihedrals and pairs...
Before cleaning: 5142 pairs
Before cleaning: 5187 dihedrals
Keeping all generated dihedrals
Making cmap torsions...
There are 5187 dihedrals, 426 impropers, 3547 angles
          5106 pairs,          1984 bonds and          0 virtual sites
Total mass 14313.193 a.m.u.
Total charge 8.000 e
Writing topology

Writing coordinate file...

----- PLEASE NOTE -----
You have successfully generated a topology from: 1AKI_clean.pdb.
The Oplsaa force field and the spce water model are used.
----- ETON ESAELP -----

GROMACS reminds you: "In this house, we OBEY the laws of thermodynamics!" (Homer
Simpson)
```

注意：这里的输出Total charge 8.000 e，带有8个电荷，后面我们加入离子时会参照这里的信息。

执行完成后，可以看到生成了3个新的文件：

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```
asp@coding_gene94_ubuntu16:~/mdtest/demol$ ls -lrt
total 884
-rw-r--r-- 1 asp asp 116316 Apr 26 10:45 laki.pdb
-rw-r--r-- 1 asp asp 109917 Apr 26 15:04 1AKI_clean.pdb
-rw-rw-r-- 1 asp asp 552955 May 12 19:02 topol.top
-rw-rw-r-- 1 asp asp 31304 May 12 19:02 posre.itp
-rw-rw-r-- 1 asp asp 88246 May 12 19:02 1AKI_processed.gro
```

分别是：

- ✓ 分子拓扑文件 .top
- ✓ 位置限制文件 .itp
- ✓ 后处理结构文件 .gro
- 1AKI_processed.gro是一个GROMACS格式的结构文件，包含力场中定义的所有原子（即蛋白质中的氨基酸添加了H原子）
- topol.top文件是分子系统的拓扑文件
- posre.itp文件包含了用来限制重原子（heavy atoms）位置的信息

备注说明：pdb2gmx命令还有一些其他有用的选项，可以执行gmx pdb2gmx -h查看帮助信息

例如：

- ✓ -ignh:忽略PDB文件中的H原子，尤其适用于核磁共振结构
- ✓ -ter:交互式指定N端和C端的电荷状态
- ✓ -inter:交互地为Glu、Asp、Lys、Arg和His分配电荷状态

完整的帮助信息参考下图：

```
Force field, interactive by default. Use -h for information.
-water <enum> (select)
      Water model to use: select, none, spc, spce, tip3p, tip4p, tip5p,
      tips3p
-[no]inter (no)
      Set the next 8 options to interactive
-[no]ss (no)
      Interactive SS bridge selection
-[no]ter (no)
      Interactive termini selection, instead of charged (default)
-[no]lys (no)
      Interactive lysine selection, instead of charged
-[no]arg (no)
      Interactive arginine selection, instead of charged
-[no]asp (no)
      Interactive aspartic acid selection, instead of charged
-[no]glu (no)
      Interactive glutamic acid selection, instead of charged
-[no]gln (no)
      Interactive glutamine selection, instead of charged
-[no]his (no)
      Interactive histidine selection, instead of checking H-bonds
-angle <real> (135)
      Minimum hydrogen-donor-acceptor angle for a H-bond (degrees)
-dist <real> (0.3)
      Maximum donor-acceptor distance for a H-bond (nm)
-[no]una (no)
      Select aromatic rings with united CH atoms on phenylalanine,
      tryptophane and tyrosine
-[no]ignh (no)
      Ignore hydrogen atoms that are in the coordinate file
-[no]missing (no)
      Continue when atoms are missing and bonds cannot be made, dangerous
-[no]v (no)
      Be slightly more verbose in messages
-posrefc <real> (1000)
      Force constant for position restraints
-vsite <enum> (none)
      Convert atoms to virtual sites: none, hydrogens, aromatics
-[no]heavyh (no)
      Make hydrogen atoms heavy
-[no]deuterate (no)
      Change the mass of hydrogens to 2 amu
-[no]chargegrp (yes)
      Use charge groups in the .rtp file
-[no]cmap (yes)
      Use cmap torsions (if enabled in the .rtp file)
-[no]renum (no)
      Renumber the residues consecutively in the output
-[no]rtpres (no)
      Use .rtp entry names as residue names
```

6.2 步骤 2-检查和理解拓扑文件

本章没有实际需要用到gromacs做的操作，却非常重要，主要是理解gromacs使用gmx

pdb2gmx生成的拓扑文件topol.top的内容，这对后续的模拟至关重要。后续结合自己的研究对象进行模拟时，如果拓扑文件存在异常，则可能出现相关严重的运行报错，因此理解拓扑文件才能解决后续报错问题。

上一章节，我们通过gmx pdb2gmx命令，输入去掉了水分子的1AKI_clean.pdb文件。并使用gmx pdb2gmx -f 1AKI_clean.pdb -o 1AKI_processed.gro -water spce生成了对应的3个文件：topol.top、posre.itp、1AKI_processed.gro

```
asp@coding_gene94_ubuntu16:~/mdtest/demo1$ ls -lrt
total 888
-rw-r--r-- 1 asp asp 116316 Apr 26 10:45 1aki.pdb
-rw-r--r-- 1 asp asp 109917 Apr 26 15:04 1AKI_clean.pdb
-rw-rw-r-- 1 asp asp 552963 May 14 10:47 topol.top
-rw-rw-r-- 1 asp asp 31304 May 14 10:47 posre.itp
-rw-rw-r-- 1 asp asp 88246 May 14 10:47 1AKI_processed.gro
asp@coding_gene94_ubuntu16:~/mdtest/demo1$
```

1AKI_processed.gro是一个GROMACS格式的结构文件，包含力场中定义的所有原子(即蛋白质中的氨基酸添加了H原子)

topol.top文件是蛋白分子系统的拓扑文件

posre.itp文件包含了用来限制重原子 (heavy atoms) 位置的信息

本章我们重点学习理解topol.top文件

6.2.1实操 5-linux 基础补充—如何查看 top 文件

可以把topol.top下载到Windows系统上, 使用一般的文本编辑器如Notepad++、UltraEdit等文本编辑器查看。

不过既然我们在linux上进行操作，还是建议习惯使用linux的命令操作；

这里我们再顺便复习和巩固一下linux基础操作命令

在Linux下查看这个top文件需要使用相关的命令，方式有多种：

■ 方式1: cat命令查看

查看topol.top文件的行数

```
cat topol.top | wc -l
```

显示有18408行

可以直接cat topol.top去查看全部内容，但是直接显示全部18408行内容不太友好，也不方便看

■ 方式2: 通过vi查看文件

vi topol.top

使用vi的好处就是可以定位到任意一个想查看的位置，可以显示行号，还可以搜索

查看行号: vi topol.top后, 执行:set nu 就可以显示文件的行编号

```
25 Protein_chain_A      3
26
27 [ atoms ]
28 ; nr      type  resnr residue  atom    cgnr      charge
29 ; residue  1 LYS  rtp  LYSH q +2.0
30      1  op1s_287      1  LYS      N        1      -0.3
31      2  op1s_290      1  LYS      H1        1       0.33
:set nu
```

搜索文件内容: vi topol.top后, 搜索某个字符或单词pdb2gmx

输入 :/pdb2gmx 回车后就开始搜索, 按n键可以查找下一个

```
9 ;      Created by:
10 ;
11 ;      :-) GROMACS - gmx pdb2gmx, 2020.7 (-:
12 ;
13 ;      Executable:  /apps/avr/gromacs/gmx2020/bin/gmx
14 ;      Data prefix: /apps/avr/gromacs/gmx2020
15 ;      Working dir: /home/asp/mdtest/demo1
16 ;      Command line:
17 ;      gmx pdb2gmx -f 1AKI_clean.pdb -o 1AKI_processed.gro -water spce
18 ;      Force field was read from the standard GROMACS share directory.
19 ;
20 ; Include forcefield parameters
21 #include "oplsaa.ff/forcefield.itp"
22
23 [ moleculetype ]
24 ; Name      nrexcl
25 Protein_chain_A      3
26
27 [ atoms ]
28 ; nr      type  resnr residue  atom    cgnr      charge      mass  typeB
29 ; residue  1 LYS  rtp  LYSH q +2.0
30      1  op1s_287      1  LYS      N        1      -0.3     14.0027
31      2  op1s_290      1  LYS      H1        1       0.33     1.008
:/pdb2gmx
```


定位到文件最后一行: vi topol.top后, 键盘输入大写字母G, 光标即可调到最后一行的位置

定位到指定行: vi topol.top后, 输入冒号+行号, 如 :20 回车后即可调到文件的第20行

■ 方式3: more命令翻页查看

```
more topol.top
```

会只显示文件前面的一页内容就不继续显示了:

想查看下一行, 就持续按“回车键”;

想查看下一页, 就持续按“空格键”

```
asp@coding_gene94_ubuntu16:~/mdtest/demo1$ more topol.top
;
;   File 'topol.top' was generated
;   By user: asp (1000)
;   On host: coding_gene94_ubuntu16
;   At date: Sun May 14 10:47:36 2023
;
;   This is a standalone topology file
;
;   Created by:
;
;               :- ) GROMACS - gmx pdb2gmx, 2020.7 (-:
;
;   Executable: /apps/avr/gromacs/gmx2020/bin/gmx
;   Data prefix: /apps/avr/gromacs/gmx2020
;   Working dir: /home/asp/mdtest/demo1
;   Command line:
;       gmx pdb2gmx -f 1AKI_clean.pdb -o 1AKI_processed.gro -water spce
;   Force field was read from the standard GROMACS share directory.
;
; Include forcefield parameters
#include "oplsaa.ff/forcefield.itp"

[ moleculetype ]
; Name          nrexcl
Protein_chain_A 3
```

关于在linux下查看topol.top文件就先说这么多

感觉是不是有点啰嗦, linux查看文件都要说半天, 可能是我年纪大了^.^

如果对linux还不怎么熟悉的同学, 就在Windows下看其实也是可以的。

6.2.2检查 top 文件的文件头

```
more topol.top
```



```
asp@coding_gene94_ubuntu16:~/mdtest/demo1$ more topol.top
;
;   File 'topol.top' was generated
;   By user: asp (1000)
;   On host: coding_gene94_ubuntu16
;   At date: Sun May 14 10:47:36 2023
;
;   This is a standalone topology file
;
;   Created by:
;               :-) GROMACS - gmx pdb2gmx, 2020.7 (-:
;
;   Executable:  /apps/svr/gromacs/gmx2020/bin/gmx
;   Data prefix: /apps/svr/gromacs/gmx2020
;   Working dir: /home/asp/mdtest/demo1
;   Command line:
;       gmx pdb2gmx -f 1AKI_clean.pdb -o 1AKI_processed.gro -water spce
;   Force field was read from the standard GROMACS share directory.
;
; Include forcefield parameters
#include "oplsaa.ff/forcefield.itp"

[ moleculetype ]
; Name          nrexcl
Protein_chain_A    3
```

前面这些以分号开头的是文件注释，

看红色框的第2行：

```
#include "oplsaa.ff/forcefield.itp"
```

这个定义了力场文件，调用了OPLS-AA力场的参数，一般位于拓扑文件的开头

其实我们可以在linux系统上查找对应的文件，参考命令如下：

```
find /apps/svr/gromacs/gmx2020 -name forcefield.itp
```

```
find /apps/svr/gromacs/gmx2020 -name forcefield.itp | grep oplsaa
```

```
asp@coding_gene94_ubuntu16:~$ find /apps/svr/gromacs/gmx2020 -name forcefield.itp
/apps/svr/gromacs/gmx2020/share/gromacs/top/amberG5.ff/forcefield.itp
/apps/svr/gromacs/gmx2020/share/gromacs/top/gromos45a3.ff/forcefield.itp
/apps/svr/gromacs/gmx2020/share/gromacs/top/gromos53a6.ff/forcefield.itp
/apps/svr/gromacs/gmx2020/share/gromacs/top/amber99sb.ff/forcefield.itp
/apps/svr/gromacs/gmx2020/share/gromacs/top/gromos54a7.ff/forcefield.itp
/apps/svr/gromacs/gmx2020/share/gromacs/top/amber03.ff/forcefield.itp
/apps/svr/gromacs/gmx2020/share/gromacs/top/amber99sb-ildn.ff/forcefield.itp
/apps/svr/gromacs/gmx2020/share/gromacs/top/amber96.ff/forcefield.itp
/apps/svr/gromacs/gmx2020/share/gromacs/top/gromos53a5.ff/forcefield.itp
/apps/svr/gromacs/gmx2020/share/gromacs/top/amber99.ff/forcefield.itp
/apps/svr/gromacs/gmx2020/share/gromacs/top/charmm27.ff/forcefield.itp
/apps/svr/gromacs/gmx2020/share/gromacs/top/oplsaa.ff/forcefield.itp
/apps/svr/gromacs/gmx2020/share/gromacs/top/gromos43a1.ff/forcefield.itp
/apps/svr/gromacs/gmx2020/share/gromacs/top/amber94.ff/forcefield.itp
/apps/svr/gromacs/gmx2020/share/gromacs/top/gromos43a2.ff/forcefield.itp
asp@coding_gene94_ubuntu16:~$
asp@coding_gene94_ubuntu16:~$
asp@coding_gene94_ubuntu16:~$
asp@coding_gene94_ubuntu16:~$ find /apps/svr/gromacs/gmx2020 -name forcefield.itp | grep oplsaa
/apps/svr/gromacs/gmx2020/share/gromacs/top/oplsaa.ff/forcefield.itp
asp@coding_gene94_ubuntu16:~$
```

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后面是moleculetype, 这个也很重要

```
[ moleculetype ]
```

```
; Name nrexcl
```

```
Protein_chain_A 3
```

```
; Include forcefield parameters
#include "oplsaa.ff/forcefield.itp"

[ moleculetype ]
; Name nrexcl
Protein_chain_A 3
```

“Protein_A” 定义了分子名称（这个蛋白质在PDB文件中被标定为A链）

nrexcl为3, nrexcl=n:表示不计算 $\leq n$ 个键相连原子间非键作用（这里是3个键以内的原子不算非键相互作用）

6.2.3检查 top 文件的原子定义部分

接下来的部分则定义了蛋白质中的原子

```
; Include forcefield parameters
#include "oplsaa.ff/forcefield.itp"

[ moleculetype ]
; Name nrexcl
Protein_chain_A 3

[ atoms ]
; nr      type  resnr residue  atom  cgnr      charge      mass  typeB      chargeB      massB
; residue 1 LYS rtp LYSH q +2.0
1  op1s_287  1  LYS      N      1      -0.3      14.0027
2  op1s_290  1  LYS      H1     1      0.33     1.008
3  op1s_290  1  LYS      H2     1      0.33     1.008
4  op1s_290  1  LYS      H3     1      0.33     1.008
5  op1s_2938 1  LYS      CA     1      0.25     12.011
6  op1s_140  1  LYS      HA     1      0.06     1.008
7  op1s_136  1  LYS      CB     2      -0.12    12.011
8  op1s_140  1  LYS      HB1    2      0.06     1.008
9  op1s_140  1  LYS      HB2    2      0.06     1.008
10 op1s_136  1  LYS      CG     3      -0.12    12.011
11 op1s_140  1  LYS      HG1    3      0.06     1.008
12 op1s_140  1  LYS      HG2    3      0.06     1.008
13 op1s_136  1  LYS      CD     4      -0.12    12.011
14 op1s_140  1  LYS      HD1    4      0.06     1.008
15 op1s_140  1  LYS      HD2    4      0.06     1.008
16 op1s_292  1  LYS      CE     5      0.19     12.011
17 op1s_140  1  LYS      HE1    5      0.06     1.008
18 op1s_140  1  LYS      HE2    5      0.06     1.008
19 op1s_287  1  LYS      NZ     6      -0.3     14.0067
20 op1s_290  1  LYS      HZ1    6      0.33     1.008
21 op1s_290  1  LYS      HZ2    6      0.33     1.008
22 op1s_290  1  LYS      HZ3    6      0.33     1.008
```

这里面列值很多，解释参考如下：

nr: 原子序号

type: 原子类型

resnr: 氨基酸残基编号

residue: 氨基酸残基名

注意这里的残基在原来的PDB文件中为“LYS”，使用.rtp中的“LYSH”项意味着这是质子化的残基(中性PH环境下会处于主导地位)。

atom: 原子名称

cgnr: 电荷组编号 (电荷组定义了整数电荷单元，可加速计算。)

charge: 无需解释，“qtot”为对分子总电荷的持续累加

mass: 也无需解释

typeB, chargeB, massB: 用于自由能微扰(这里不讨论)

下面几节包括[bonds], [pairs], [angles]和[dihedrals]。其中一些无需解释：bonds, angles, and dihedrals(键, 键角, 二面角)，主要是和蛋白质有关的一些基础化学概念。

可以参考gromacs手册进行查阅

6.2.4检查 top 文件的位置限制配置

看拓扑文件的最后几行，本例子大约在第18384行文件开始

```

18384 ; Include Position restraint file
18385 #ifdef POSRES
18386 #include "posre.itp"
18387 #endif
18388
18389 ; Include water topology
18390 #include "oplsaa.ff/spce.itp"
18391
18392 #ifdef POSRES_WATER
18393 ; Position restraint for each water oxygen
18394 [ position_restraints ]
18395 ; i funct      fcx      fcy      fcz
18396   1      1      1000      1000      1000
18397 #endif
18398
18399 ; Include topology for ions
18400 #include "oplsaa.ff/ions.itp"
18401
18402 [ system ]
18403 ; Name
18404 LYSOZYME
18405
18406 [ molecules ]
18407 ; Compound      #mols
18408 Protein_chain_A      1
    
```

```
; Include Position restraint file
```

```
#ifdef POSRES
```

```
#include "posre.itp"
```

```
#endif
```

该部分定义了位置限制文件

(posre.itp由pdb2gmx生成，此文件包含了用来限制重原子（heavy atoms）位置的信息，定义了平衡时用于维持原子位置的力常数)

到这里Protein_A分子的定义就结束了。剩下的是定义其他分子并提供体系级别的说明

6.3 步骤 3-定义模拟体系的尺寸并加水

6.3.1 晶胞容器说明

本节主题是：Defining the Unit Cell & Adding Solvent 我们说的“模拟体系”有多种不

同的说法，比如盒子、晶胞、容器等，大家理解意思就行。

上一章熟悉了拓扑文件，下面开始定义我们要模拟的系统，要做两件事情：

使用gmx程序的editconf模块定义晶胞容器的尺寸

二、使用gmx程序的solvate模块向晶胞容器中填充水溶剂

那么该使用哪种形状的容器？本实验这里先使用一个简单的立方体形状作为容器。后续如果熟悉了周期性的边界条件与晶胞类型，可以使用菱形十二面体晶胞。因为在周期性距离相同的情况下，它的体积大约只有立方体晶胞的71%，因此可以减少需要加入的溶剂水分子的数目。

6.3.2实操 6-editconf 定义容器大小

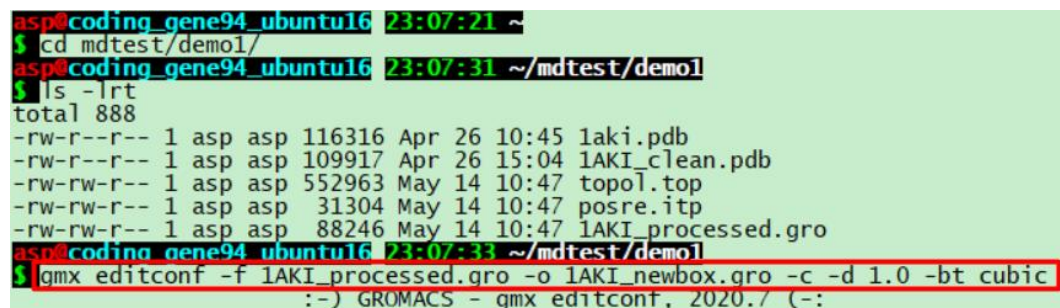
下面开始实际操作：

一、使用editconf模块定义晶胞容器的尺寸

Linux执行：

```
gmx editconf -f lAKI_processed.gro -o lAKI_newbox.gro -c -d 1.0 -bt cubic
```

实操截图如下：



```
asp@coding_gene94_ubuntu16 23:07:21 ~  
$ cd mdtest/demo1/  
asp@coding_gene94_ubuntu16 23:07:31 ~/mdtest/demo1  
$ ls -lrt  
total 888  
-rw-r--r-- 1 asp asp 116316 Apr 26 10:45 laki.pdb  
-rw-r--r-- 1 asp asp 109917 Apr 26 15:04 lAKI_clean.pdb  
-rw-rw-r-- 1 asp asp 552963 May 14 10:47 topol.top  
-rw-rw-r-- 1 asp asp 31304 May 14 10:47 posre.itp  
-rw-rw-r-- 1 asp asp 88246 May 14 10:47 lAKI_processed.gro  
asp@coding_gene94_ubuntu16 23:07:33 ~/mdtest/demo1  
$ gmx editconf -f lAKI_processed.gro -o lAKI_newbox.gro -c -d 1.0 -bt cubic  
:-) GROMACS - gmx editconf, 2020.7 (-:
```



```
GROMACS:      gmx editconf, version 2020.7
Executable:    /apps/svr/gromacs/gmx2020/bin/gmx
Data prefix:   /apps/svr/gromacs/gmx2020
Working dir:   /home/asp/mdtest/demo1
Command line:  gmx editconf -f 1AKI_processed.gro -o 1AKI_newbox.gro -c -d 1.0 -bt cubic

Note that major changes are planned in future for editconf, to improve usability and utility.
Read 1960 atoms
Volume: 123.376 nm^3, corresponds to roughly 55500 electrons
No velocities found
  system size :  3.817  4.234  3.454 (nm)
  diameter    :  5.010                      (nm)
  center      :  2.781  2.488  0.017 (nm)
  box vectors :  5.906  6.845  3.052 (nm)
  box angles  :  90.00  90.00  90.00 (degrees)
  box volume  : 123.38                      (nm^3)
  shift       :  0.724  1.017  3.488 (nm)
new center    :  3.505  3.505  3.505 (nm)
new box vectors :  7.010  7.010  7.010 (nm)
new box angles :  90.00  90.00  90.00 (degrees)
new box volume : 344.48                      (nm^3)
```

执行后，会把原先的拓扑文件做一个备份：/#topol.top.1

输出的关键文本信息如下：

```
GROMACS:      gmx editconf, version 2020.7
Executable:    /apps/svr/gromacs/gmx2020/bin/gmx
Data prefix:   /apps/svr/gromacs/gmx2020
Working dir:   /home/asp/mdtest/demo1
Command line:
  gmx editconf -f 1AKI_processed.gro -o 1AKI_newbox.gro -c -d 1.0 -bt cubic
```

```
Note that major changes are planned in future for editconf, to improve usability and
utility.
```

```
Read 1960 atoms
```

```
Volume: 123.376 nm^3, corresponds to roughly 55500 electrons
```

```
No velocities found
```

```
  system size :  3.817  4.234  3.454 (nm)
```

```
  diameter    :  5.010                      (nm)
```

```
  center      :  2.781  2.488  0.017 (nm)
```

```
  box vectors :  5.906  6.845  3.052 (nm)
```

```
  box angles  :  90.00  90.00  90.00 (degrees)
```

```
  box volume  : 123.38                      (nm^3)
```

```
shift      : 0.724 1.017 3.488 (nm)
new center : 3.505 3.505 3.505 (nm)
new box vectors : 7.010 7.010 7.010 (nm)
new box angles : 90.00 90.00 90.00 (degrees)
new box volume : 344.48 (nm^3)
```

执行后新生成了文件1AKI_newbox.gro，我们需要理解这个命令的参数和用途：

将蛋白质置于容器的中心 (-c 参数)

到盒子边缘的距离至少为1.0 nm (-d 1.0)

盒子类型是立方体 (-bt cubic)

到盒子边缘的距离参数比较重要，要使用周期性边界条件 (periodic boundary conditions)，

必须满足最小构象约定 (minimum image convention)，一个蛋白质永远不能“看到”它

自身的周期性构象(不能与其自身有相互作用)，否则计算的力就不准确。

指定溶质与盒子之间的距离为1.0nm意味着，蛋白质分子的任意两个周期性构象之间的距离

至少是2.0nm

6.3.3 实操 7-solvate 加入水溶剂

二、使用solvate调用溶剂模块向晶胞容器中填充水

Linux执行：

```
gmx solvate -cp 1AKI_newbox.gro -cs spc216.gro -o 1AKI_solv.gro -p topol.top
```

实操截图如下：



```
asp@coding_gene94_ubuntu16 23:21:26 ~/mdtest/demo1
$ ls
1AKI_clean.pdb 1AKI_newbox.gro laki.pdb 1AKI_processed.gro posre.itp topol.top
asp@coding_gene94_ubuntu16 23:21:27 ~/mdtest/demo1
$ gmx solvate -cp 1AKI_newbox.gro -cs spc216.gro -o 1AKI_solv.gro -p topol.top
:-) GROMACS - gmx solvate, 2020.7 (-:
GROMACS is written by:
```



```
GROMACS:      gmx solvate, version 2020.7
Executable:    /apps/avr/gromacs/gmx2020/bin/gmx
Data prefix:   /apps/avr/gromacs/gmx2020
Working dir:   /home/asp/mdtest/demo1
Command line:  gmx solvate -cp 1AKI_newbox.gro -cs spc216.gro -o 1AKI_solv.gro -p topol.top

Reading solute configuration
Reading solvent configuration

Initialising inter-atomic distances...

WARNING: Masses and atomic (Van der Waals) radii will be guessed
         based on residue and atom names, since they could not be
         definitively assigned from the information in your input
         files. These guessed numbers might deviate from the mass
         and radius of the atom type. Please check the output
         files if necessary.

NOTE: From version 5.0 gmx solvate uses the Van der Waals radii
      from the source below. This means the results may be different
      compared to previous GROMACS versions.

++++ PLEASE READ AND CITE THE FOLLOWING REFERENCE ++++
A. Bondi
van der Waals Volumes and Radii
J. Phys. Chem. 68 (1964) pp. 441-451
----- Thank You -----

Generating solvent configuration
Will generate new solvent configuration of 4x4x4 boxes
Solvent box contains 39252 atoms in 13084 residues
Removed 5451 solvent atoms due to solvent-solvent overlap
Removed 1869 solvent atoms due to solute-solvent overlap
Sorting configuration
Found 1 molecule type:
  SOL ( 3 atoms): 10644 residues
Generated solvent containing 31932 atoms in 10644 residues
Writing generated configuration to 1AKI_solv.gro

Output configuration contains 33892 atoms in 10773 residues
Volume      : 344.484 (nm^3)
Density     : 997.935 (g/l)
Number of solvent molecules: 10644

Processing topology
Adding line for 10644 solvent molecules with resname (SOL) to topology file (topol.top)
Back Off! I just backed up topol.top to ./#topol.top.1#
```

输出的关键文本信息如下：

```
GROMACS:      gmx solvate, version 2020.7
```

```
Executable:    /apps/avr/gromacs/gmx2020/bin/gmx
```

```
Data prefix:   /apps/avr/gromacs/gmx2020
```

```
Working dir:   /home/asp/mdtest/demo1
```

```
Command line:
```

```
gmx solvate -cp 1AKI_newbox.gro -cs spc216.gro -o 1AKI_solv.gro -p topol.top
```

```
Reading solute configuration
```

```
Reading solvent configuration
```

Initialising inter-atomic distances...

WARNING: Masses and atomic (Van der Waals) radii will be guessed based on residue and atom names, since they could not be definitively assigned from the information in your input files. These guessed numbers might deviate from the mass and radius of the atom type. Please check the output files if necessary.

NOTE: From version 5.0 gmx solvate uses the Van der Waals radii from the source below. This means the results may be different compared to previous GROMACS versions.

++++ PLEASE READ AND CITE THE FOLLOWING REFERENCE +++++

A. Bondi

van der Waals Volumes and Radii

J. Phys. Chem. 68 (1964) pp. 441-451

----- Thank You -----

Generating solvent configuration

Will generate new solvent configuration of 4x4x4 boxes

Solvent box contains 39252 atoms in 13084 residues

Removed 5451 solvent atoms due to solvent-solvent overlap

Removed 1869 solvent atoms due to solute-solvent overlap

Sorting configuration

Found 1 molecule type:

SOL (3 atoms): 10644 residues

Generated solvent containing 31932 atoms in 10644 residues

Writing generated configuration to 1AKI_solv.gro

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```
Output configuration contains 33892 atoms in 10773 residues
```

```
Volume           :      344.484 (nm^3)
```

```
Density          :      997.935 (g/l)
```

```
Number of solvent molecules: 10644
```

```
Processing topology
```

```
Adding line for 10644 solvent molecules with resname (SOL) to topology file  
(topol.top)
```

```
Back Off! I just backed up topol.top to ./#topol.top.1#
```

同样也需要理解这个命令的参数和作用：

使用的蛋白质拓扑文件来自editconf步骤中的输出文件1AKI_newbox.gro (-cp参数)

溶剂的构型文件来自标准安装的GROMACS文件(-cs参数)，这个spc216.gro是通用的已平衡的三位点溶剂模型，可以使用spc216.gro作为SPC, SPC/E或TIP3P水模型的溶剂构型，因为它们都是三位点的水模型。输出文件的名称为1AKI_solv.gro。

输出的文件叫1AKI_solv.gro，我们为solvate模块指定了拓扑文件topol.top，这样它就能修改拓扑文件（在操作的时候原来的拓扑文件会备份一份）

注意新生成的topol.top中[molecules]的变化:

```
$ tail topol.top  
  
[ system ]  
; Name  
LYSOZYME in water  
  
[ molecules ]  
; Compound      #mols  
Protein_chain_A    1  
SOL                10636
```

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可以在linux下使用diff命令对比下区别：

```
asp@coding_gene94_ubuntu16 23:29:37 ~/mdtest/demo1
$ diff topol.top \#topol.top.1#
18404c18404
< LYSOZYME in water
---
> LYSOZYME
18409d18408
< SOL                      10644
```

或者直接看最后几行的区别：新文件多了一行SOL溶剂水分子的数量

<pre>asp@coding_gene94_ubuntu16 23:30:04 ~/mdtest/demo1 \$ tail \#topol.top.1# ; Include topology for ions #include "oplsaa.ff/ions.itp" [system] ; Name LYSOZYME [molecules] ; Compound #mols Protein_chain_A 1 asp@coding_gene94_ubuntu16 23:30:08 ~/mdtest/demo1</pre>	<pre>asp@coding_gene94_ubuntu16 23:29:58 ~/mdtest/demo1 \$ tail topol.top #include "oplsaa.ff/ions.itp" [system] ; Name LYSOZYME in water [molecules] ; Compound #mols Protein_chain_A 1 SOL 10644</pre>
---	---

另外如果你使用的是其他的非水溶剂，solvate不会在拓扑文件中写入这些信息。因为自动记录更新水分子的功能是直接写在原始代码中的（也就是说自动记录更新水分子的功能代码写死了）

6.3.4 检查 top 文件中溶剂拓扑定义

```
; Include Position restraint file
#ifdef POSRES
#include "posre.itp"
#endif

; Include water topology
#include "oplsaa.ff/spce.itp"

#ifdef POSRES_WATER
; Position restraint for each water oxygen
[ position_restraints ]
; i funct      fcx      fcy      fcz
  1  1      1000      1000      1000
#endif

; Include topology for ions
#include "oplsaa.ff/ions.itp"

[ system ]
; Name
LYSOZYME

[ molecules ]
; Compound      #mols
Protein_chain_A      1
```

接下来默认定义的是溶剂分子类型（一般就是水），在本例中为SPC/E模型的水分子（其他水分子模型还有SPC, TIP3P和TIP4P等）。水也可以被位置约束，使用 $1000 \text{ kJ mol}^{-1} \text{ nm}^{-2}$ 的力常数(kpr)

通过上一步生成拓扑的时候，在pdb2gmx命令中使用“-water spce”选项（`gmx pdb2gmx -f 1AKI_clean.pdb -o 1AKI_processed.gro -water spce`），gromacs程序选择了默认的SPC/E水模型。（顺便说一下，GROMACS并没有包含所有的水模型，关于不同类型水模型的解释，可以参考资料https://water.lsbu.ac.uk/water/water_models.html，下图是网站首页的截图，简要说明了水分子模型概况）

WATER STRUCTURE AND SCIENCE

Martin Chaplin



Water Models

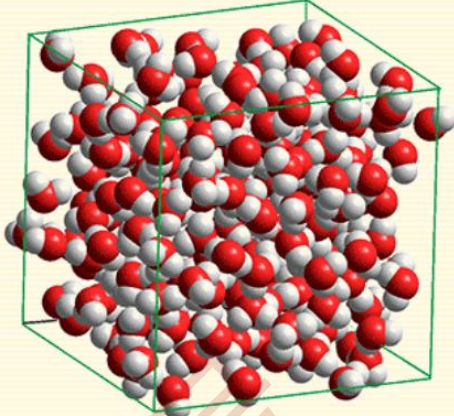
Water molecular models are computational techniques that have been developed to help discover the structure of water.

- ✔ Parameterization of the water models
- ✔ Water model descriptions
- ✔ Water model properties
- ✔ Aqueous solutions
- ✔ The Lennard-Jones relationship

*'with four parameters I can fit an elephant,
and with five I can make him wiggle his trunk'*
Enrico Fermi as stated in [2438]

*'although the model can be very accurate in many respects,
it should not be confused with the real liquid'*
Nilsson, Schlesinger and Pettersson [2569]

A computational model of a small periodic box containing 216 water molecules



For reviews of the development of water models, see [275,2537], for a review of the development of analytical potential energy functions, see [2641], for a review of their use in supercritical water see [433], for an appraisal of their accomplishments see [400], for a comparison of some important nonpolarizable models see [1478], for a review of progress in *ab initio* methods see [1611], and for a discussion of their use in biomolecular simulations see [3395]. It should be noted that most simulations using models can only investigate structural changes and interactions over very short timescales, generally less than a few microseconds. Although computers have become more powerful, simulations often involve more complicated calculations with greater numbers of constituent molecules and exponentially growing interactions. Reliable experimental data are always to be preferred over results obtained by simulation methods

查看拓扑文件末尾配置

```
; Include Position restraint file
#ifdef POSRES
#include "posre.itp"
#endif

; Include water topology
#include "oplsaa.ff/spce.itp"

#ifdef POSRES_WATER
; Position restraint for each water oxygen
[ position_restraints ]
; i funct      fcx      fcy      fcz
  1      1      1000      1000      1000
#endif

; Include topology for ions
#include "oplsaa.ff/ions.itp"

[ system ]
; Name
LYSOZYME

[ molecules ]
; Compound      #mols
Protein_chain_A      1
```

; Include topology for ions

#include "oplsaa.ff/ions.itp"

上面这个是离子相关的拓扑配置。

最后是体系级别的定义。

[system]指令给出了体系的名称, 在模拟中此名称将被写入到输出文件中. 例子中的

LYSOZYME指的是我们本次模拟的物质对象: 溶菌酶

[molecules]模块列出了体系中的所有分子, 有几个关键注意点需要注意:

列出分子的顺序必须与坐标(本例中为.gro)文件中的分子顺序保持一致

对每一个研究体系, [molecules]列出的名称必须与[moleculetype]中的名称一致, 而不是

残基名称或其他名称

下面是文件前面定义的

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```
[ moleculetype ]
: Name          nrexcl
Protein_chain_A 3
```

这是文件最后定义的

```
[ molecules ]
: Compound      #mols
Protein_chain_A 1
```

本节非常重要，主要是理解gromacs pdb2gmx生成的拓扑文件内容。

6.4 步骤 4-为模拟体系添加离子

6.4.1 关于离子加入与 mdp 参数

现在我们已经有了一个带电荷的溶液体系。

之前pdb2gmx程序处理后，我们所用的蛋白质带有+8e的净电荷（根据它的氨基酸残基计算得到）

我们记录了之前的输出，参照如下截图：

```
Linking CYS-6 SG-48 and CYS-127 SG-981...
Linking CYS-30 SG-238 and CYS-115 SG-889...
Linking CYS-64 SG-513 and CYS-80 SG-630...
Linking CYS-76 SG-601 and CYS-94 SG-724...
Start terminus LYS-1: NH3+
End terminus LEU-129: COO-
Checking for duplicate atoms....
Generating any missing hydrogen atoms and/or adding termini.
Now there are 129 residues with 1960 atoms
Making bonds...
Number of bonds was 1984, now 1984
Generating angles, dihedrals and pairs...
Before cleaning: 5142 pairs
Before cleaning: 5187 dihedrals
Keeping all generated dihedrals
Making cmap torsions...
There are 5187 dihedrals, 426 impropers, 3547 angles
          5106 pairs,          1984 bonds and          0 virtual sites
Total mass 14313.193 a.m.u.
Total charge 8.000 e
writing topology

Writing coordinate file...
----- PLEASE NOTE -----
You have successfully generated a topology from: 1AKI_clean.pdb.
The Oplsaa force field and the spce water model are used.
----- ETON ESAELP -----
```

由于正常生命体不存在净电荷, 所以我们需要往里加入离子, 以保证我们的模拟体系总电荷为零。GROMACS中添加离子的工具是genion, genion的功能是读取拓扑信息, 然后将体系中的一些水分子替换为指定的离子。

genion需要的输入文件称为运行输入文件, 扩展名为.tpr. 这个文件可使用GROMACS的grompp(GROMacs Pre- Processor)模块产生, 后面我们运行模拟时也会用到。grompp的功能是处理坐标文件和分子结构拓扑以产生“原子级别”的输入文件(.tpr), 说它是原子级别的文件, 是因为.tpr文件包含了体系中所有原子的相关参数。

为了用grompp产生.tpr文件, 我们还需要一个扩展名为.mdp(molecular dynamics parameter)的参数输入文件, grompp会将坐标和拓扑信息与.mdp文件中设定的参数组合起来生成.tpr文件。.mdp文件通常用于运行能量最小化(EM)或分子动力学模拟(MD)--后面进行能量最小化和分子动力学模拟的时候我们会用到不同的mdp参数文件。

一个离子的mdp参数文件ions.mdp示例如下:

(对应的原始文件为<http://www.mdtutorials.com/gmx/lysozyme/Files/ions.mdp>)

```
; ions.mdp - used as input into grompp to generate ions.tpr
; Parameters describing what to do, when to stop and what to save
integrator      = steep          ; Algorithm (steep = steepest descent minimization)
emtol           = 1000.0         ; Stop minimization when the maximum force < 1000.0
                                kJ/mol/nm
emstep          = 0.01          ; Minimization step size
nsteps          = 50000         ; Maximum number of (minimization) steps to perform

; Parameters describing how to find the neighbors of each atom and how to calculate
the interactions
nstlist         = 1             ; Frequency to update the neighbor list and long range
forces
cutoff-scheme   = Verlet        ; Buffered neighbor searching
ns_type         = grid          ; Method to determine neighbor list (simple, grid)
coulombtype     = cutoff        ; Treatment of long range electrostatic interactions
rcoulomb        = 1.0           ; Short-range electrostatic cut-off
rvdw            = 1.0           ; Short-range Van der Waals cut-off
pbc             = xyz           ; Periodic Boundary Conditions in all 3 dimensions
```

```
; ions.mdp - used as input into grompp to generate ions.tpr
; Parameters describing what to do, when to stop and what to save
integrator = steep          ; Algorithm (steep = steepest descent minimization)
emtol      = 1000.0         ; Stop minimization when the maximum force < 1000.0 kJ/mol/nm
emstep     = 0.01          ; Minimization step size
nsteps     = 50000         ; Maximum number of (minimization) steps to perform

; Parameters describing how to find the neighbors of each atom and how to calculate the interactions
nstlist    = 1             ; Frequency to update the neighbor list and long range forces
cutoff-scheme = Verlet     ; Buffered neighbor searching
ns_type    = grid         ; Method to determine neighbor list (simple, grid)
coulombtype = cutoff       ; Treatment of long range electrostatic interactions
rcoulomb    = 1.0         ; Short-range electrostatic cut-off
rvdw        = 1.0         ; Short-range Van der Waals cut-off
pbc        = xyz          ; Periodic Boundary Conditions in all 3 dimensions
```

6.4.2 实操 8-生成 tpr 文件

使用下面的命令来产生ions.tpr这个.tpr文件:

```
gmx grompp -f ions.mdp -c 1AKI_solv.gro -p topol.top -o ions.tpr
```

```
asp@coding_gene94_ubuntu16:00:11:27 ~/mdtest/demo1
$ ls
1AKI_clean.pdb  laki.pdb          1AKI_solv.gro  md.mdp          npt.mdp        posre.itp      #topol.top.1#
1AKI_newbox.gro 1AKI_processed.gro ions.mdp       minim.mdp       nvt.mdp        topol.top
asp@coding_gene94_ubuntu16:00:11:28 ~/mdtest/demo1
$ gmx grompp -f ions.mdp -c 1AKI_solv.gro -p topol.top -o ions.tpr
:~) GROMACS - gmx grompp, 2020.7 (-:
```

```
GROMACS:      gmx grompp, version 2020.7
Executable:    /apps/avr/gromacs/gmx2020/bin/gmx
Data prefix:    /apps/avr/gromacs/gmx2020
Working dir:    /home/asp/mdtest/demo1
Command line:
  gmx grompp -f ions.mdp -c 1AKI_solv.gro -p topol.top -o ions.tpr

Ignoring obsolete mdp entry 'ns_type'

NOTE 1 [file ions.mdp]:
  With Verlet lists the optimal nstlist is >= 10, with GPUs >= 20. Note
  that with the Verlet scheme, nstlist has no effect on the accuracy of
  your simulation.

Setting the LD random seed to -705167667
Generated 330891 of the 330891 non-bonded parameter combinations
Generating 1-4 interactions: fudge = 0.5
Generated 330891 of the 330891 1-4 parameter combinations
Excluding 3 bonded neighbours molecule type 'Protein_chain_A'
Excluding 2 bonded neighbours molecule type 'SOL'

NOTE 2 [file topol.top, line 18409]:
  System has non-zero total charge: 8.000000
  Total charge should normally be an integer. See
  http://www.gromacs.org/Documentation/Floating_Point_Arithmetic
  for discussion on how close it should be to an integer.

Analysing residue names:
There are: 129      Protein residues
There are: 10644    Water residues
Analysing Protein...
Number of degrees of freedom in T-Coupling group rest is 69741.00

NOTE 3 [file ions.mdp]:
  You are using a plain Coulomb cut-off, which might produce artifacts.
  You might want to consider using PME electrostatics.

This run will generate roughly 3 Mb of data
There were 3 notes
```

执行输出的结果显示如下:

Command line:

```
gmx grompp -f ions.mdp -c 1AKI_solv.gro -p topol.top -o ions.tpr
```

Ignoring obsolete mdp entry 'ns_type'

NOTE 1 [file ions.mdp]:

With Verlet lists the optimal nstlist is >= 10, with GPUs >= 20. Note
that with the Verlet scheme, nstlist has no effect on the accuracy of
your simulation.

Setting the LD random seed to -339223553

Generated 330891 of the 330891 non-bonded parameter combinations

Generating 1-4 interactions: fudge = 0.5

Generated 330891 of the 330891 1-4 parameter combinations

Excluding 3 bonded neighbours molecule type 'Protein_chain_A'

Excluding 2 bonded neighbours molecule type 'SOL'

NOTE 2 [file topol.top, line 18409]:

System has non-zero total charge: 8.000000

Total charge should normally be an integer. See

http://www.gromacs.org/Documentation/Floating_Point_Arithmetic

for discussion on how close it should be to an integer.

Analysing residue names:

There are: 129 Protein residues

There are: 10644 Water residues

Analysing Protein...

Number of degrees of freedom in T-Coupling group rest is 69741.00

NOTE 3 [file ions.mdp]:

You are using a plain Coulomb cut-off, which might produce artifacts.

You might want to consider using PME electrostatics.

This run will generate roughly 3 Mb of data

There were 3 notes

6.4.3 实操 9-genion 加入离子

现在我们得到了一个二进制的.tpr文件, 它提供了我们体系的原子级别的描述, 将此文件用于genion执行:

```
gmx genion -s ions.tpr -o lAKI_solv_ions.gro -p topol.top -pname NA -nname CL -neutral
```

根据提示选择13 “SOL” 也就是用离子替换一些水分子

```
Command line:
gmx genion -s ions.tpr -o lAKI_solv_ions.gro -p topol.top -pname NA -nname CL -neutral

Reading file ions.tpr, VERSION 2020.7 (single precision)
Reading file ions.tpr, VERSION 2020.7 (single precision)
Will try to add 0 NA ions and 8 CL ions.
Select a continuous group of solvent molecules
Group 0 (      System) has 33892 elements
Group 1 (      Protein) has 1960 elements
Group 2 (    Protein-H) has 1001 elements
Group 3 (      C-alpha) has 129 elements
Group 4 (    Backbone) has 387 elements
Group 5 (    MainChain) has 517 elements
Group 6 (  MainChain+Cb) has 634 elements
Group 7 (  MainChain+H) has 646 elements
Group 8 (    SideChain) has 1314 elements
Group 9 (    SideChain-H) has 484 elements
Group 10 (  Prot-Masses) has 1960 elements
Group 11 (  non-Protein) has 31932 elements
Group 12 (      water) has 31932 elements
Group 13 (      SOL) has 31932 elements
Group 14 (  non-Water) has 1960 elements
Select a group: 13
Selected 13: 'SOL'
Number of (3-atomic) solvent molecules: 10644

Processing topology
Replacing 8 solute molecules in topology file (topol.top) by 0 NA and 8 CL ions.

Back Off! I just backed up topol.top to ./#topol.top.2#
Using random seed -268714006.
Replacing solvent molecule 1744 (atom 7192) with CL
Replacing solvent molecule 9947 (atom 31801) with CL
Replacing solvent molecule 3527 (atom 12541) with CL
Replacing solvent molecule 8792 (atom 28336) with CL
Replacing solvent molecule 2892 (atom 10636) with CL
Replacing solvent molecule 4376 (atom 15088) with CL
Replacing solvent molecule 882 (atom 4606) with CL
Replacing solvent molecule 3521 (atom 12523) with CL
```

关键输出文本如下:

```
Select a group: 13
```

```
Selected 13: 'SOL'
```

```
Number of (3-atomic) solvent molecules: 10644
```

```
Processing topology
```

```
Replacing 8 solute molecules in topology file (topol.top) by 0 NA and 8 CL ions.
```

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```
Back Off! I just backed up topol.top to ./#topol.top.2#
```

```
Using random seed -17825805.
```

```
Replacing solvent molecule 5584 (atom 18712) with CL
```

```
Replacing solvent molecule 1421 (atom 6223) with CL
```

```
Replacing solvent molecule 6259 (atom 20737) with CL
```

```
Replacing solvent molecule 4884 (atom 16612) with CL
```

```
Replacing solvent molecule 1956 (atom 7828) with CL
```

```
Replacing solvent molecule 5631 (atom 18853) with CL
```

```
Replacing solvent molecule 137 (atom 2371) with CL
```

```
Replacing solvent molecule 3672 (atom 12976) with CL
```

关键信息在这: Replacing 8 solute molecules in topology file (topol.top) by 0 NA and 8 CL ions.

注意看命令: `gmx genion -s ions.tpr -o 1AKI_solv_ions.gro -p topol.top -pname NA -nname CL -neutral`, 在gmx genion命令中, 我们以结构/状态文件(-s)作为输入, 以.gro文件作为输出(-o), 以拓扑文件(-p)来反映去除的水分子和增加的离子, 并且定义了阳离子和阴离子的名称(分别使用-pname和-nname)。

这个-neutral参数很关键, -neutral参数告诉genion只需要添加必要的离子来中和蛋白质所带的净电荷 (-neutral, which in this case will add 8 Cl- ions to offset the +8 charge on the protein), 也就是说系统自动添加了8个Cl-氯离子, 来中和系统所带的8个正电荷, 并没有添加NA离子。另外对于genion, 除了简单地中和体系所带的净电荷以外, 你也可以同时指定-neutral和-conc选项来添加指定浓度的离子

linux执行tail topol.top就可以看到，目前分子拓扑中的配置如下（新增了溶剂分子SOL和负离子CL）

```
asp@coding_gene94_ubuntu16 00:27:34 ~/mdtest/demo1
$ tail topol.top

[ system ]
; Name
LYSOZYME in water

[ molecules ]
; Compound      #mols
Protein_chain_A    1
SOL                10636
CL                 8
```

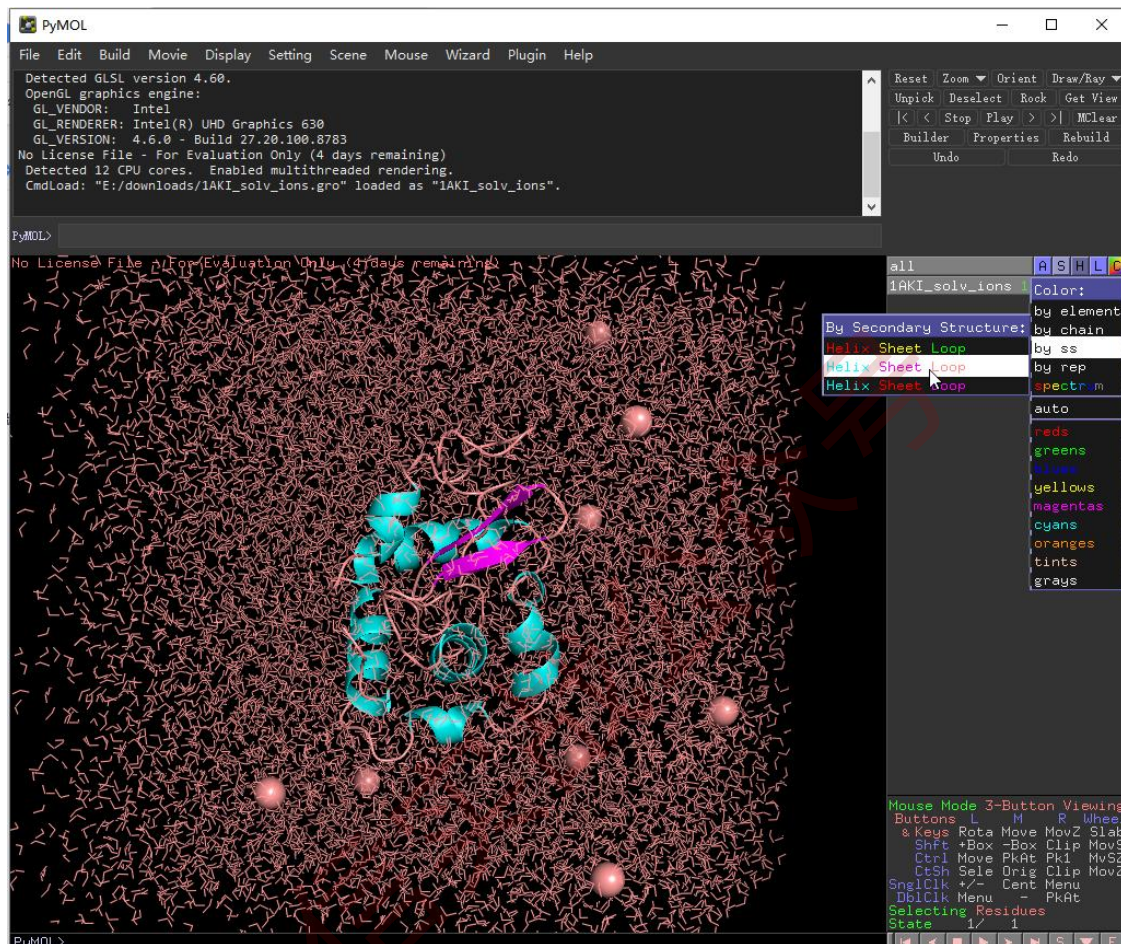
solvate记录了增加的水分子数目，并将其写入拓扑文件中 即：SOL这一行
原先10644个SOL水分子，被替换了8个CL例子，剩下10636个SOL水分子

6. 4. 4实操 10-使用 pymol 或 vmd 等查看体系

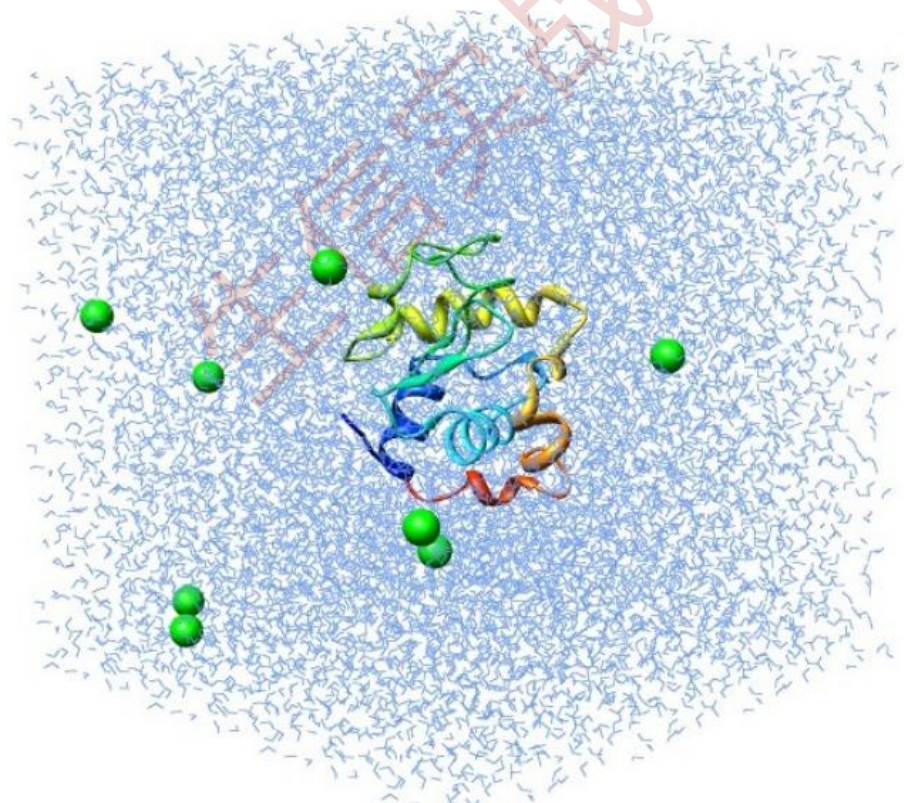
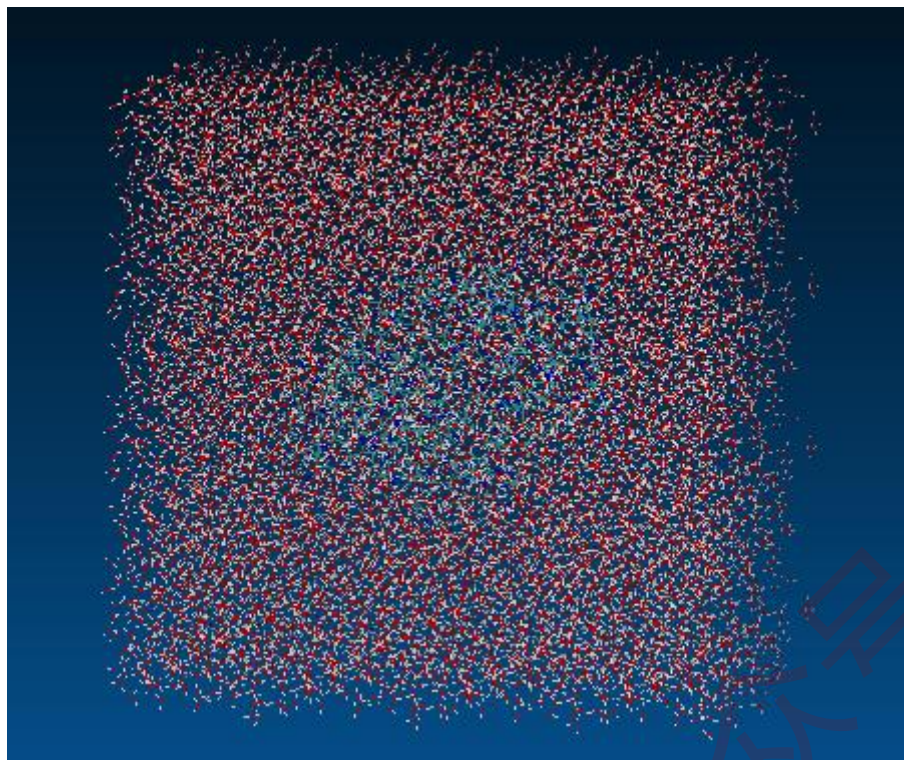
现在我们可以使用pymol或vmd等可视化软件查看最新生成的1AKI_solv_ions.gro

```
asp@coding_gene94_ubuntu16 11:55:06 ~/mdtest/demo1
$ ls -lrt
total 6320
-rw-r--r-- 1 asp asp 116316 Apr 26 10:45 1aki.pdb
-rw-r--r-- 1 asp asp 109917 Apr 26 15:04 1AKI_clean.pdb
-rw-rw-r-- 1 asp asp 552964 May 22 10:10 #topol.top.1#
-rw-rw-r-- 1 asp asp 31304 May 22 10:10 posre.itp
-rw-rw-r-- 1 asp asp 88246 May 22 10:10 1AKI_processed.gro
-rw-rw-r-- 1 asp asp 88246 May 22 10:34 1AKI_newbox.gro
-rw-rw-r-- 1 asp asp 1525186 May 22 10:48 1AKI_solv.gro
-rw----- 1 asp asp 552995 May 22 10:49 #topol.top.2#
-rw-rw-r-- 1 asp asp 2369 May 22 10:53 nvt.mdp
-rw-rw-r-- 1 asp asp 2591 May 22 10:53 npt.mdp
-rw-rw-r-- 1 asp asp 1045 May 22 10:53 minim.mdp
-rw-rw-r-- 1 asp asp 2741 May 22 10:53 md.mdp
-rw-rw-r-- 1 asp asp 1045 May 22 10:53 ions.mdp
-rw-rw-r-- 1 asp asp 10775 May 22 11:19 mdout.mdp
-rw-rw-r-- 1 asp asp 1266912 May 22 11:19 ions.tpr
-rw----- 1 asp asp 553010 May 22 11:20 topol.top
-rw-rw-r-- 1 asp asp 1524475 May 22 11:20 1AKI_solv_ions.gro
```

使用PyMOL, 打开1AKI_solv_ions.gro (选择使用二级结构模式查看 Color by ss, 可以清晰得看到体系的水和离子)



使用VMD查看1AKI_solv_ions.gro



6.5 步骤 5-体系能量最小化

6.5.1 关于能力最小化的理论

可以参照之前的一篇文章：

用理论方法如同源模建等构建出来的分子结构往往还不稳定，要用更系统的方法优化分子内的原子坐标，降低分子势能，使得核骨架进一步逼近稳定的结构(对于一个分子体系，其中的核质量远大于电子质量，核骨架指分子体系中的原子核)，这类方法称之为能量最小化。

简单的说，分子模拟时希望得到分子的低能构象，需要求解与其对应的分子势能面上的极小值，并力求实现全局最小值。分子力学的最小化算法能较快实现能量优化，但是求的往往是局部最小值，而要寻求全局最小值只能采用系统搜索法或者分子动力学方法。

就本例子而言，我们目前得到了一个电中性溶剂+离子的蛋白质溶剂体系，但是还不能直接开始动力学模拟，需要先优化体系中的空间冲突或不适当的几何形状，使得结构更稳定，也就是能量最小化。

6.5.2 实操 11-准备能量最小化参数和新 tpr 文件

和添加离子的过程类似，能量最小化过程我们也是使用grompp将结构信息、拓扑信息、模拟参数等写入一个二进制的.tpr文件，但这次我们不使用genion的相关参数，而是使用

GROMACS的MD引擎的mdrun模块来进行能量最小化。

能量最小化的minim.mdp参数如下：

```
; minim.mdp - used as input into grompp to generate em.tpr
; Parameters describing what to do, when to stop and what to save

integrator    = steep           ; Algorithm (steep = steepest descent minimization)
emtol         = 1000.0          ; Stop minimization when the maximum force < 1000.0
                                kJ/mol/nm
emstep        = 0.01           ; Minimization step size
nsteps        = 50000           ; Maximum number of (minimization) steps to perform

; Parameters describing how to find the neighbors of each atom and how to calculate
the interactions

nstlist       = 1              ; Frequency to update the neighbor list and long range
forces
cutoff-scheme = Verlet         ; Buffered neighbor searching
ns_type       = grid           ; Method to determine neighbor list (simple, grid)
coulombtype    = PME           ; Treatment of long range electrostatic interactions
rcoulomb       = 1.0           ; Short-range electrostatic cut-off
rvdw           = 1.0           ; Short-range Van der Waals cut-off
pbc            = xyz            ; Periodic Boundary Conditions in all 3 dimensions
```

确保拓扑文件topol.top是上一步使用genbox和genion添加了溶剂和离子更新后的信息，要不然执行下面的能量最小化可能得到一些错误信息，如“number of coordinates in coordinate file does not match topology”，即坐标文件中的坐标与拓扑不匹配等等错误。

Linux执行:

```
gmx grompp -f minim.mdp -c 1AKI_solv_ions.gro -p topol.top -o em.tpr
```

新生成了em.tpr文件, 执行输出如下:

```
GROMACS:      gmx grompp, version 2020.7
Executable:    /apps/avr/gromacs/gmx2020/bin/gmx
Data prefix:    /apps/avr/gromacs/gmx2020
Working dir:    /home/asp/mdtest/demo1
Command line:   gmx grompp -f minim.mdp -c 1AKI_solv_ions.gro -p topol.top -o em.tpr

Ignoring obsolete mdp entry 'ns_type'

NOTE 1 [file minim.mdp]:
  With Verlet lists the optimal nstlist is >= 10, with GPUs >= 20. Note
  that with the Verlet scheme, nstlist has no effect on the accuracy of
  your simulation.

Setting the LD random seed to -58878020
Generated 330891 of the 330891 non-bonded parameter combinations
Generating 1-4 interactions: fudge = 0.5
Generated 330891 of the 330891 1-4 parameter combinations
Excluding 3 bonded neighbours molecule type 'Protein_chain_A'
Excluding 2 bonded neighbours molecule type 'SOL'
Excluding 1 bonded neighbours molecule type 'CL'
Analysing residue names:
There are: 129      Protein residues
There are: 10636    Water residues
There are: 8        Ion residues
Analysing Protein...
Analysing residues not classified as Protein/DNA/RNA/Water and splitting into groups...
Number of degrees of freedom in T-Coupling group rest is 69717.00
Calculating fourier grid dimensions for X Y Z
Using a fourier grid of 60x60x60, spacing 0.117 0.117 0.117
Estimate for the relative computational load of the PME mesh part: 0.26
This run will generate roughly 3 Mb of data

There was 1 note
```

关键日志文本信息:

Command line:

```
gmx grompp -f minim.mdp -c 1AKI_solv_ions.gro -p topol.top -o em.tpr
```

Ignoring obsolete mdp entry 'ns_type'

NOTE 1 [file minim.mdp]:

```
With Verlet lists the optimal nstlist is >= 10, with GPUs >= 20. Note
that with the Verlet scheme, nstlist has no effect on the accuracy of
your simulation.
```



```
Setting the LD random seed to -58878020
Generated 330891 of the 330891 non-bonded parameter combinations
Generating 1-4 interactions: fudge = 0.5
Generated 330891 of the 330891 1-4 parameter combinations
Excluding 3 bonded neighbours molecule type 'Protein_chain_A'
Excluding 2 bonded neighbours molecule type 'SOL'
Excluding 1 bonded neighbours molecule type 'CL'
Analysing residue names:
There are:   129   Protein residues
There are: 10636   Water residues
There are:    8   Ion residues
Analysing Protein...
Analysing residues not classified as Protein/DNA/RNA/Water and splitting into
groups...
Number of degrees of freedom in T-Coupling group rest is 69717.00
Calculating fourier grid dimensions for X Y Z
Using a fourier grid of 60x60x60, spacing 0.117 0.117 0.117
Estimate for the relative computational load of the PME mesh part: 0.26
This run will generate roughly 3 Mb of data
```

6.5.3 实操 12-调用 mdrun 执行能量最小化

根据上一步产生的em.tpr文件执行能量最小化，Linux下执行：

```
gmx mdrun -v -deffnm em
```

执行后会输出一堆信息：

```
Steepest Descents:
Tolerance (Fmax)   = 1.00000e+03
```

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```
Number of steps      =      50000

Step=    0, Dmax= 1.0e-02 nm, Epot= -3.31916e+05 Fmax= 2.28198e+05, atom= 1905
Step=    1, Dmax= 1.0e-02 nm, Epot= -3.47978e+05 Fmax= 8.64638e+04, atom= 19877
Step=    2, Dmax= 1.2e-02 nm, Epot= -3.70296e+05 Fmax= 3.30123e+04, atom= 19877
Step=    3, Dmax= 1.4e-02 nm, Epot= -3.98673e+05 Fmax= 1.32088e+04, atom= 24182
Step=    4, Dmax= 1.7e-02 nm, Epot= -4.27120e+05 Fmax= 5.66426e+03, atom= 21851
.....
.....

Step=  858, Dmax= 6.3e-03 nm, Epot= -5.87436e+05 Fmax= 5.31422e+03, atom= 736
Step=  859, Dmax= 7.6e-03 nm, Epot= -5.87449e+05 Fmax= 6.33299e+03, atom= 736
Step=  860, Dmax= 9.1e-03 nm, Epot= -5.87458e+05 Fmax= 7.67593e+03, atom= 736
Step=  861, Dmax= 1.1e-02 nm, Epot= -5.87465e+05 Fmax= 9.09571e+03, atom= 736
Step=  863, Dmax= 6.5e-03 nm, Epot= -5.87517e+05 Fmax= 9.81522e+02, atom= 736

writing lowest energy coordinates.

Steepest Descents converged to Fmax < 1000 in 864 steps
Potential Energy  = -5.8751662e+05
Maximum force     =  9.8152197e+02 on atom 736
Norm of force     =  2.0807849e+01
```

中间的step信息我省略了，输出了这些每一步Step信息，是因为执行的时候使用了-v参数

-deffnm 选项定义了输入文件和输出文件的名称，这里指定了em，因此输出的文件都是em开头的：

```
-rw----- 1 asp asp 553010 May 22 11:20 topol.top
-rw-rw-r-- 1 asp asp 1524475 May 22 11:20 1AKI_solv_ions.gro
-rw-rw-r-- 1 asp asp 10776 May 22 19:30 mdout.mdp
-rw-rw-r-- 1 asp asp 1267416 May 22 19:30 em.tpr
-rw-rw-r-- 1 asp asp 134808 May 22 19:40 em.edr
-rw-rw-r-- 1 asp asp 406632 May 22 19:40 em.trr
-rw-rw-r-- 1 asp asp 1524475 May 22 19:40 em.gro
-rw-rw-r-- 1 asp asp 313371 May 22 19:40 em.log
```

输出的这些文件包括：

em.edr: 二进制能量文件

em.trr: 二进制轨迹文件

em.gro: 能量最小化后的结构拓扑

em.log: 日志文件, 记录了能量最小化过程

根据minim.mdp的参数, 可以看出, 这里正是使用了“最陡下降法”

```
; minim.mdp - used as input into grompp to generate em.tpr
; Parameters describing what to do, when to stop and what to save
integrator = steep ; Algorithm (steep = steepest descent minimization)
emtol = 1000.0 ; Stop minimization when the maximum force < 1000.0
kJ/mol/nm
emstep = 0.01 ; Minimization step size
nsteps = 50000 ; Maximum number of (minimization) steps to perform
```

最终经过了864步的模拟达到了能量最小化（就这个实验而言, 不同的环境运行时间需要的

step可能不一样, 但差别不大)

6.5.4实操 13-评估能量最小化

如何评估和确定能量最小化 (EM) 有没有完成呢? 有两个重要的因素:

第1个是势能Epot (potential energy) 这个信息会在能量最小化的最后输出。势能Epot是一个负值, 对于一个简单的水中的蛋白体系来说, 这个势能大约在105-106之间, 具体的还是取决于系统的大小和水分子的数量。

第2个是最大力值Fmax,也就是我们在minim.mdp参数中设置的emtol = 1000.0, 意思是说Fmax 不大于1000 kJ/mol/nm。

能量最小化后会得到一个合适的势能Epot, 但如果最终Fmax 还是大于emtol, 那么你的系统可能还不稳定, 还不能去直接模拟, 可以尝试调整能量最小化的参数如integrator、emstep来重新进行能量最小化。

6.5.5 实操 14-能量最小化结果分析

二进制的能量文件em.edr, 包含了GROMACS在能量最小化过程中记录的能量项, 我们可以使用GROMACS的gmx energy模块来分析:

Linux下执行: `gmx energy -f em.edr -o potential.xvg`

提示时, 输入10 0来选择势能Potential(10), 并用零(0)来结束输入。屏幕上会显示Epot的平均值, 得到的能量值会写入potential.xvg文件

Command line:

```
gmx energy -f em.edr -o potential.xvg
```

```
Opened em.edr as single precision energy file
```

```
Select the terms you want from the following list by
```

selecting either (part of) the name or the number or a combination.

End your selection with an empty line or a zero.

1	Bond	2	Angle	3	Proper-Dih.	4	Ryckaert-Bell.
5	LJ-14	6	Coulomb-14	7	LJ-(SR)	8	Coulomb-(SR)
9	Coul.-recip.	10	Potential	11	Pressure	12	Vir-XX
13	Vir-XY	14	Vir-XZ	15	Vir-YX	16	Vir-YY
17	Vir-YZ	18	Vir-ZX	19	Vir-ZY	20	Vir-ZZ
21	Pres-XX	22	Pres-XY	23	Pres-XZ	24	Pres-YX
25	Pres-YY	26	Pres-YZ	27	Pres-ZX	28	Pres-ZY
29	Pres-ZZ	30	#Surf*SurfTen	31	T-rest		

输入:

10 0

Last energy frame read 725 time 916.000

Statistics over 917 steps [0.0000 through 916.0000 ps], 1 data sets

All statistics are over 726 points (frames)

Energy	Average	Err.Est.	RMSD	Tot-Drift
Potential	-566593	11000	28032.5	-73338.4 (kJ/mol)

可以简单查看一下这个文件: [more potential.xvg](#)

```
@ title "GROMACS Energies"
@ xaxis label "Time (ps)"
@ yaxis label "(kJ/mol)"
@TYPE xy
@ view 0.15, 0.15, 0.75, 0.85
@ legend on
```

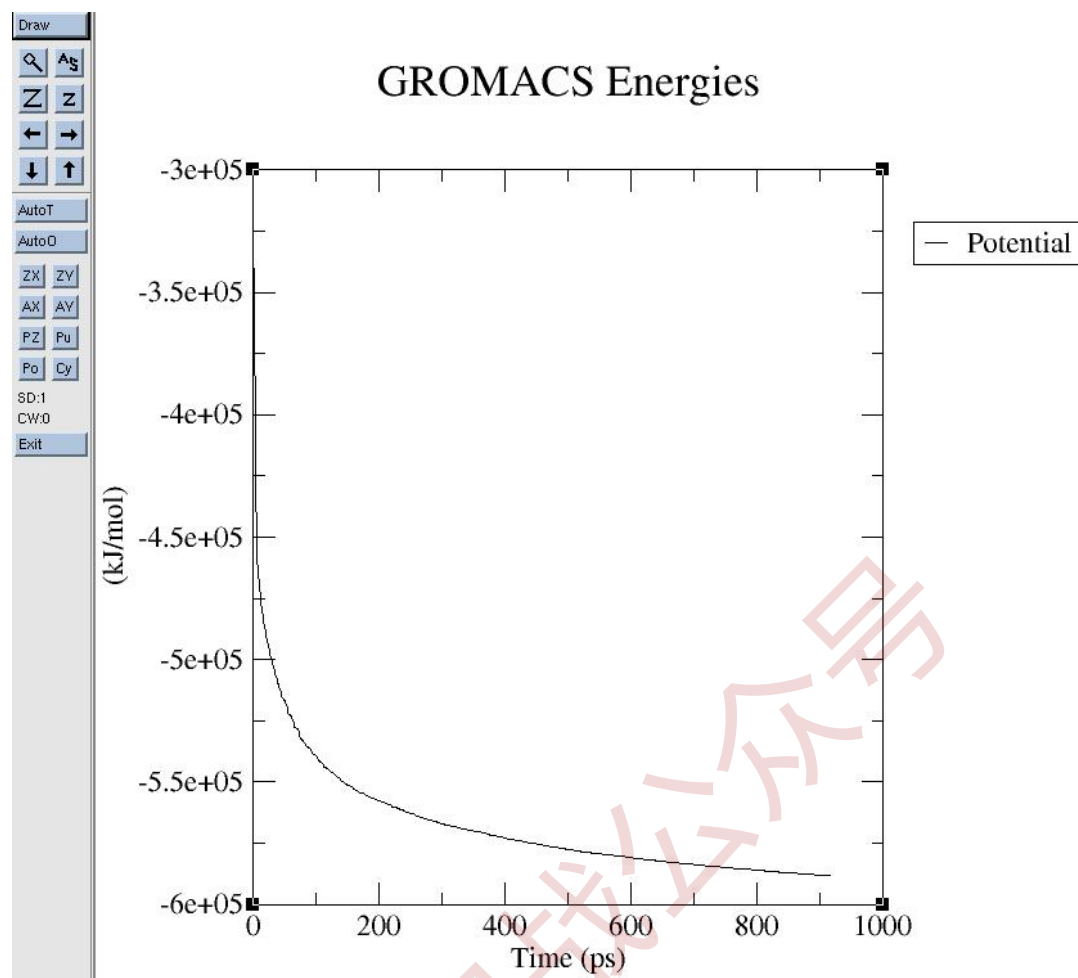
```
@ legend box on
@ legend loctype view
@ legend 0.78, 0.8
@ legend length 2
@ s0 legend "Potential"
0.000000 -331915.562500
1.000000 -347978.500000
2.000000 -370295.875000
3.000000 -398672.937500
4.000000 -427119.781250
5.000000 -452107.750000
.....
.....
```

里面就是有两列数据：第一列是时间（皮秒），第二列是能量最小化的势能（kJ/mol）

我们可以安装xmgrace表格工具进行查看，执行：`apt-get install grace`

在图形界面下执行xgrace potential.xvg

```
Terminal
File Edit View Search Terminal Help
asp@coding_gene94_ubuntu16 22:24:13 ~
$ xmgrace /home/asp/mdtest/demol/potential.xvg
asp@coding_gene94_ubuntu16 22:25:15 ~
```

从中可看到Epot收敛得不错，随着时间的变化，势能逐步趋于稳定。

【关于这里使用图形界面调用xmgrace工具，见下面的说明】

6.5.6ubuntu 图形界面补充说明

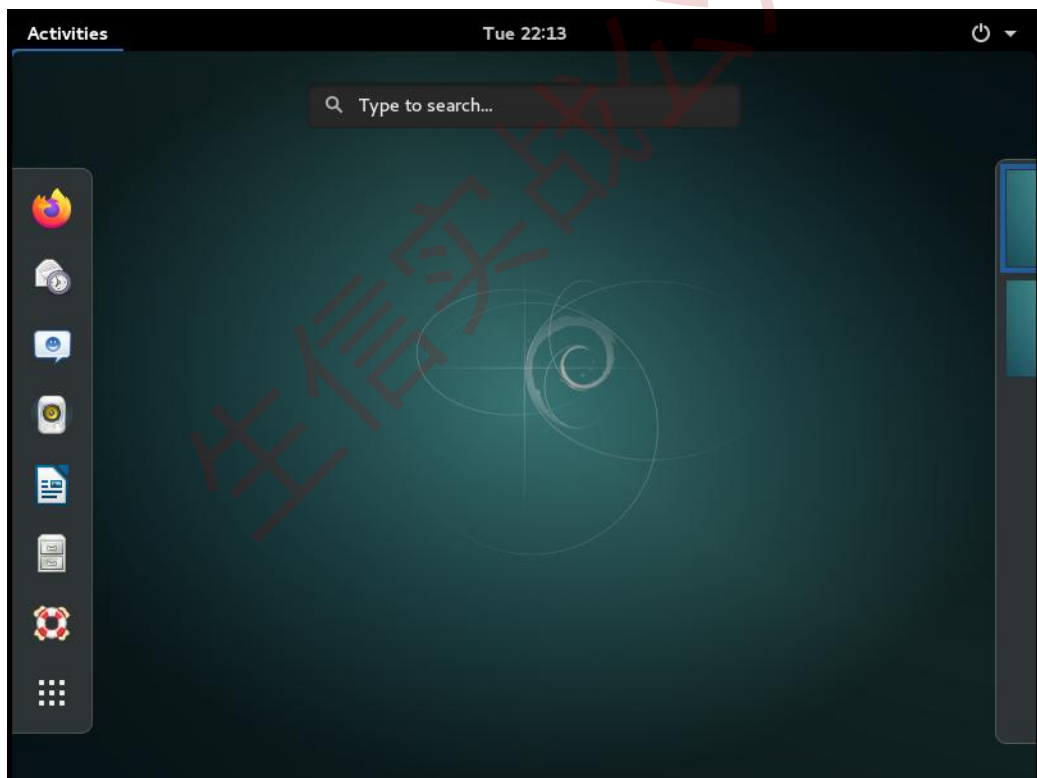
使用图形界面调用xmgrace工具查看能量最小化的曲线图时，涉及图形界面，可以参考如下。

其实也可以通过其他工具查看，甚至使用Excel作图也可以看，就看随着横坐标（时间）的变化，对应的纵坐标（势能）的变化。

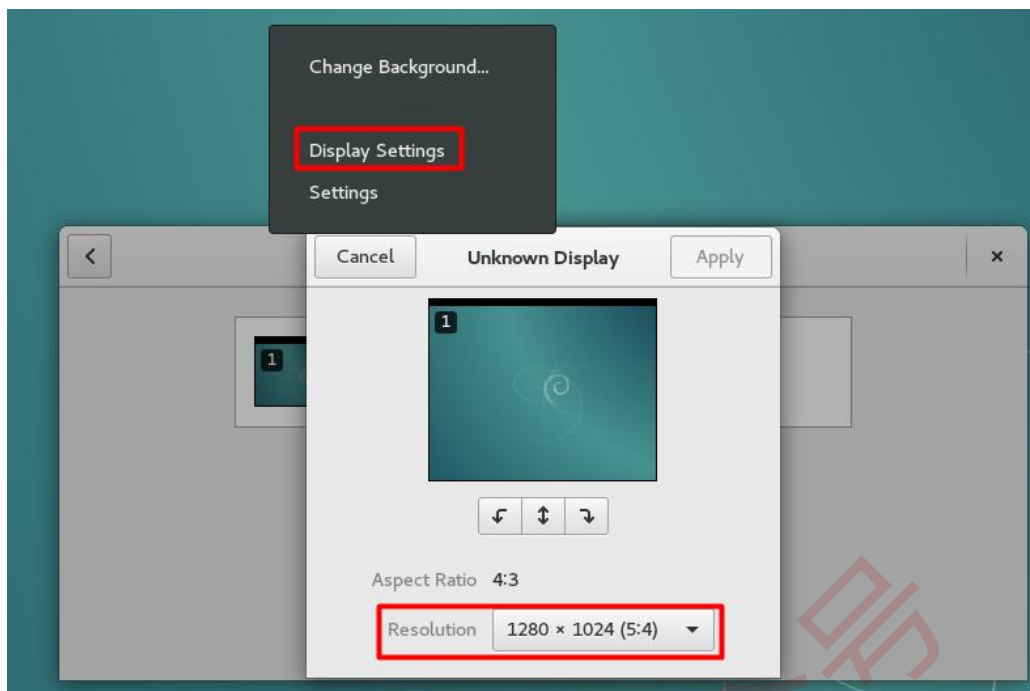
- 1、如果服务器或虚拟机没有安装图形界面，执行：apt-get install xinit 然后在命令行startx启动一个简易的图形化界面；

```
asp@coding_gene94_ubuntu16 20:35:08
$
asp@coding_gene94_ubuntu16 20:41:14
$ cd mdtest/demo1/
asp@coding_gene94_ubuntu16 20:41:19
$ xmgrace potential.xvg
asp@coding_gene94_ubuntu16 20:42:18
$
```

- 2、如果嫌这个xinit界面不好用，也可以安装专业的图形化桌面gnome。先执行apt-get update 再安装apt-get install gnome，需要的空间也多一些，2-3G的样子，安装后也是执行startx启动图形界面，如下：



- 3、如果使用gnome图形界面，屏幕比较小，显示不完整，可以调整一下桌面分辨率



如果是使用远程服务器，也没法使用图形界面，可以考虑使用VNCServer进行连接图形界面环境。如果远程环境如果不方便，也下载这个potential.xvg文件，在本地带图形界面的Linux上查看

以上是对图形化界面查看的补充说明，有相关问题的可以参考。

完成能量最小化后，下一步要进行“NVT平衡”

6.6 步骤 6-NVT 温度平衡

6.6.1 NVT 平衡是做什么

能量最小化 (EM) 的目的是促使体系初始结构在几何构型和溶剂分子取向等方面都合理，为

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了开始真正的动力学模拟, 我们还需要对蛋白质周围的溶剂和离子进行平衡。如果未经平衡就直接尝试进行非限制的动力学模拟, 体系可能会崩溃, 原因在于目前我们只是优化了溶剂分子本身, 而没有考虑溶质。我们需要将体系置于设定的模拟温度下, 以确定蛋白质溶质的合理结构取向。达到目标温度后, 再对体系施加压力, 达到合适的密度。

之前我们用pdb2gmx生成了posre.itp文件, 此文件的目的在于对蛋白质中的非氢原子类的重原子施加位置限制(position restraining)力。这些原子一般不会移动, 除非增加非常大的能量。位置限制的作用在于平衡蛋白质周围的溶剂分子, 而不引起蛋白质结构的变化。

平衡往往分两个阶段进行:

第一个阶段在NVT系统包括粒子数, 体积和温度都恒定的情况下进行。这个系统也被称为等温、等容系综。这个过程的需要的时间与体系的构成有关, 但在NVT系统中, 体系的温度应达到预期值并基本保持不变, 如果温度仍然没有稳定, 那就需要更多的时间。一般50-100 ps就足够了, 因此在本实验中我们进行100ps的NVT平衡。

6.6.2nvt mdp 参数

运行之前需要准备的.mdp文件如下:

获取链接 <http://www.mdtutorials.com/gmx/lysozyme/Files/nvt.mdp>

```
title                      = OPLS Lysozyme NVT equilibration
define                    = -DPOSRES ; position restrain the protein
; Run parameters
```

```

integrator          = md          ; leap-frog integrator
nsteps             = 50000        ; 2 * 50000 = 100 ps
dt                 = 0.002        ; 2 fs

; Output control
nstxout            = 500          ; save coordinates every 1.0 ps
nstvout            = 500          ; save velocities every 1.0 ps
nstenergy          = 500          ; save energies every 1.0 ps
nstlog             = 500          ; update log file every 1.0 ps

; Bond parameters
continuation       = no          ; first dynamics run
constraint_algorithm = lincs      ; holonomic constraints
constraints        = h-bonds     ; bonds involving H are constrained
lincs_iter         = 1           ; accuracy of LINCS
lincs_order        = 4           ; also related to accuracy

; Nonbonded settings
cutoff-scheme      = Verlet      ; Buffered neighbor searching
ns_type            = grid        ; search neighboring grid cells
nstlist            = 10          ; 20 fs, largely irrelevant with Verlet
rcoulomb           = 1.0         ; short-range electrostatic cutoff (in nm)
rvdw               = 1.0         ; short-range van der Waals cutoff (in nm)
DispCorr           = EnerPres    ; account for cut-off vdW scheme

; Electrostatics
coulombtype        = PME         ; Particle Mesh Ewald for long-range
electrostatics

pme_order          = 4           ; cubic interpolation
fourierspacing     = 0.16        ; grid spacing for FFT

; Temperature coupling is on
tcoupl             = V-rescale    ; modified Berendsen thermostat
tc-grps            = Protein Non-Protein ; two coupling groups - more

```

```
accurate
tau_t          = 0.1    0.1          ; time constant, in ps
ref_t          = 300    300          ; reference temperature, one for
each group, in K
; Pressure coupling is off
pcoupl         = no      ; no pressure coupling in NVT
; Periodic boundary conditions
pbc            = xyz      ; 3-D PBC
; Velocity generation
gen_vel        = yes      ; assign velocities from Maxwell distribution
gen_temp       = 300      ; temperature for Maxwell distribution
gen_seed       = -1       ; generate a random seed
```



```

title                = OPLS Lysozyme NVT equilibration
define               = -DPOSRES ; position restrain the protein
; Run parameters
integrator           = md        ; leap-frog integrator
nsteps               = 50000     ; 2 * 50000 = 100 ps
dt                   = 0.002     ; 2 fs
; Output control
nstxout              = 500       ; save coordinates every 1.0 ps
nstvout              = 500       ; save velocities every 1.0 ps
nstenergy            = 500       ; save energies every 1.0 ps
nstlog               = 500       ; update log file every 1.0 ps
; Bond parameters
continuation         = no        ; first dynamics run
constraint_algorithm = lincs     ; holonomic constraints
constraints          = h-bonds   ; bonds involving H are constrained
lincs_iter           = 1         ; accuracy of LINCS
lincs_order          = 4         ; also related to accuracy
; Nonbonded settings
cutoff-scheme        = Verlet    ; Buffered neighbor searching
ns_type              = grid      ; search neighboring grid cells
nstlist              = 10        ; 20 fs, largely irrelevant with Verlet
rcoulomb             = 1.0       ; short-range electrostatic cutoff (in nm)
rvdw                 = 1.0       ; short-range van der Waals cutoff (in nm)
DispCorr             = EnerPres  ; account for cut-off vdW scheme
; Electrostatics
coulombtype          = PME       ; Particle Mesh Ewald for long-range electrostatics
pme_order            = 4         ; cubic interpolation
fourierspacing       = 0.16     ; grid spacing for FFT
; Temperature coupling is on
tcoupl               = V-rescale ; modified Berendsen thermostat
tc-grps              = Protein Non-Protein ; two coupling groups - more accurate
tau_t                = 0.1 0.1  ; time constant, in ps
ref_t                = 300 300   ; reference temperature, one for each group, in K
; Pressure coupling is off
pcoupl               = no        ; no pressure coupling in NVT
; Periodic boundary conditions
pbc                  = xyz       ; 3-D PBC
; Velocity generation
gen_vel              = yes       ; assign velocities from Maxwell distribution
gen_temp             = 300       ; temperature for Maxwell distribution
gen_seed             = -1        ; generate a random seed

```

6.6.3 实操 15-执行 NVT 平衡操作

Linux下执行:

第1步: `gmx grompp -f nvt.mdp -c em.gro -r em.gro -p topol.top -o nvt.tpr`

第2步: `gmx mdrun -deffnm nvt`

(根据机器的不同, 运行需要一段时间, 我这里12核CPU大概10分钟运行完成)

第1步执行截图:

```
asp@coding_gene94_ubuntu16 10:01:13 ~/mdtest/demo1
$ ls -lrt
total 9912
-rw-r--r-- 1 asp asp 116316 Apr 26 10:45 laki.pdb
-rw-r--r-- 1 asp asp 109917 Apr 26 15:04 lAKI_clean.pdb
-rw-rw-r-- 1 asp asp 552964 May 22 10:10 #topol.top.1#
-rw-rw-r-- 1 asp asp 31304 May 22 10:10 posre.itp
-rw-rw-r-- 1 asp asp 88246 May 22 10:10 lAKI_processed.gro
-rw-rw-r-- 1 asp asp 88246 May 22 10:34 lAKI_newbox.gro
-rw-rw-r-- 1 asp asp 1525186 May 22 10:48 lAKI_solv.gro
-rw----- 1 asp asp 552995 May 22 10:49 #topol.top.2#
-rw-rw-r-- 1 asp asp 2369 May 22 10:53 nvt.mdp
-rw-rw-r-- 1 asp asp 2591 May 22 10:53 npt.mdp
-rw-rw-r-- 1 asp asp 1045 May 22 10:53 minim.mdp
-rw-rw-r-- 1 asp asp 2741 May 22 10:53 md.mdp
-rw-rw-r-- 1 asp asp 1045 May 22 10:53 ions.mdp
-rw-rw-r-- 1 asp asp 1266912 May 22 11:19 ions.tpr
-rw----- 1 asp asp 553010 May 22 11:20 topol.top
-rw-rw-r-- 1 asp asp 1524475 May 22 11:20 lAKI_solv_ions.gro
-rw-rw-r-- 1 asp asp 10776 May 22 19:30 mdout.mdp
-rw-rw-r-- 1 asp asp 1267416 May 22 19:30 em.tpr
-rw-rw-r-- 1 asp asp 134808 May 22 19:40 em.edr
-rw-rw-r-- 1 asp asp 406632 May 22 19:40 em.trr
-rw-rw-r-- 1 asp asp 1524475 May 22 19:40 em.gro
-rw-rw-r-- 1 asp asp 313371 May 22 19:40 em.log
-rw-rw-r-- 1 asp asp 20480 May 22 20:24 potential.xvg
asp@coding_gene94_ubuntu16 10:01:15 ~/mdtest/demo1
$ gmx grompp -f nvt.mdp -c em.gro -r em.gro -p topol.top -o nvt.tpr
:~) GROMACS - gmx grompp, 2020.7 (-:
```

第1步得到 nvt.tpr文件

第2步执行截图

```

Command line:
gmX grompp -f nvt.mdp -c em.gro -r em.gro -p topol.top -o nvt.tpr

Ignoring obsolete mdp entry 'title'
Ignoring obsolete mdp entry 'ns_type'
Setting the LD random seed to 996917246
Generated 330891 of the 330891 non-bonded parameter combinations
Generating 1-4 interactions: fudge = 0.5
Generated 330891 of the 330891 1-4 parameter combinations
Excluding 3 bonded neighbours molecule type 'Protein_chain_A'
turning H bonds into constraints...
Excluding 2 bonded neighbours molecule type 'SOL'
turning H bonds into constraints...
Excluding 1 bonded neighbours molecule type 'CL'
turning H bonds into constraints...
Setting gen_seed to -302024739
Velocities were taken from a Maxwell distribution at 300 K
Analysing residue names:
There are: 129 Protein residues
There are: 10636 Water residues
There are: 8 Ion residues
Analysing Protein...
Analysing residues not classified as Protein/DNA/RNA/Water and splitting into groups..
Number of degrees of freedom in T-Coupling group Protein is 4920.79
Number of degrees of freedom in T-Coupling group non-Protein is 63837.21
Determining Verlet buffer for a tolerance of 0.005 kJ/mol/ps at 300 K
Calculated rlist for 1x1 atom pair-list as 1.035 nm, buffer size 0.035 nm
Set rlist, assuming 4x4 atom pair-list, to 1.000 nm, buffer size 0.000 nm
Note that mdrun will redetermine rlist based on the actual pair-list setup

NOTE 1 [file nvt.mdp]:
  Removing center of mass motion in the presence of position restraints
  might cause artifacts. When you are using position restraints to
  equilibrate a macro-molecule, the artifacts are usually negligible.

Calculating fourier grid dimensions for X Y Z
Using a fourier grid of 44x44x44, spacing 0.159 0.159 0.159
Estimate for the relative computational load of the PME mesh part: 0.22
This run will generate roughly 81 Mb of data

There was 1 note

GROMACS reminds you: "You always pass failure on the way to success." (Mickey Rooney)

asp@coding_gene94_ubuntu16 10:01:51 ~/mdtest/demo1
$ nohup gmX mdrun -defnm nvt &
[1] 2168
  
```

第2步是执行NVT平衡。可以查看输出提示需要进行100ps的运行

```

asp@coding_gene94_ubuntu16 10:06:26 ~/mdtest/demo1
$ tail -f nohup.out
Changing nstlist from 10 to 50, rlist from 1 to 1.115

Using 1 MPI thread

Non-default thread affinity set, disabling internal thread affinity

Using 12 OpenMP threads

starting mdrun 'LYSOZYME in water'
50000 steps, 100.0 ps.
  
```


6.6.4 查看运行的详细进度

现在通过linux的top命令可以看到CPU使用率比较高

```
root@coding_gene94_ubuntu16:~# top
top - 10:08:14 up 1:43, 3 users, load average: 10.20, 7.18, 3.29
Tasks: 248 total, 1 running, 247 sleeping, 0 stopped, 0 zombie
%Cpu0  : 96.6 us, 1.0 sy, 0.0 ni, 2.4 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu1  : 96.9 us, 0.3 sy, 0.0 ni, 2.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu2  : 96.6 us, 0.7 sy, 0.0 ni, 2.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu3  : 96.6 us, 1.0 sy, 0.0 ni, 2.4 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu4  : 96.6 us, 0.3 sy, 0.0 ni, 3.1 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu5  : 96.9 us, 0.3 sy, 0.0 ni, 2.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu6  : 96.3 us, 0.7 sy, 0.0 ni, 3.1 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu7  : 96.9 us, 0.0 sy, 0.0 ni, 3.1 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu8  : 96.2 us, 0.7 sy, 0.0 ni, 3.1 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu9  : 97.2 us, 0.0 sy, 0.0 ni, 2.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu10 : 96.9 us, 0.0 sy, 0.0 ni, 3.1 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu11 : 97.3 us, 0.3 sy, 0.0 ni, 2.4 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
```

```
asp@coding_gene94_ubuntu16 10:02:42 ~/mdtest/demol
$ ls -lrt
total 16880
-rw-r--r-- 1 asp asp 116316 Apr 26 10:45 laki.pdb
-rw-r--r-- 1 asp asp 109917 Apr 26 15:04 lAKI_clean.pdb
-rw-rw-r-- 1 asp asp 552964 May 22 10:10 #topol.top.1#
-rw-rw-r-- 1 asp asp 31304 May 22 10:10 posre.itp
-rw-rw-r-- 1 asp asp 88246 May 22 10:10 lAKI_processed.gro
-rw-rw-r-- 1 asp asp 88246 May 22 10:34 lAKI_newbox.gro
-rw-rw-r-- 1 asp asp 1525186 May 22 10:48 lAKI_solv.gro
-rw-rw-r-- 1 asp asp 552995 May 22 10:49 #topol.top.2#
-rw-rw-r-- 1 asp asp 2369 May 22 10:53 nvt.mdp
-rw-rw-r-- 1 asp asp 2591 May 22 10:53 npt.mdp
-rw-rw-r-- 1 asp asp 1045 May 22 10:53 minim.mdp
-rw-rw-r-- 1 asp asp 2741 May 22 10:53 md.mdp
-rw-rw-r-- 1 asp asp 1045 May 22 10:53 ions.mdp
-rw-rw-r-- 1 asp asp 1266912 May 22 11:19 ions.tpr
-rw-rw-r-- 1 asp asp 553010 May 22 11:20 topol.top
-rw-rw-r-- 1 asp asp 1524475 May 22 11:20 lAKI_solv_ions.gro
-rw-rw-r-- 1 asp asp 1267416 May 22 19:30 em.tpr
-rw-rw-r-- 1 asp asp 134808 May 22 19:40 em.edr
-rw-rw-r-- 1 asp asp 406632 May 22 19:40 em.trr
-rw-rw-r-- 1 asp asp 1524475 May 22 19:40 em.gro
-rw-rw-r-- 1 asp asp 313371 May 22 19:40 em.log
-rw-rw-r-- 1 asp asp 20480 May 22 20:24 potential.xvg
-rw-rw-r-- 1 asp asp 10835 May 24 10:01 mdout.mdp
-rw-rw-r-- 1 asp asp 1408388 May 24 10:01 nvt.tpr
-rw-rw-r-- 1 asp asp 2070 May 24 10:02 nohup.out
-rw-rw-r-- 1 asp asp 5692008 May 24 10:02 nvt.trr
-rw-rw-r-- 1 asp asp 19541 May 24 10:02 nvt.log
-rw-rw-r-- 1 asp asp 4732 May 24 10:02 nvt.edr
asp@coding_gene94_ubuntu16 10:02:44 ~/mdtest/demol
$ tail -f nvt.log
Energies (kJ/mol)
Bond          Angle      Proper Dih. Ryckaert-Bell.      LJ-14
1.46803e+03    3.74631e+03 2.44773e+02 1.80479e+03 2.74070e+03
Coulomb-14     LJ (SR)     Disper. corr. Coulomb (SR)  Coul. recip.
7.78086e+03    9.25005e+04 -4.44920e+03 -6.31353e+05 3.06222e+03
Position Rest. Potential Kinetic En. Total Energy Conserved En.
4.77663e+02    -5.21976e+05 8.59694e+04 -4.36007e+05 -5.04852e+05
Temperature Pres. DC (bar) Pressure (bar) Constr. rmsd
3.00757e+02    -2.14729e+02 -3.97178e+02 2.97086e-06
```

使用tail -f nvt.log可以实时查看输出结果：

```

Step      Time
49000    98.00000

Energies (kJ/mol)
Bond      Angle      Proper Dih.  Ryckaert-Bell.  LJ-14
1.53470e+03 3.81054e+03 2.31966e+02 1.88328e+03 2.69510e+03
Coulomb-14  LJ (SR)    Disper. corr. Coulomb (SR)  Coul. recip.
7.77762e+03 9.31963e+04 -4.44920e+03 -6.32913e+05 3.00062e+03
Position Rest. Potential Kinetic En. Total Energy Conserved En.
5.45535e+02 -5.22687e+05 8.58673e+04 -4.36819e+05 -5.04910e+05
Temperature Pres. DC (bar) Pressure (bar) Constr. rmsd
3.00400e+02 -2.14729e+02 -3.36890e+02 2.84918e-06

Step      Time
49500    99.00000

Energies (kJ/mol)
Bond      Angle      Proper Dih.  Ryckaert-Bell.  LJ-14
1.38633e+03 3.82553e+03 2.35599e+02 1.83542e+03 2.68473e+03
Coulomb-14  LJ (SR)    Disper. corr. Coulomb (SR)  Coul. recip.
7.68576e+03 9.27240e+04 -4.44920e+03 -6.32113e+05 3.06604e+03
Position Rest. Potential Kinetic En. Total Energy Conserved En.
4.86636e+02 -5.22633e+05 8.49902e+04 -4.37642e+05 -5.04924e+05
Temperature Pres. DC (bar) Pressure (bar) Constr. rmsd
2.97332e+02 -2.14729e+02 -1.68533e+02 2.89803e-06

Step      Time
50000    100.00000

Writing checkpoint, step 50000 at Wed May 24 10:08:34 2023

Energies (kJ/mol)
Bond      Angle      Proper Dih.  Ryckaert-Bell.  LJ-14
1.39555e+03 3.66444e+03 2.46549e+02 1.87505e+03 2.70540e+03
Coulomb-14  LJ (SR)    Disper. corr. Coulomb (SR)  Coul. recip.
7.77597e+03 9.22897e+04 -4.44920e+03 -6.32177e+05 3.01019e+03
Position Rest. Potential Kinetic En. Total Energy Conserved En.
4.92577e+02 -5.23170e+05 8.52713e+04 -4.37899e+05 -5.04927e+05
Temperature Pres. DC (bar) Pressure (bar) Constr. rmsd
2.98315e+02 -2.14729e+02 -4.66463e+02 2.66376e-06

<===== ##### ==>
<===== A V E R A G E S =====>
<== ##### ==>

Statistics over 50001 steps using 501 frames

```

我这个服务器大概10分钟完成了（12核CPU）

运行结束后，会给出相关的汇总信息：

MEGA - FLOPS ACCOUNTING			
NB=Group-cutoff nonbonded kernels NxN=N-by-N cluster Verlet kernels RF=Reaction-Field Vdw=Van der Waals QSTab=quadratic-spline table W3=SPC/TIP3p W4=TIP4p (single or pairs) V&F=Potential and force V=Potential only F=Force only			
Computing:	M-Number	M-Flops	% Flops
Pair Search distance check	14832.415764	133491.742	0.2
NxN Ewald Elec. + LJ [F]	522443.961728	34481301.474	52.1
NxN Ewald Elec. + LJ [V&F]	5287.984496	565814.341	0.9
NxN Ewald Elec. [F]	460100.147488	28066108.997	42.4
NxN Ewald Elec. [V&F]	4656.815632	391172.513	0.6
1,4 nonbonded interactions	255.305106	22977.460	0.0
Calc Weights	5081.501628	182934.059	0.3
Spread Q Bspline	108405.368064	216810.736	0.3
Gather F Bspline	108405.368064	650432.208	1.0
3D-FFT	139519.590336	1116156.723	1.7
Solve PME	96.801936	6195.324	0.0
Shift-X	33.909876	203.459	0.0
Bonds	51.251025	3023.810	0.0
Angles	177.353547	29795.396	0.0
Propers	21.300426	4877.798	0.0
RB-Dihedrals	197.503950	48783.476	0.1
Pos. Restr.	50.051001	2502.550	0.0
Virial	16.994421	305.900	0.0
Stop-CM	17.005752	170.058	0.0
Calc-Ekin	338.827752	9148.349	0.0
Lincs	47.952877	2877.173	0.0
Lincs-Mat	238.814328	955.257	0.0
Constraint-V	1691.367652	13530.941	0.0
Constraint-Vir	16.466367	395.193	0.0
Settle	531.831908	171781.706	0.3
Total		66121746.642	100.0

REAL CYCLE AND TIME ACCOUNTING						
On 1 MPI rank, each using 12 OpenMP threads						
Computing:	Num Ranks	Num Threads	Call Count	Wall time (s)	Giga-Cycles total sum	%
Neighbor search	1	12	1001	6.700	256.625	1.8
Force	1	12	50001	177.203	6787.739	48.6
PME mesh	1	12	50001	114.912	4401.703	31.5
NB X/F buffer ops.	1	12	99001	34.215	1310.599	9.4
Write traj.	1	12	101	0.732	28.042	0.2
Update	1	12	50001	14.467	554.162	4.0
Constraints	1	12	50003	13.784	528.004	3.8
Rest				2.727	104.447	0.7
Total				364.740	13971.321	100.0
Breakdown of PME mesh computation						
PME spread	1	12	50001	44.942	1721.491	12.3
PME gather	1	12	50001	32.519	1245.625	8.9
PME 3D-FFT	1	12	100002	32.704	1252.732	9.0
PME solve Elec	1	12	50001	0.937	35.894	0.3
Time: Core t (s) 4376.882 Wall t (s) 364.740 (%) 1200.0						
(ns/day) (hour/ns)						
Performance: 23.689 1.013						
Finished mdrun on rank 0 Wed May 24 10:08:34 2023						

6.6.5实操 16-NVT 平衡结果分析

Linux执行:

```
gmx energy -f nvt.edr -o temperature.xvg
```



```

Command line:
gmx energy -f nvt.edr -o temperature.xvg
Opened nvt.edr as single precision energy file

Select the terms you want from the following list by
selecting either (part of) the name or the number or a combination.
End your selection with an empty line or a zero.
-----
 1 Bond                      2 Angle                     3 Proper-Dih.             4 Ryckaert-Bell.
 5 LJ-14                     6 Coulomb-14                7 LJ-(SR)                 8 Disper.-corr.
 9 Coulomb-(SR)              10 Coul.-recip.             11 Position-Rest.         12 Potential
13 Kinetic-En.               14 Total-Energy             15 Conserved-En.          16 Temperature
17 Pres.-DC                  18 Pressure                  19 Constr.-rmsd           20 Vir-XX
21 Vir-XY                    22 Vir-XZ                   23 Vir-YX                  24 Vir-YY
25 Vir-YZ                    26 Vir-ZX                   27 Vir-ZY                  28 Vir-ZZ
29 Pres-XX                   30 Pres-XY                   31 Pres-XZ                 32 Pres-YX
33 Pres-YY                   34 Pres-YZ                   35 Pres-ZX                 36 Pres-ZY
37 Pres-ZZ                   38 #Surf*SurfTen            39 T-Protein               40 T-non-Protein
41 Lamb-Protein              42 Lamb-non-Protein
-----
16 0
Last energy frame read 100 time 100.000

Statistics over 50001 steps [ 0.0000 through 100.0000 ps ], 1 data sets
All statistics are over 501 points

Energy
-----
Temperature                Average    Err.Est.    RMSD    Tot-Drift
299.667      0.082    2.98461    0.643795 (K)

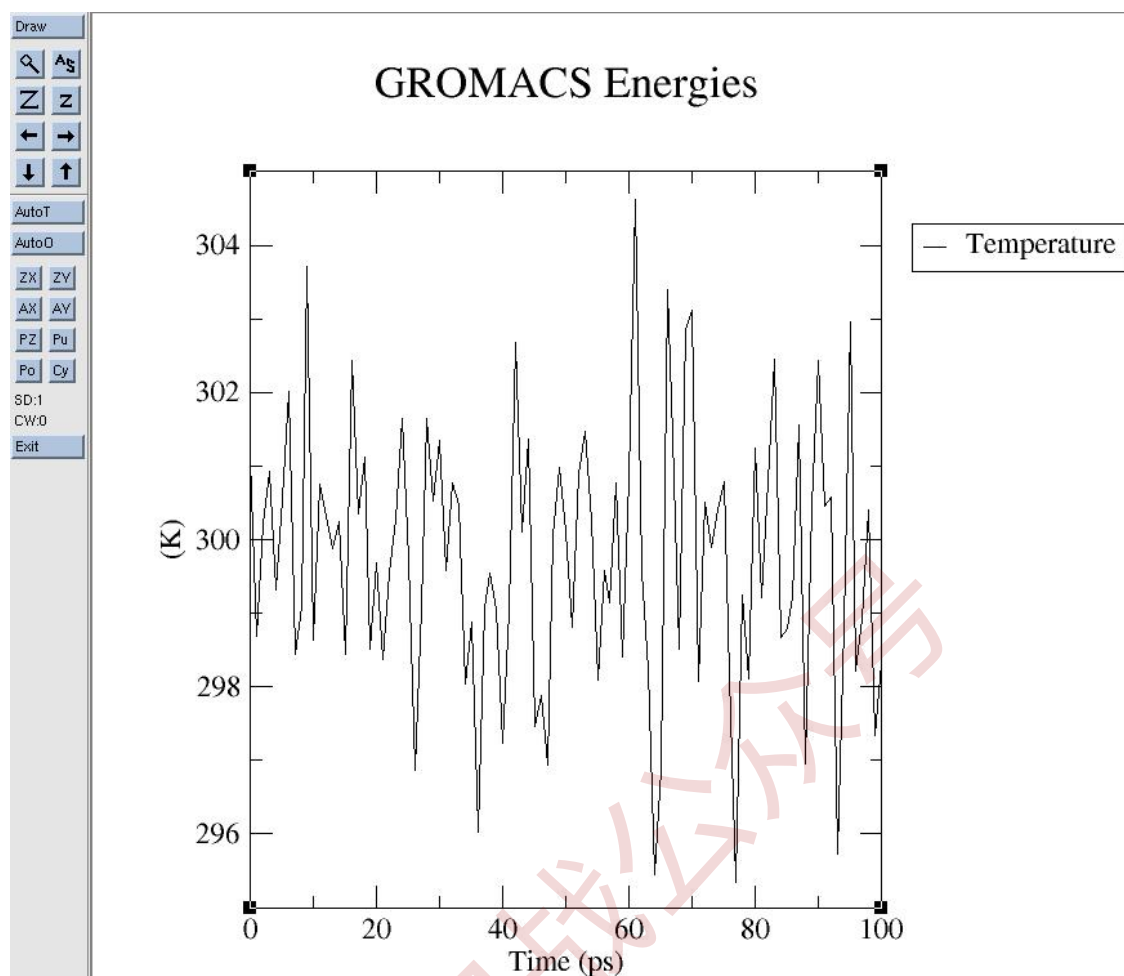
GROMACS reminds you: "Kick the Dog and You will Die" (Magnapop)
asp@coding_gene94_ubuntu16 10:31:05 ~/mdtest/demo1
$

```

在提示符下键入“16 0”，选择温度代号16，选择0退出

继续使用xmgrace工具查看变化图：

```
xmgrace temperature.xvg
```



可以看出，系统的温度迅速达到目标值300K左右，并在平衡的剩余时间内保持稳定（按这个图的结果，总体波动不大，其实较短的平衡时间(大约50 ps)应该就足够了。

6.7 步骤 7-NPT 压力平衡

6.7.1 NPT 平衡是做什么

上一步NVT平衡建立了温度系统，这一步需要建立压力和密度系统（稳定系统的压力，以及保证合适的密度）。压力的平衡是在NPT系统下进行的，其粒子数，压力和温度都保持不变，这个系统也被称为等温、等压系统，比较接近实验环境。

6.7.2npt mdp 参数

100-ps NPT equilibration 配置参数链接:

<http://www.mdtutorials.com/gmx/lysozyme/Files/npt.mdp>

这个参数文件与NVT平衡时所用的参数文件类似。

注意添加的压力耦合部分, 其中使用了Parrinello-Rahman控压器. 其他几项改动如下:

continuation = yes: 我们将从NVT平衡阶段开始继续进行模拟

gen_vel =no: 从轨迹中读取速度 (Velocities are read from the trajectory)

```
$ cat npt.mdp
title                      = OPLS Lysozyme NPT equilibration
define                    = -DPOSRES   ; position restrain the protein
; Run parameters
integrator                = md          ; leap-frog integrator
nsteps                    = 50000       ; 2 * 50000 = 100 ps
dt                        = 0.002       ; 2 fs
; Output control
nstxout                   = 500          ; save coordinates every 1.0 ps
nstvout                   = 500          ; save velocities every 1.0 ps
nstenergy                 = 500          ; save energies every 1.0 ps
nstlog                    = 500          ; update log file every 1.0 ps
; Bond parameters
```

```

continuation          = yes          ; Restarting after NVT
constraint_algorithm   = lincs       ; holonomic constraints
constraints           = h-bonds     ; bonds involving H are constrained
lincs_iter            = 1           ; accuracy of LINCS
lincs_order           = 4           ; also related to accuracy
; Nonbonded settings
cutoff-scheme         = Verlet      ; Buffered neighbor searching
ns_type               = grid        ; search neighboring grid cells
nstlist               = 10          ; 20 fs, largely irrelevant with Verlet scheme
rcoulomb              = 1.0         ; short-range electrostatic cutoff (in nm)
rvdw                  = 1.0         ; short-range van der Waals cutoff (in nm)
DispCorr              = EnerPres   ; account for cut-off vdW scheme
; Electrostatics
coulombtype           = PME         ; Particle Mesh Ewald for long-range
electrostatics
pme_order             = 4           ; cubic interpolation
fourierspacing        = 0.16       ; grid spacing for FFT
; Temperature coupling is on
tcoupl                = V-rescale    ; modified Berendsen thermostat
tc-grps               = Protein Non-Protein ; two coupling groups - more
accurate
tau_t                 = 0.1 0.1     ; time constant, in ps
ref_t                 = 300 300     ; reference temperature, one for
each group, in K
; Pressure coupling is on
pcoupl                = Parrinello-Rahman ; Pressure coupling on in NPT
pcoupltype            = isotropic   ; uniform scaling of box vectors
tau_p                 = 2.0         ; time constant, in ps
ref_p                 = 1.0         ; reference pressure, in bar

```

```
compressibility      = 4.5e-5          ; isothermal compressibility of
water, bar^-1
refcoord_scaling      = com
; Periodic boundary conditions
pbc                   = xyz            ; 3-D PBC
; Velocity generation
gen_vel               = no            ; Velocity generation is off
```

6.7.3 实操 17-执行 NPT 平衡操作

NPT平衡也是使用gmx的grompp和mdrun模块

- 1、-t选项，指定NPT平衡过程中的产生的检查点文件名（这个文件包含了继续模拟所需要的所有状态变量）
- 2、-c选项，指定NPT模拟最终输出的坐标文件

第1步: `gmx grompp -f npt.mdp -c nvt.gro -r nvt.gro -t nvt.cpt -p topol.top -o npt.tpr`

第2步: `gmx mdrun -deffnm npt`

```

Command line:
  gmx grompp -f npt.mdp -c nvt.gro -r nvt.gro -t nvt.cpt -p topol.top -o npt.tpr

Ignoring obsolete mdp entry 'title'
Ignoring obsolete mdp entry 'ns_type'
Setting the LD random seed to 1649261567
Generated 330891 of the 330891 non-bonded parameter combinations
Generating 1-4 interactions: fudge = 0.5
Generated 330891 of the 330891 1-4 parameter combinations
Excluding 3 bonded neighbours molecule type 'Protein_chain_A'
turning H bonds into constraints...
Excluding 2 bonded neighbours molecule type 'SOL'
turning H bonds into constraints...
Excluding 1 bonded neighbours molecule type 'CL'
turning H bonds into constraints...

NOTE 1 [file topol.top, line 18410]:
  You are combining position restraints with Parrinello-Rahman pressure
  coupling, which can lead to instabilities. If you really want to combine
  position restraints with pressure coupling, we suggest to use Berendsen
  pressure coupling instead.

The center of mass of the position restraint coord's is 3.487 3.517 3.503
The center of mass of the position restraint coord's is 3.487 3.517 3.503
Analysing residue names:
There are: 129 Protein residues
There are: 10636 Water residues
There are: 8 Ion residues
Analysing Protein...
Analysing residues not classified as Protein/DNA/RNA/Water and splitting into groups...
Number of degrees of freedom in T-Coupling group Protein is 4920.79
Number of degrees of freedom in T-Coupling group non-Protein is 63837.21
Determining Verlet buffer for a tolerance of 0.005 kJ/mol/ps at 300 K
Calculated rlist for 1x1 atom pair-list as 1.035 nm, buffer size 0.035 nm
Set rlist, assuming 4x4 atom pair-list, to 1.000 nm, buffer size 0.000 nm
Note that mdrun will redetermine rlist based on the actual pair-list setup

NOTE 2 [file npt.mdp]:
  Removing center of mass motion in the presence of position restraints
  might cause artifacts. When you are using position restraints to
  equilibrate a macro-molecule, the artifacts are usually negligible.

Reading Coordinates, Velocities and Box size from old trajectory
Will read whole trajectory
Last frame -1 time 100.000
Using frame at t = 100 ps
Starting time for run is 0 ps
Calculating fourier grid dimensions for X Y Z
Using a fourier grid of 44x44x44, spacing 0.159 0.159 0.159
Estimate for the relative computational load of the PME mesh part: 0.22
This run will generate roughly 81 Mb of data
  
```

```

asp@coding_gene94_ubuntu16 12:27:40 ~/mdtest/demo1
$ nohup gmx mdrun -deffnm npt &
[1] 33613
asp@coding_gene94_ubuntu16 12:27:48 ~/mdtest/demo1
$ nohup: ignoring input and appending output to 'nohup.out'
  
```

执行gmx mdrun后给出的提示:

```
Reading file npt.tpr, VERSION 2020.7 (single precision)
```

```
Changing nstlist from 10 to 50, rlist from 1 to 1.115
```

```
Using 1 MPI thread
```



```
Non-default thread affinity set, disabling internal thread affinity
```

```
Using 12 OpenMP threads
```

```
starting mdrun 'LYSOZYME in water'
```

```
50000 steps, 100.0 ps.
```

使用了1个并行线程，使用12核CPU线程进行计算，需要执行5万步，完成100皮秒的模拟。

6.7.4 查看运行的详细进度

Linux下可以使用top命令，再按1，查看12核CPU使用率，基本跑满

```
Tasks: 319 total, 2 running, 316 sleeping, 1 stopped, 0 zombie
%Cpu0  : 96.9 us, 0.3 sy, 0.0 ni, 2.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu1  : 96.6 us, 0.3 sy, 0.0 ni, 3.1 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu2  : 96.3 us, 0.7 sy, 0.0 ni, 3.1 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu3  : 96.9 us, 0.3 sy, 0.0 ni, 2.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu4  : 96.9 us, 0.3 sy, 0.0 ni, 2.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu5  : 96.6 us, 1.0 sy, 0.0 ni, 2.4 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu6  : 95.3 us, 0.0 sy, 0.0 ni, 4.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu7  : 96.9 us, 0.7 sy, 0.0 ni, 2.4 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu8  : 96.6 us, 0.7 sy, 0.0 ni, 2.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu9  : 97.3 us, 0.3 sy, 0.0 ni, 2.4 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu10 : 96.6 us, 0.7 sy, 0.0 ni, 2.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu11 : 97.3 us, 1.0 sy, 0.0 ni, 1.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
```

可以通过日志tail -100f npt.log 实时查看进行的进度：

```
-rw-rw-r-- 1 asp asp 87486 May 24 10:08 nvt.log
-rw-rw-r-- 1 asp asp 58876 May 24 10:08 nvt.edr
-rw-rw-r-- 1 asp asp 3159 May 24 10:31 temperature.xvg
-rw-rw-r-- 1 asp asp 10875 May 24 12:26 mdout.mdp
-rw-rw-r-- 1 asp asp 1408388 May 24 12:26 npt.tpr
-rw----- 1 asp asp 4419 May 24 12:27 nohup.out
-rw-rw-r-- 1 asp asp 1626288 May 24 12:27 npt.trr
-rw-rw-r-- 1 asp asp 16029 May 24 12:27 npt.log
-rw-rw-r-- 1 asp asp 2244 May 24 12:27 npt.edr
asp@coding_gene94_ubuntu16 12:27:53 ~/mdtest/demo1
$ tail -100f npt.log
```

Step	Time			
37500	75.00000			
Energies (kJ/mol)				
Bond	Angle	Proper Dih.	Ryckaert-Bell.	LJ-14
1.44490e+03	3.81945e+03	2.27700e+02	1.77314e+03	2.64399e+03
Coulomb-14	LJ (SR)	Disper. corr.	Coulomb (SR)	Coul. recip.
7.77840e+03	9.33115e+04	-4.57180e+03	-6.34879e+05	2.96922e+03
Position Rest.	Potential	Kinetic En.	Total Energy	Conserved En.
4.91872e+02	-5.24990e+05	8.57048e+04	-4.39286e+05	-4.38381e+05
Temperature	Pres. DC (bar)	Pressure (bar)	Constr. rmsd	
2.99832e+02	-2.26711e+02	-1.50014e+02	3.01263e-06	
Step	Time			
38000	76.00000			
Energies (kJ/mol)				
Bond	Angle	Proper Dih.	Ryckaert-Bell.	LJ-14
1.34843e+03	3.89662e+03	2.39776e+02	1.88039e+03	2.63906e+03
Coulomb-14	LJ (SR)	Disper. corr.	Coulomb (SR)	Coul. recip.
7.65135e+03	9.42223e+04	-4.57149e+03	-6.35122e+05	2.95190e+03
Position Rest.	Potential	Kinetic En.	Total Energy	Conserved En.
4.64161e+02	-5.24399e+05	8.57171e+04	-4.38682e+05	-4.38370e+05
Temperature	Pres. DC (bar)	Pressure (bar)	Constr. rmsd	
2.99875e+02	-2.26680e+02	1.66540e+02	2.51474e-06	
Step	Time			
38500	77.00000			
Energies (kJ/mol)				
Bond	Angle	Proper Dih.	Ryckaert-Bell.	LJ-14
1.41418e+03	3.73298e+03	2.54245e+02	1.74413e+03	2.68757e+03
Coulomb-14	LJ (SR)	Disper. corr.	Coulomb (SR)	Coul. recip.
7.76366e+03	9.49267e+04	-4.55869e+03	-6.37348e+05	3.02564e+03
Position Rest.	Potential	Kinetic En.	Total Energy	Conserved En.
4.80654e+02	-5.25877e+05	8.59643e+04	-4.39913e+05	-4.38367e+05
Temperature	Pres. DC (bar)	Pressure (bar)	Constr. rmsd	
3.00740e+02	-2.25414e+02	1.25484e+02	2.62535e-06	

我的测试服务器Intel(R) Core(TM) i7-8700 CPU @ 3.20GHz 12核，大概10分钟跑完

tail -100f npt.log最后显示的日志：

M E G A - F L O P S A C C O U N T I N G			
NB=Group-cutoff nonbonded kernels NxN=N-by-N cluster Verlet kernels RF=Reaction-Field VdW=Van der Waals QSTab=quadratic-spline table W3=SPC/TIP3p W4=TIP4p (single or pairs) V&F=Potential and force V=Potential only F=Force only			
Computing:	M-Number	M-Flops	% Flops
Pair Search distance check	15119.006954	136071.063	0.2
NxN Ewald Elec. + LJ [F]	534317.936464	35264983.807	52.2
NxN Ewald Elec. + LJ [V&F]	5407.637040	578617.163	0.9
NxN Ewald Elec. [F]	470783.333872	28717783.366	42.5
NxN Ewald Elec. [V&F]	4765.051376	400264.316	0.6
1,4 nonbonded interactions	255.305106	22977.460	0.0
Calc Weights	5081.501628	182934.059	0.3
Spread Q Bspline	108405.368064	216810.736	0.3
Gather F Bspline	108405.368064	650432.208	1.0
3D-FFT	139519.590336	1116156.723	1.7
Solve PME	96.801936	6195.324	0.0
Shift-X	33.909876	203.459	0.0
Bonds	51.251025	3023.810	0.0
Angles	177.353547	29795.396	0.0
Proprs	21.300426	4877.798	0.0
RB-Dihedrals	197.503950	48783.476	0.1
Pos. Restr.	50.051001	2502.550	0.0
Virial	169.638921	3053.501	0.0
Stop-CM	16.971876	169.719	0.0
Calc-Ekin	338.827752	9148.349	0.0
Lincs	47.950959	2877.058	0.0
Lincs-Mat	238.804776	955.219	0.0
Constraint-V	1691.333826	13530.671	0.0
Constraint-Vir	164.367867	3944.829	0.0
Settle	531.810636	171774.835	0.3
Total		67587866.893	100.0

R E A L C Y C L E A N D T I M E A C C O U N T I N G						
On 1 MPI rank, each using 12 OpenMP threads						
Computing:	Num Ranks	Num Threads	Call Count	Wall time (s)	Giga-Cycles total sum	%
Neighbor search	1	12	1001	7.341	281.179	1.3
Force	1	12	50001	225.612	8641.111	40.9
PME mesh	1	12	50001	195.056	7470.804	35.3
NB X/F buffer ops.	1	12	99001	66.718	2555.332	12.1
Write traj.	1	12	101	0.703	26.930	0.1
Update	1	12	50001	25.206	965.417	4.6
Constraints	1	12	50001	24.564	940.801	4.4
Rest				6.883	263.617	1.2
Total				552.083	21145.190	100.0
Breakdown of PME mesh computation						
PME spread	1	12	50001	77.093	2952.723	14.0
PME gather	1	12	50001	57.604	2206.270	10.4
PME 3D-FFT	1	12	100002	51.132	1958.379	9.3
PME solve Elec	1	12	50001	1.190	45.569	0.2

Time:	Core t (s)	Wall t (s)	(%)
	6624.997	552.083	1200.0
	(ns/day)	(hour/ns)	
Performance:	15.650	1.534	

Finished mdrun on rank 0 Wed May 24 12:37:01 2023

生成的文件：

```
asp@coding_gene94_ubuntu16 12:45:35 ~/mdtest/demo1
$ ls -lrt
total 179576
-rw-r--r-- 1 asp asp 116316 Apr 26 10:45 laki.pdb
-rw-r--r-- 1 asp asp 109917 Apr 26 15:04 lAKI_clean.pdb
-rw-rw-r-- 1 asp asp 552964 May 22 10:10 #topol.top.1#
-rw-rw-r-- 1 asp asp 31304 May 22 10:10 posre.itp
-rw-rw-r-- 1 asp asp 88246 May 22 10:10 lAKI_processed.gro
-rw-rw-r-- 1 asp asp 88246 May 22 10:34 lAKI_newbox.gro
-rw-rw-r-- 1 asp asp 1525186 May 22 10:48 lAKI_solv.gro
-rw----- 1 asp asp 552995 May 22 10:49 #topol.top.2#
-rw-rw-r-- 1 asp asp 2369 May 22 10:53 nvt.mdp
-rw-rw-r-- 1 asp asp 2591 May 22 10:53 npt.mdp
-rw-rw-r-- 1 asp asp 1045 May 22 10:53 minim.mdp
-rw-rw-r-- 1 asp asp 2741 May 22 10:53 md.mdp
-rw-rw-r-- 1 asp asp 1045 May 22 10:53 ions.mdp
-rw-rw-r-- 1 asp asp 1266912 May 22 11:19 ions.tpr
-rw----- 1 asp asp 553010 May 22 11:20 topol.top
-rw-rw-r-- 1 asp asp 1524475 May 22 11:20 lAKI_solv_ions.gro
-rw-rw-r-- 1 asp asp 1267416 May 22 19:30 em.tpr
-rw-rw-r-- 1 asp asp 134808 May 22 19:40 em.edr
-rw-rw-r-- 1 asp asp 406632 May 22 19:40 em.trr
-rw-rw-r-- 1 asp asp 1524475 May 22 19:40 em.gro
-rw-rw-r-- 1 asp asp 313371 May 22 19:40 em.log
-rw-rw-r-- 1 asp asp 20480 May 22 20:24 potential.xvg
-rw-rw-r-- 1 asp asp 1408388 May 24 10:01 nvt.tpr
-rw-rw-r-- 1 asp asp 814692 May 24 10:08 nvt.cpt
-rw-rw-r-- 1 asp asp 82127544 May 24 10:08 nvt.trr
-rw-rw-r-- 1 asp asp 2337499 May 24 10:08 nvt.gro
-rw-rw-r-- 1 asp asp 87486 May 24 10:08 nvt.log
-rw-rw-r-- 1 asp asp 58876 May 24 10:08 nvt.edr
-rw-rw-r-- 1 asp asp 3159 May 24 10:31 temperature.xvg
-rw-rw-r-- 1 asp asp 10875 May 24 12:26 mdout.mdp
-rw-rw-r-- 1 asp asp 1408388 May 24 12:26 npt.tpr
-rw-rw-r-- 1 asp asp 815064 May 24 12:37 npt.cpt
-rw-rw-r-- 1 asp asp 82127544 May 24 12:37 npt.trr
-rw-rw-r-- 1 asp asp 2337499 May 24 12:37 npt.gro
-rw-rw-r-- 1 asp asp 71148 May 24 12:37 npt.edr
-rw-rw-r-- 1 asp asp 87382 May 24 12:37 npt.log
-rw----- 1 asp asp 4742 May 24 12:37 nohup.out
```

6.7.5实操 18-压力分析

Linux执行：

```
gmx energy -f npt.edr -o pressure.xvg
```

根据提示输入18 0 获得压力的数据结果


```

gmx energy -f npt.edr -o pressure.xvg
Opened npt.edr as single precision energy file
Select the terms you want from the following list by
selecting either (part of) the name or the number or a combination.
End your selection with an empty line or a zero.
-----
 1 Bond                2 Angle                3 Proper-Dih.          4 Ryckaert-Bell.
 5 LJ-14               6 Coulomb-14          7 LJ-(SR)             8 Disper.-corr.
 9 Coulomb-(SR)        10 Coul.-recip.       11 Position-Rest.     12 Potential
13 Kinetic-En.         14 Total-Energy       15 Conserved-En.      16 Temperature
17 Pres.-DC            18 Pressure            19 Constr.-rmsd       20 Box-X
21 Box-Y              22 Box-Z              23 Volume              24 Density
25 pV                 26 Enthalpy           27 Vir-XX              28 Vir-XY
29 Vir-XZ             30 Vir-YX             31 Vir-YY              32 Vir-YZ
33 Vir-ZX             34 Vir-ZY             35 Vir-ZZ              36 Pres-XX
37 Pres-XY            38 Pres-XZ            39 Pres-YX             40 Pres-YY
41 Pres-YZ            42 Pres-ZX            43 Pres-ZY             44 Pres-ZZ
45 #Surf*SurfTen      46 Box-Vel-XX         47 Box-Vel-YY         48 Box-Vel-ZZ
49 T-Protein          50 T-non-Protein       51 Lamb-Protein        52 Lamb-non-Protein
18 0
Last energy frame read 100 time 100.000

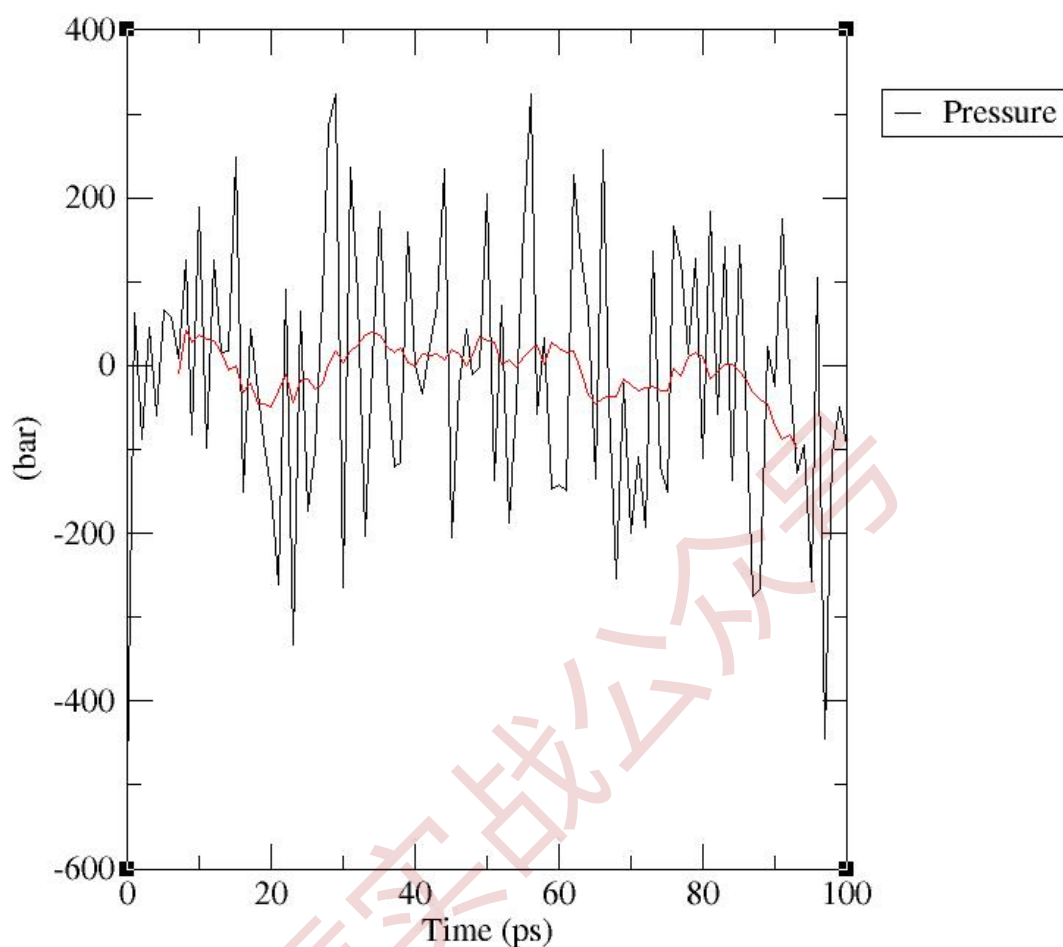
Statistics over 50001 steps [ 0.0000 through 100.0000 ps ], 1 data sets
All statistics are over 501 points

Energy                Average    Err.Est.    RMSD    Tot-Drift
-----
Pressure              -7.34034    3.3    163.406   -12.7503 (bar)

```

查看压力变化曲线图: [xmgrace pressure.xvg](#)

GROMACS Energies



在100 ps的平衡过程中压力值涨落范围较大, 这其实也是意料中的. 图中的红线为数据的移动平均值. 在整个平衡过程中, 压力的平均值为 7.5 ± 160.5 bar (参考压力值设置的是1 bar) 这个结果是否可靠? 从这个160.5 bar的大均方根波动可以看出, 模拟过程中, 压力指标变化范围是比较大的, 从统计学上讲, 并不能区分 $(7.5 \pm 160.5 \text{ bar})$ 与目标参考值 (1 bar)之间的差异。

6.7.6实操 19-密度分析

```
gmx energy -f npt.edr -o density.xvg
```

根据提示 输入 24 0 获得结果

```
Command line:
gmx energy -f npt.edr -o density.xvg

Opened npt.edr as single precision energy file

Select the terms you want from the following list by
selecting either (part of) the name or the number or a combination.
End your selection with an empty line or a zero.
-----
 1 Bond                2 Angle                3 Proper-Dih.          4 Ryckaert-Bell.
 5 LJ-14               6 Coulomb-14          7 LJ-(SR)              8 Disper.-corr.
 9 Coulomb-(SR)        10 Coul.-recip.       11 Position-Rest.      12 Potential
13 Kinetic-En.         14 Total-Energy       15 Conserved-En.       16 Temperature
17 Pres.-DC            18 Pressure           19 Constr.-rmsd        20 Box-X
21 Box-Y              22 Box-Z              23 Volume               24 Density
25 pV                 26 Enthalpy           27 Vir-XX               28 Vir-XY
29 Vir-XZ             30 Vir-YX             31 Vir-YY               32 Vir-YZ
33 Vir-ZX             34 Vir-ZY             35 Vir-ZZ               36 Pres-XX
37 Pres-XY            38 Pres-XZ            39 Pres-YX              40 Pres-YY
41 Pres-YZ            42 Pres-ZX            43 Pres-ZY              44 Pres-ZZ
45 #Surf*SurfTen      46 Box-Vel-XX         47 Box-Vel-YY          48 Box-Vel-ZZ
49 T-Protein          50 T-non-Protein
51 Lamb-Protein       52 Lamb-non-Protein

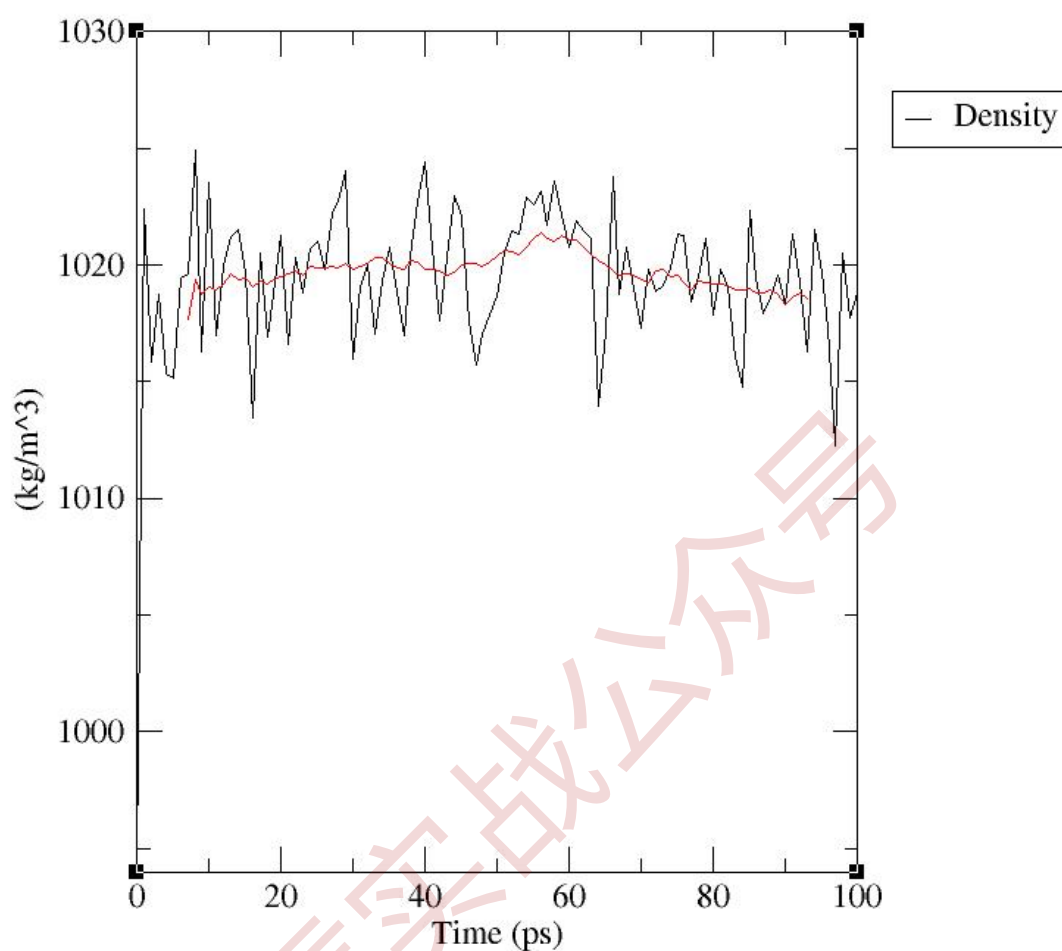
24 0
Last energy frame read 100 time 100.000

Statistics over 50001 steps [ 0.0000 through 100.0000 ps ], 1 data sets
All statistics are over 501 points

Energy                Average    Err.Est.    RMSD    Tot-Drift
-----
Density              1019.31    0.44    3.00823    1.05614 (kg/m^3)
```

```
xmgrace density.xvg
```

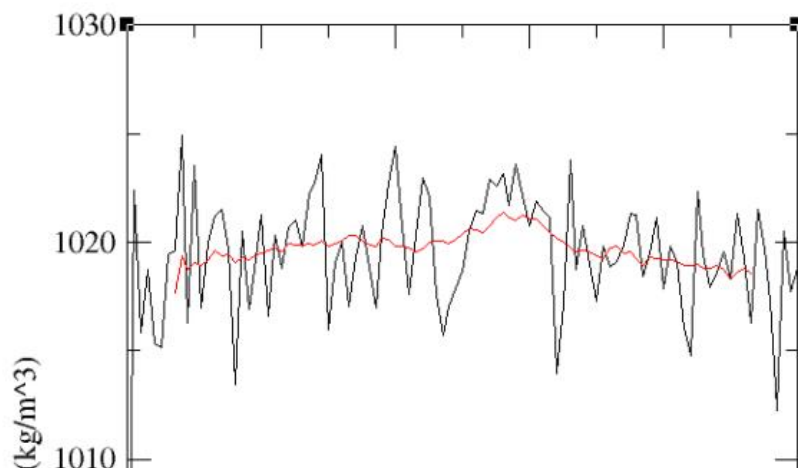
GROMACS Energies



跟压力一样，红线是密度的移动平均值。100ps过程中密度的平均值为 1019 ± 3 kg每立方米，比较接近实验值1000 kg每立方米，SPC/E水模型的密度预期值1008 kg每立方米。SPC/E水模型的参数给出的密度值接近水的实验值。在整个过程中密度值都很稳定，随着时间推移，意味着体系的压力和密度都达到了较好的平衡

6.7.7 xmgrace 显示补充

中间这条红色的线条如何画出：



在xmgrace中, 点击菜单Data -> Transformations -> Running averages, 在弹出的对话框中设定Length of average, Accept

6.8 步骤 8-正式的动力学模拟

6.8.1 实操 20-模拟准备

随着NVT和NPT两个平衡过程的完成, 体系已经在需要的温度和压强下平衡了。我们现在可以放开位置限制, 并进行正式的分子动力学模拟来采集信息。这个过程跟前面的类似, 运行grompp时, 我们还要用到检查点文件(包含了压力、耦合信息的文件)。

我们要进行1ns的分子动力学模拟, 需要使用的mdp参数参考

<http://www.mdtutorials.com/gmx/lysozyme/Files/md.mdp>

这个1ns的模拟如何指定呢

```
; Run parameters
```

```
integrator          = md          ; leap-frog integrator
```

```
nsteps              = 500000      ; 2 * 500000 = 1000 ps (1 ns)
```

```
dt = 0.002 ; 2 fs
```

模拟时间等于模拟步长乘以步数

$2 * 500000 = 1000 \text{ ps (1 ns)}$

(1秒=109纳秒=1012皮秒=1015飞秒)

6.8.2实操 21-生成模拟前的 tpr 文件

执行grompp生成模拟的tpr文件:

```
gmX grompp -f md.mdp -c npt.gro -t npt.cpt -p topol.top -o md_0_1.tpr
```

```
Command line:
gmX grompp -f md.mdp -c npt.gro -t npt.cpt -p topol.top -o md_0_1.tpr

Ignoring obsolete mdp entry 'title'
Ignoring obsolete mdp entry 'ns_type'
Setting the LD random seed to -553649379
Generated 330891 of the 330891 non-bonded parameter combinations
Generating 1-4 interactions: fudge = 0.5
Generated 330891 of the 330891 1-4 parameter combinations
Excluding 3 bonded neighbours molecule type 'Protein_chain_A'
turning H bonds into constraints...
Excluding 2 bonded neighbours molecule type 'SOL'
turning H bonds into constraints...
Excluding 1 bonded neighbours molecule type 'CL'
turning H bonds into constraints...
Analysing residue names:
There are: 129 Protein residues
There are: 10636 Water residues
There are: 8 Ion residues
Analysing Protein...
Analysing residues not classified as Protein/DNA/RNA/Water and splitting into groups...
Number of degrees of freedom in T-Coupling group Protein is 4920.79
Number of degrees of freedom in T-Coupling group non-Protein is 63837.21
Determining Verlet buffer for a tolerance of 0.005 kJ/mol/ps at 300 K
Calculated rlist for 1x1 atom pair-list as 1.035 nm, buffer size 0.035 nm
Set rlist, assuming 4x4 atom pair-list, to 1.000 nm, buffer size 0.000 nm
Note that mdrun will redetermine rlist based on the actual pair-list setup
Reading Coordinates, Velocities and Box size from old trajectory
Will read whole trajectory
Last frame -1 time 100.000
Using frame at t = 100 ps
Starting time for run is 0 ps
Calculating fourier grid dimensions for X Y Z
Using a fourier grid of 44x44x44, spacing 0.158 0.158 0.158
Estimate for the relative computational load of the PME mesh part: 0.22
This run will generate roughly 19 Mb of data
```

grompp将打印PME负载的预测值,PME负载预测可指示应该使用多少处理器进行PME计算,

多少处理器进行PP计算。例如我这里的结果是: Estimate for the relative computational

load of the PME mesh part: 0.22 (如上图红色框的)。对立方体型的晶胞容器, 最佳设置

的PME负载为0.25(3:1 PP:PME, 我们已经比较接近这个理想值); 对十二面体晶胞容器, 最佳PME负载为0.33(2:1 PP:PME)。当执行mdrun的时候, 程序会自动分配用于PP和PME计算的处理器数目。

关于PME和PP, 大家可以参考网络资料了解:

Gromacs 中主要耗时的两类任务:

1, 粒子-粒子相互作用 (particle-particle, PP)

在实空间中计算原子之间的两两相互作用力, 这些作用主要包括两部分: 非键原子间短程相互作用 (NB), 以及成键原子之间的相互作用 (BF)。在并行计算时, 对于这部分任务采用 Domain Decomposition的方法, 也就是把整个胞像切蛋糕一样切成小块, 每块交给一个Rank, 在这个Rank中, 几个Thread共同计算这个小块中的PP作用力。这里需要注意的是, 如果你的体系本身就比较小, 如果使用的Rank太多, 或者不合适, 会出现no domain decomposition compatible with the given box 的错误。

2, 粒子网格埃瓦尔德 (particle-mesh Ewald, PME)

在倒空间中使用FFT对长程作用力进行计算, 这里的“长程”和上面的“短程”是由mdp参数中的rcoulomb和rvdw参数决定的。这部分计算在倒空间中进行, 不能像实空间中的PP那样, 可以切成小块, 因此Rank并行效率较低。所以如果总Rank数较少, 那么这些Rank可以既做PP任务又做PME任务, 随着Rank数的增多, 更为合适的做法是分出少量的Rank专门做PME任务(当Rank数超过16之后, Gromacs会自动分配PME Rank, 在此之前, 可以用-npme 选项手动分配PME Rank数), 如果用显卡算PME, 目前最多只能分配一个Rank(一块显卡)来做PME计算。

6.8.3 实操 22-执行模拟

```
gmx mdrun -deffnm md_0_1
```

《生信实战》微信公众号

输出提示信息：

Command line:

```
gmx mdrun -deffnm md_0_1
```

Reading file md_0_1.tpr, VERSION 2020.7 (single precision)

Changing nstlist from 10 to 50, rlist from 1 to 1.116

Using 1 MPI thread

Non-default thread affinity set, disabling internal thread affinity

Using 12 OpenMP threads

starting mdrun 'LYSOZYME in water'

500000 steps, 1000.0 ps.

6.8.4 查看模拟的进度

CPU依然跑满

```
Tasks: 318 total, 2 running, 315 sleeping, 1 stopped, 0 zombie
%Cpu0 : 96.2 us, 0.3 sy, 0.0 ni, 3.5 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu1 : 95.5 us, 0.3 sy, 0.0 ni, 4.2 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu2 : 95.8 us, 0.3 sy, 0.0 ni, 3.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu3 : 94.8 us, 0.7 sy, 0.0 ni, 4.5 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu4 : 94.7 us, 0.7 sy, 0.0 ni, 4.6 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu5 : 95.8 us, 0.0 sy, 0.0 ni, 4.2 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu6 : 95.5 us, 0.3 sy, 0.0 ni, 4.2 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu7 : 94.8 us, 0.7 sy, 0.0 ni, 4.5 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu8 : 95.2 us, 0.3 sy, 0.0 ni, 4.5 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu9 : 94.1 us, 1.4 sy, 0.0 ni, 4.5 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu10 : 95.1 us, 0.7 sy, 0.0 ni, 4.2 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
%Cpu11 : 96.1 us, 0.0 sy, 0.0 ni, 3.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
```

我们可以估算一下运行的时间

开始运行的时间是：Started mdrun on rank 0 Wed May 24 22:37:40 2023

运行到10万步的时间是22:52:42，大概15分钟。因此要完成500000的运算，总共需要大约75分钟。

Writing checkpoint, step 92700 at Wed May 24 22:52:42 2023

Step	Time			
95000	190.00000			
Energies (kJ/mol)				
Bond	Angle	Proper Dih.	Ryckaert-Bell.	LJ-14
1.41238e+03	3.71802e+03	2.35226e+02	1.80019e+03	2.69403e+03
Coulomb-14	LJ (SR)	Disper. corr.	Coulomb (SR)	Coul. recip.
7.83002e+03	9.31370e+04	-4.54833e+03	-6.33210e+05	3.01976e+03
Potential	Kinetic En.	Total Energy	Conserved En.	Temperature
-5.23912e+05	8.56906e+04	-4.38222e+05	-4.37507e+05	2.99782e+02
Pres. DC (bar)	Pressure (bar)	Constr. rmsd		
-2.24391e+02	-1.77112e+00	3.20048e-06		
Step	Time			
100000	200.00000			
Energies (kJ/mol)				
Bond	Angle	Proper Dih.	Ryckaert-Bell.	LJ-14
1.48668e+03	3.70917e+03	2.55160e+02	1.80716e+03	2.65911e+03
Coulomb-14	LJ (SR)	Disper. corr.	Coulomb (SR)	Coul. recip.
7.62501e+03	9.42829e+04	-4.55689e+03	-6.34947e+05	2.87386e+03
Potential	Kinetic En.	Total Energy	Conserved En.	Temperature
-5.24805e+05	8.61945e+04	-4.38610e+05	-4.37469e+05	3.01545e+02
Pres. DC (bar)	Pressure (bar)	Constr. rmsd		
-2.25236e+02	6.10548e+01	2.67612e-06		

继续等待计算完成

6.8.5 模拟完成

花了76分钟完成了1纳秒的模拟，我们估算的时间75分钟还是比较准的。

最后模拟的统计结果如下：

REAL CYCLE AND TIME ACCOUNTING						
On 1 MPI rank, each using 12 OpenMP threads						
Computing:	Num Ranks	Num Threads	Call Count	Wall time (s)	Giga-Cycles total sum	%
Neighbor search	1	12	10001	78.338	3000.525	1.7
Force	1	12	500001	2086.182	79905.795	45.7
PME mesh	1	12	500001	1190.986	45617.634	26.1
NB X/F buffer ops.	1	12	990001	958.756	36722.640	21.0
Write traj.	1	12	106	0.364	13.923	0.0
Update	1	12	500001	115.236	4413.814	2.5
Constraints	1	12	500001	101.900	3903.033	2.2
Rest				35.279	1351.282	0.8
Total				4567.041	174928.646	100.0
Breakdown of PME mesh computation						
PME spread	1	12	500001	501.596	19212.344	11.0
PME gather	1	12	500001	295.631	11323.382	6.5
PME 3D-FFT	1	12	1000002	348.648	13354.046	7.6
PME solve Elec	1	12	500001	12.818	490.954	0.3
Time: Core t (s) 54804.492 Wall t (s) 4567.041 (%) 1200.0 1h16:07 (ns/day) 18.918 (hour/ns) 1.269 Performance: Finished mdrun on rank 0 Wed May 24 23:53:47 2023						

按这个计算能力，我的服务器12核CPU，1小时可以运行1.2纳米的模拟，一天可以完成18纳秒的模拟。后面就可以进行模拟轨迹数据的分析了。

6.9 步骤 9—分子动力学模拟结果分析

完成模拟之后，就到了最重要的部分：数据分析。这才是我们研究和模拟的目的。其实分析的指标很多，我们参照这个基础实验，先学习一些基础的分析

模拟后的文件如下：

```
-rw-rw-r-- 1 asp asp 10865 May 24 22:36 mdout.mdp
-rw-rw-r-- 1 asp asp 1301288 May 24 22:36 md_0_1.tpr
-rw-rw-r-- 1 asp asp 815052 May 24 23:53 md_0_1_prev.cpt
-rw-rw-r-- 1 asp asp 815052 May 24 23:53 md_0_1.cpt
```

```
-rw-rw-r-- 1 asp asp 12524284 May 24 23:53 md_0_1.xtc  
-rw-rw-r-- 1 asp asp 2337499 May 24 23:53 md_0_1.gro  
-rw-rw-r-- 1 asp asp 69912 May 24 23:53 md_0_1.edr
```

6.9.1 trjconv 优化轨迹

第一个工具处理模块是trjconv, 这是一个后处理工具, 用于处理轨迹中的坐标、去除周期性边界条件、还可以手动调整轨迹的时间单位和帧频率等。这里我们使用trjconv来处理体系中的周期性边界问题。其实在模拟的过程中, 蛋白质在晶胞中扩散运动的过程中, 可能看起来会出现“断裂”或“跳跃”到体系的另外一边的情况。

使用下面的工具来进行修复:

```
gmx trjconv -s md_0_1.tpr -f md_0_1.xtc -o md_0_1_noPBC.xtc -pbc mol -center
```

```

Command line:
gmX trjconv -s md_0_1.tpr -f md_0_1.xtc -o md_0_1_noPBC.xtc -pbc mol -center

Note that major changes are planned in future for trjconv, to improve usability and utility.
Will write xtc: Compressed trajectory (portable xdr format): xtc
Reading file md_0_1.tpr, VERSION 2020.7 (single precision)
Reading file md_0_1.tpr, VERSION 2020.7 (single precision)
Select group for centering
Group 0 (      System) has 33876 elements
Group 1 (      Protein) has 1960 elements
Group 2 (    Protein-H) has 1001 elements
Group 3 (      C-alpha) has 129 elements
Group 4 (    Backbone) has 387 elements
Group 5 (    MainChain) has 517 elements
Group 6 (  MainChain+Cb) has 634 elements
Group 7 (  MainChain+H) has 646 elements
Group 8 (    SideChain) has 1314 elements
Group 9 (  SideChain-H) has 484 elements
Group 10 (  Prot-Masses) has 1960 elements
Group 11 (   non-Protein) has 31916 elements
Group 12 (      Water) has 31908 elements
Group 13 (      SOL) has 31908 elements
Group 14 (   non-Water) has 1968 elements
Group 15 (      Ion) has 8 elements
Group 16 (      CL) has 8 elements
Group 17 (Water and ions) has 31916 elements
Select a group: 1
Selected 1: 'Protein'
Select group for output
Group 0 (      System) has 33876 elements
Group 1 (      Protein) has 1960 elements
Group 2 (    Protein-H) has 1001 elements
Group 3 (      C-alpha) has 129 elements
Group 4 (    Backbone) has 387 elements
Group 5 (    MainChain) has 517 elements
Group 6 (  MainChain+Cb) has 634 elements
Group 7 (  MainChain+H) has 646 elements
Group 8 (    SideChain) has 1314 elements
Group 9 (  SideChain-H) has 484 elements
Group 10 (  Prot-Masses) has 1960 elements
Group 11 (   non-Protein) has 31916 elements
Group 12 (      Water) has 31908 elements
Group 13 (      SOL) has 31908 elements
Group 14 (   non-Water) has 1968 elements
Group 15 (      Ion) has 8 elements
Group 16 (      CL) has 8 elements
Group 17 (Water and ions) has 31916 elements
Select a group: 0
Selected 0: 'System'
Reading frame 0 time 0.000
Precision of md_0_1.xtc is 0.001 (nm)
Using output precision of 0.001 (nm)
Last frame 100 time 1000.000 --> frame 100 time 1000.000
    
```

Select 1 ("Protein") as the group to be centered and 0 ("System") for output

完成后生成md_0_1_noPBC.xtc, 我们将基于这个修复的数据进行后续的数据分析

6.9.2 计算 RMSD 与分析

关于RMSD的统计学意义Root Mean Square Deviation均方根偏差, 模拟中计算这个值,

主要是为了衡量模拟过程中结构波动情况。

可以先看看模拟结构的稳定性:

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```
gmx rms -s md_0_1.tpr -f md_0_1_noPBC.xtc -o rmsd.xvg -tu ns
```

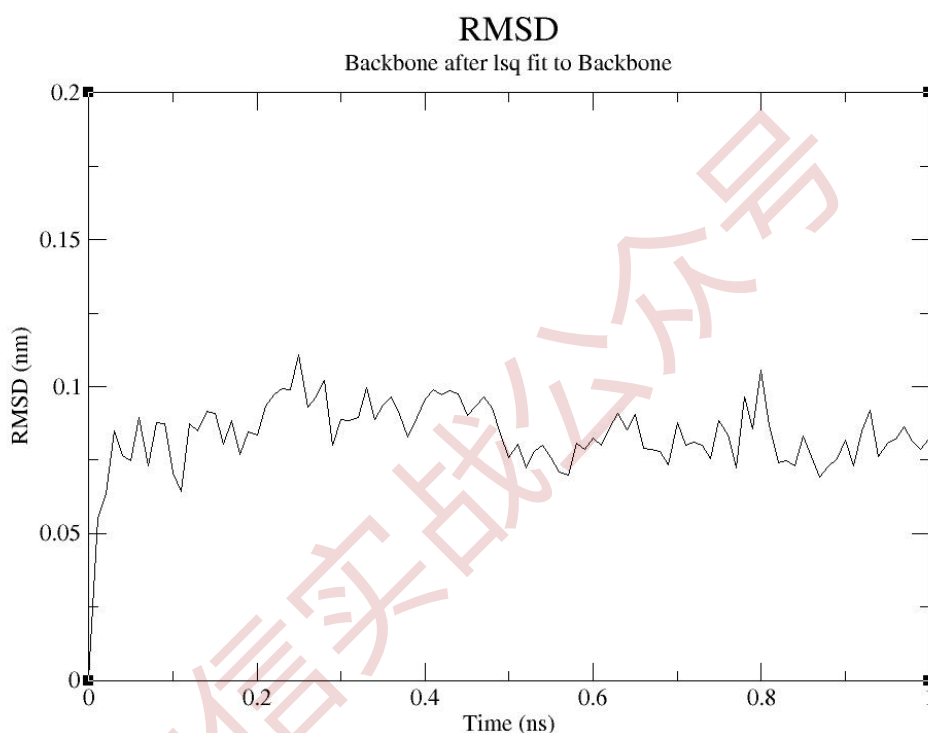
```
Command line:
gmx rms -s md_0_1.tpr -f md_0_1_noPBC.xtc -o rmsd.xvg -tu ns

Reading file md_0_1.tpr, VERSION 2020.7 (single precision)
Reading file md_0_1.tpr, VERSION 2020.7 (single precision)
Select group for least squares fit
Group 0 (      System) has 33876 elements
Group 1 (      Protein) has 1960 elements
Group 2 (      Protein-H) has 1001 elements
Group 3 (      C-alpha) has 129 elements
Group 4 (      Backbone) has 387 elements
Group 5 (      MainChain) has 517 elements
Group 6 (      MainChain+Cb) has 634 elements
Group 7 (      MainChain+H) has 646 elements
Group 8 (      SideChain) has 1314 elements
Group 9 (      SideChain-H) has 484 elements
Group 10 (     Prot-Masses) has 1960 elements
Group 11 (     non-Protein) has 31916 elements
Group 12 (      Water) has 31908 elements
Group 13 (      SOL) has 31908 elements
Group 14 (     non-Water) has 1968 elements
Group 15 (      Ion) has 8 elements
Group 16 (      CL) has 8 elements
Group 17 (     Water and ions) has 31916 elements
Select a group: 4
Selected 4: 'Backbone'
Select group for RMSD calculation
Group 0 (      System) has 33876 elements
Group 1 (      Protein) has 1960 elements
Group 2 (      Protein-H) has 1001 elements
Group 3 (      C-alpha) has 129 elements
Group 4 (      Backbone) has 387 elements
Group 5 (      MainChain) has 517 elements
Group 6 (      MainChain+Cb) has 634 elements
Group 7 (      MainChain+H) has 646 elements
Group 8 (      SideChain) has 1314 elements
Group 9 (      SideChain-H) has 484 elements
Group 10 (     Prot-Masses) has 1960 elements
Group 11 (     non-Protein) has 31916 elements
Group 12 (      Water) has 31908 elements
Group 13 (      SOL) has 31908 elements
Group 14 (     non-Water) has 1968 elements
Group 15 (      Ion) has 8 elements
Group 16 (      CL) has 8 elements
Group 17 (     Water and ions) has 31916 elements
Select a group: 4
Selected 4: 'Backbone'
Last frame      100 time      1.000
```

Choose 4 ("Backbone") for both the least-squares fit and the group for RMSD calculation

-tu 选项设定输出结果的时间单位为ns, 即便轨迹文件以ps为单位输出的。主要是为了使得输出文件更清晰(比如当模拟时间很长时, 100 ns显示起来比比起100000ps更美观一点)。

完成后生成了rmsd.xvg, 使用命令xmgrace rmsd.xvg查看



大概在0.05ns达到稳定状态后, 整体波动性范围在0.075左右, 波动比较小, 可以认为是比较稳定的。

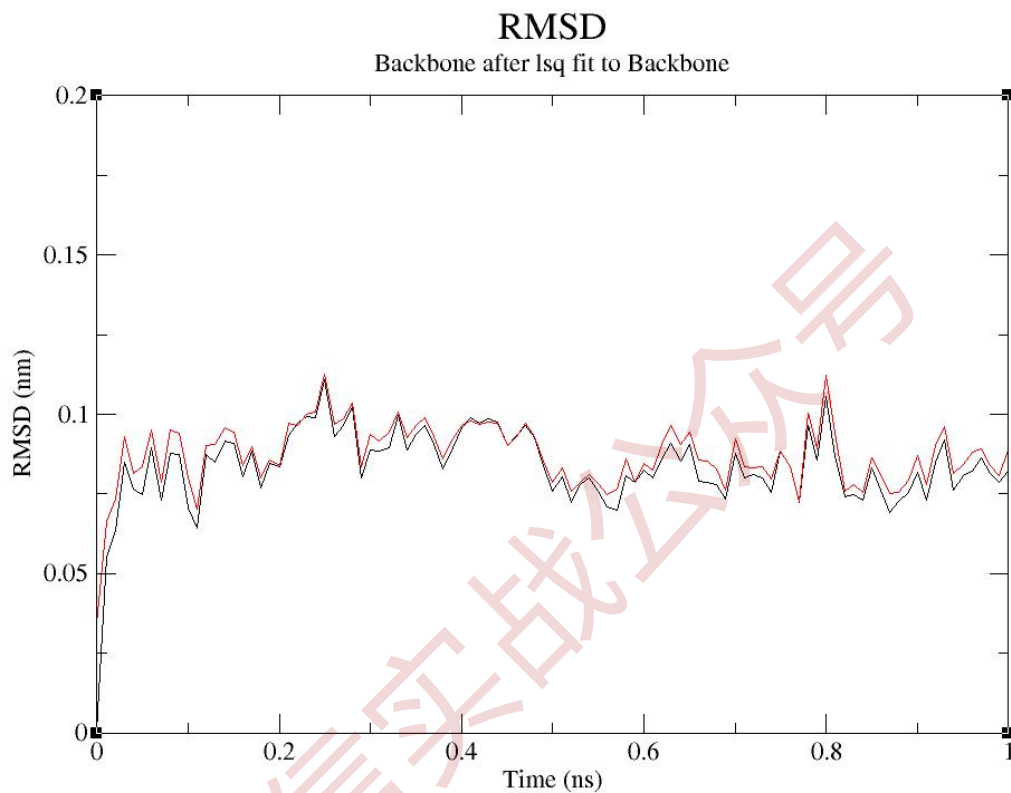
如果想计算相对于模拟之前晶体的结构差异, 可以使用下面的命令:

```
gmx rms -s em.tpr -f md_0_1_noPBC.xtc -o rmsd_xtal.xvg -tu ns
```

也是选择4 ("Backbone")

可以把两个RMSD拟合一下查看：

```
xmgrace rmsd.xvg rmsd_xtal.xvg
```



上面两个图都显示出RMSD大约是0.1nm，这表示蛋白质的结构非常稳定。

红色曲线是原来晶体的，黑色曲线是模拟后的。

6.9.3gyrate 计算回旋半径 Rg

蛋白质的回旋半径Rg可衡量其紧密程度，如果蛋白质的折叠很稳定，其回旋半径Rg将保持一个相对稳定的值。如果蛋白质去折叠，那么它的Rg将随时间变化。

```
Linux执行：gmx gyrate -s md_0_1.tpr -f md_0_1_noPBC.xtc -o gyrate.xvg
```

```

Command line:
  gmx gyrate -s md_0_1.tpr -f md_0_1_noPBC.xtc -o gyrate.xvg

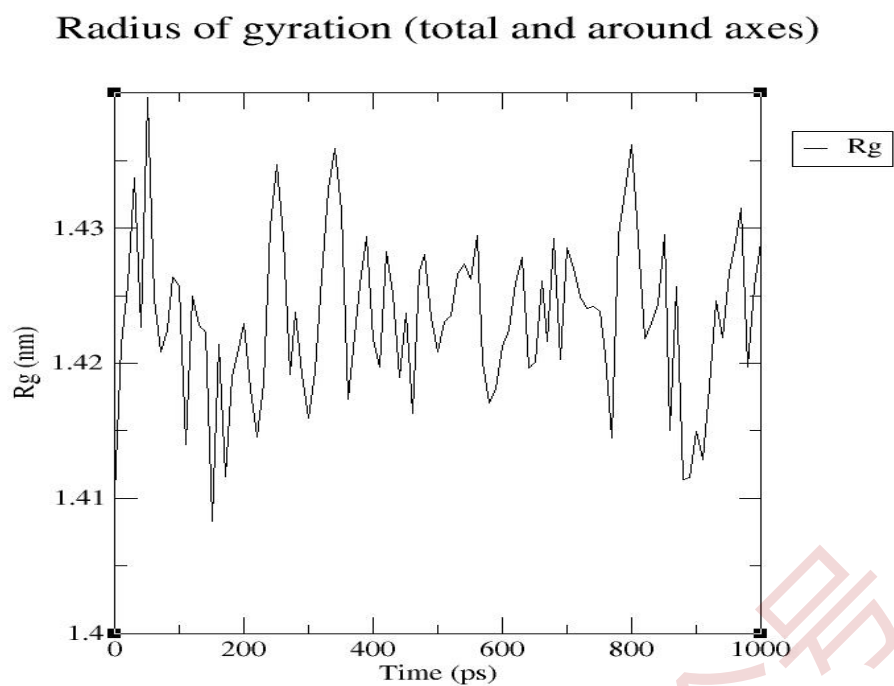
Reading file md_0_1.tpr, VERSION 2020.7 (single precision)
Reading file md_0_1.tpr, VERSION 2020.7 (single precision)
Group      0 (      System) has 33876 elements
Group      1 (      Protein) has  1960 elements
Group      2 (    Protein-H) has  1001 elements
Group      3 (      C-alpha) has   129 elements
Group      4 (    Backbone) has   387 elements
Group      5 (    MainChain) has   517 elements
Group      6 (  MainChain+Cb) has   634 elements
Group      7 (    MainChain+H) has   646 elements
Group      8 (    SideChain) has  1314 elements
Group      9 (    SideChain-H) has   484 elements
Group     10 (    Prot-Masses) has  1960 elements
Group     11 (    non-Protein) has 31916 elements
Group     12 (      Water) has 31908 elements
Group     13 (      SOL) has 31908 elements
Group     14 (    non-Water) has  1968 elements
Group     15 (      Ion) has     8 elements
Group     16 (      CL) has     8 elements
Group     17 (  Water and ions) has 31916 elements
Select a group: 1
Selected 1: 'Protein'
Last frame      100 time 1000.000
  
```

选择group 1 (Protein)

执行完成后新生成了gyrate.xvg

```

asp@coding_gene94_ubuntu16 10:25:37 ~/mdtest/demol
$ ls
1AKI_clean.pdb      gyrate.xvg          minim.mdp           nvt.mdp
1AKI_newbox.gro     ions.mdp             nohup.out           nvt.tpr
1aki.pdb            ions.tpr              npt.cpt             nvt.trr
1AKI_processed.gro  md_0_1.cpt           npt.edr             posre.itp
1AKI_solv.gro       md_0_1.edr           npt.gro             potential.xvg
1AKI_solv_ions.gro  md_0_1.gro           npt.log             pressure.xvg
density2.xvg        md_0_1.log           npt.mdp             rmsd_xtal.xvg
density.xvg         md_0_1_noPBC.xtc     npt.tpr             rmsd.xvg
em.edr              md_0_1_prev.cpt      npt.trr             temperature.xvg
em.gro              md_0_1.tpr           nvt.cpt             topol.top
em.log              md_0_1.xtc           nvt.edr             #topol.top.1#
em.tpr              md.mdp               nvt.gro             #topol.top.2#
em.trr              mdout.mdp            nvt.log
asp@coding_gene94_ubuntu16 10:25:39 ~/mdtest/demol
$ xmgrace gyrate.xvg
  
```

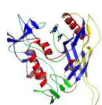


从这个图可以看出 R_g 值基本不变,总体在1.41nm和1.43nm左右,这说明在温度为300K时, 1ns的时间内, 蛋白质整体结构比较稳定, 处于紧密折叠的状态。



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上海 黄浦



扫一扫上面的二维码图案，加我为朋友。